

Gestion de portefeuille

« Investing in commodity markets »

R.Feraud



1. Overview of commodity markets
2. Understanding commodity indices
3. Different types of investment vehicles

1. Overview

There are currently about 50 major commodity markets worldwide.

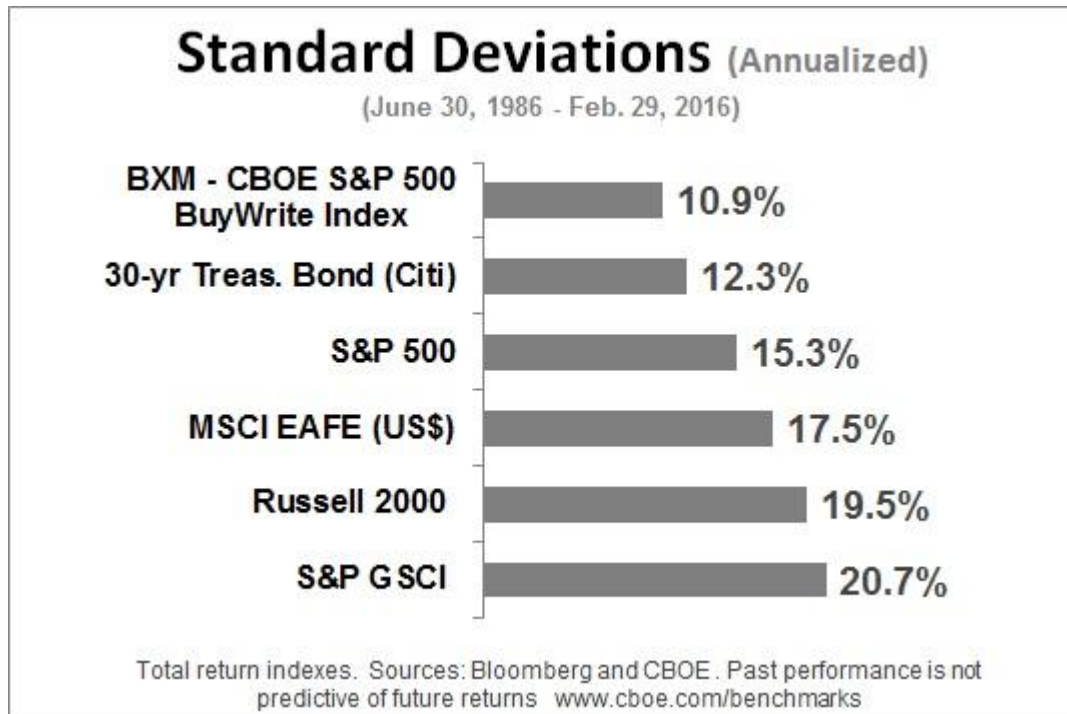
Major commodity prices are denominated in dollar (USD).

Commodities are split into two types: hard and soft commodities

Hard commodities are typically natural resources that must be mined or extracted (copper, gold, oil, natural gas, etc...).

Soft commodities are agricultural products or livestock (corn, wheat, coffee, sugar, soybeans, hog, etc...).

Commodities markets are generally subject to more extreme price movements than other traditional asset classes, such as equities or fixed income



Especially energy markets (ex crude oil)



Source: BBG

Investing in commodity markets

During crisis (ex crude oil : 500% volatility)



Source: BBG

Investing in commodity markets

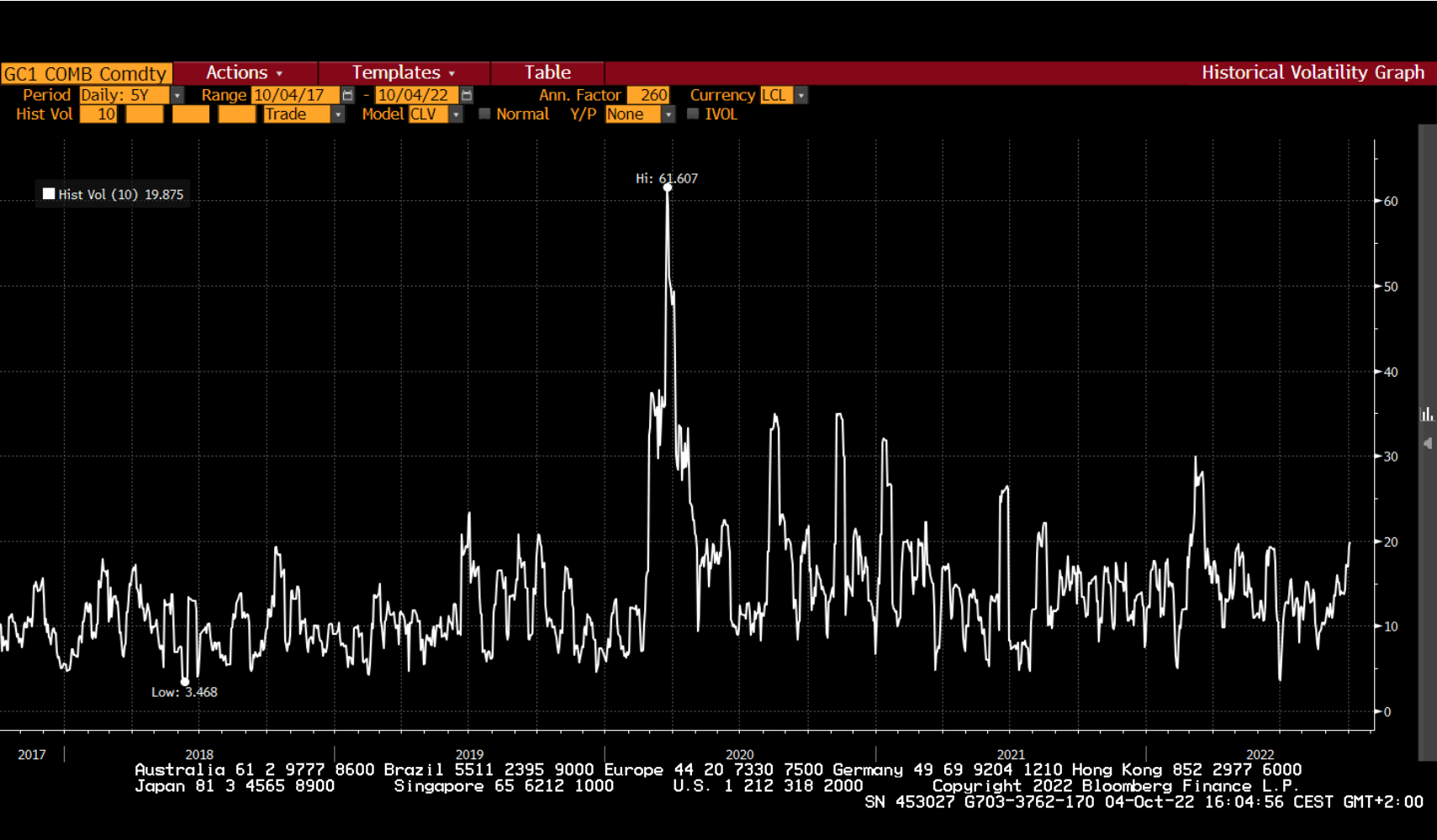
Metals are less volatile (ex gold)



Source: BBG

Investing in commodity markets

During crisis (ex gold)

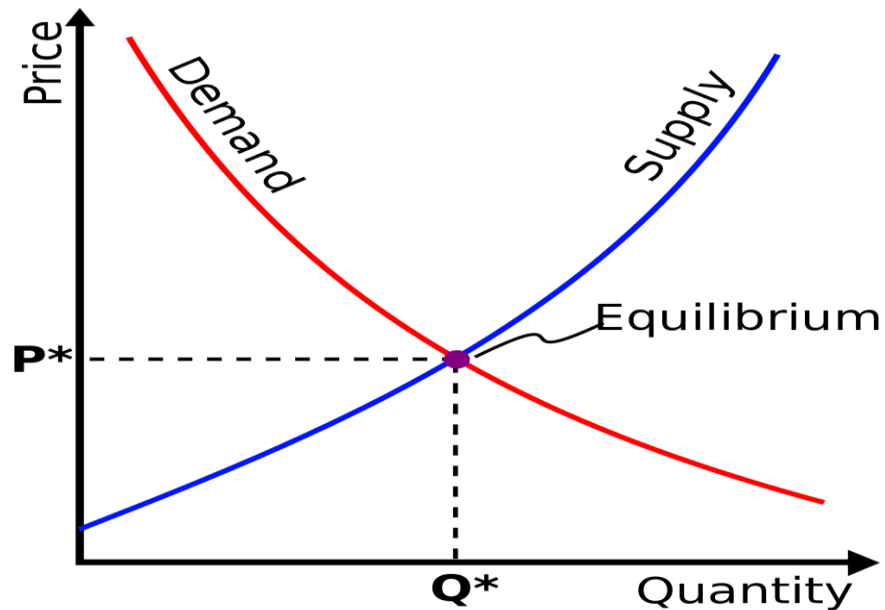


Source: BBG

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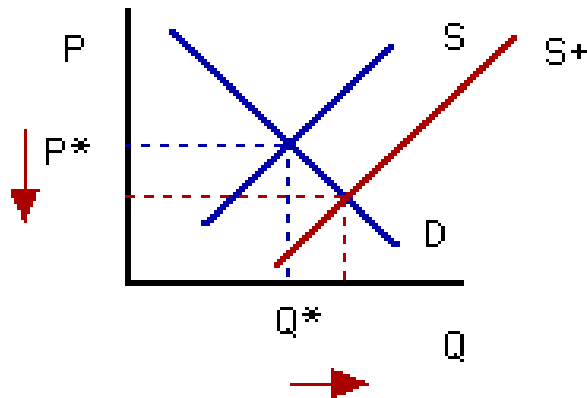
1.1 Main price drivers

Main price drivers have a direct impact on supply and demand

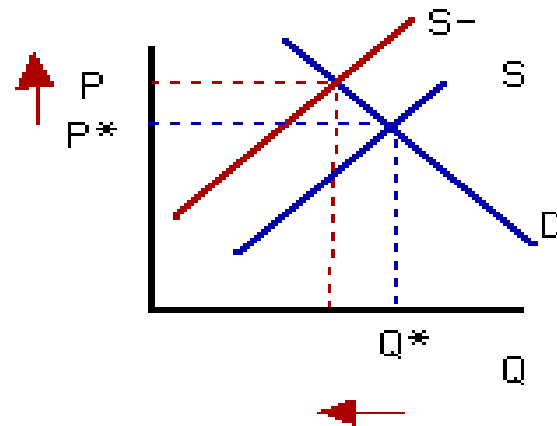


Supply shifts

Increase in Supply

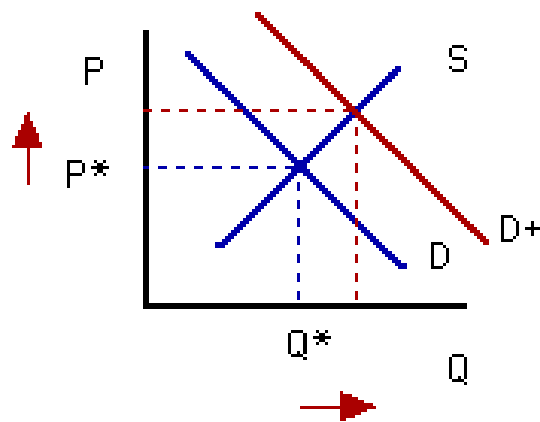


Decrease in Supply

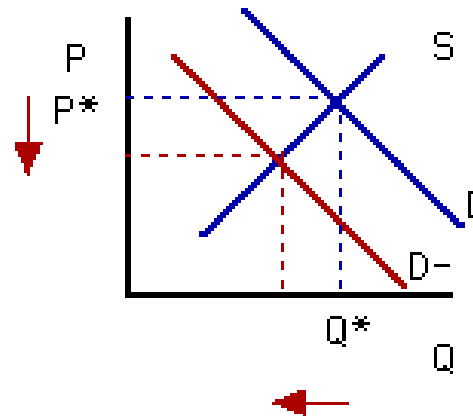


Demand shifts

Increase in Demand



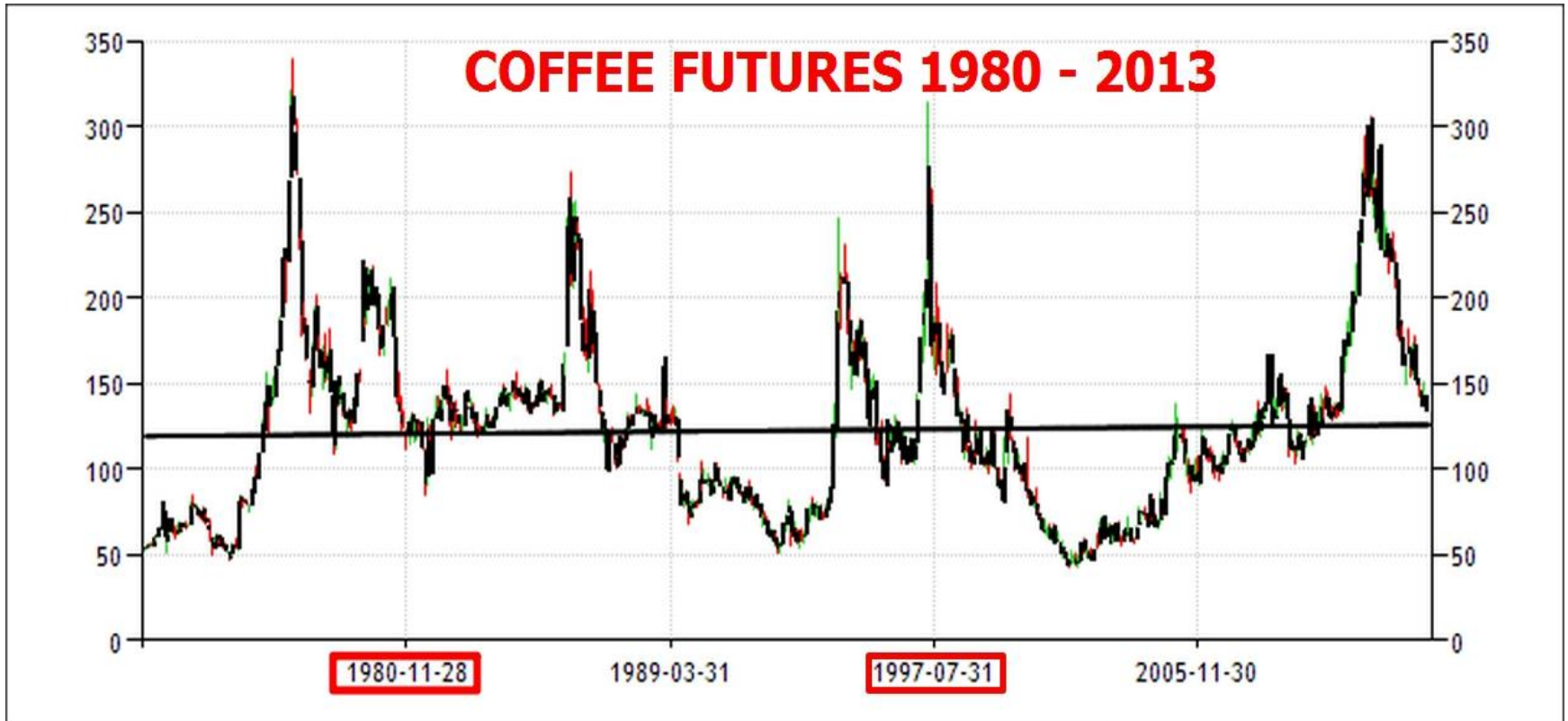
Decrease in Demand



Supply and Demand shifts

1. Weather factor
2. Geopolitical factor
3. Economic and demographic factors
4. Currency and speculation factors

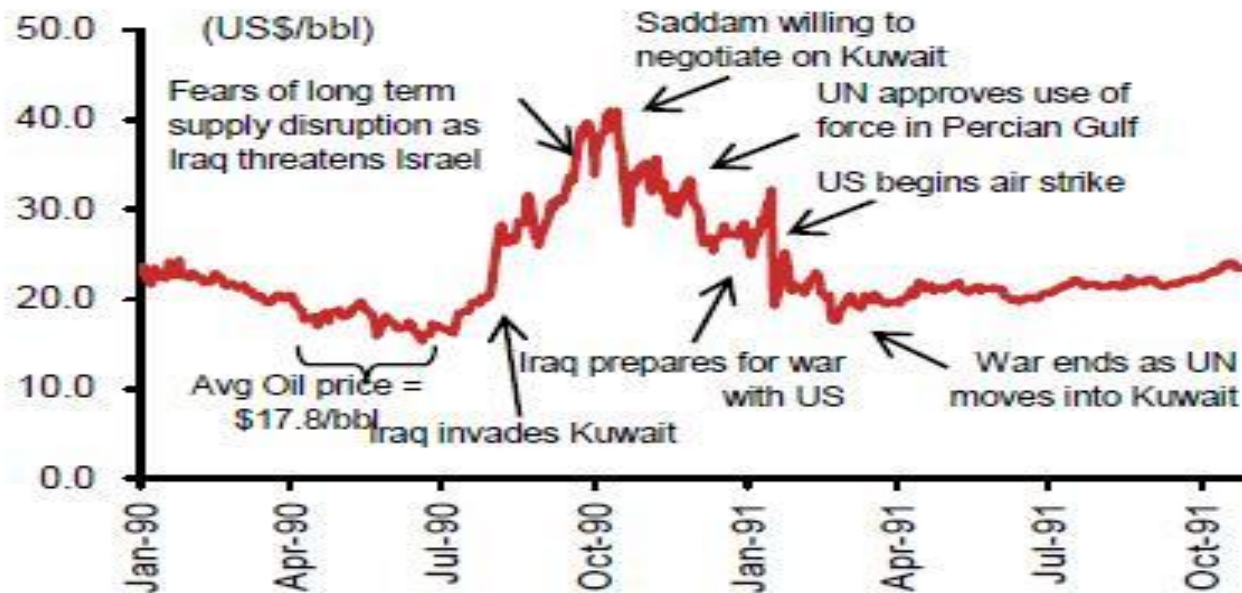
Coffee prices (in Cents/lb)



Source: CQG

WTI* oil prices (in USD/bbl)

Exhibit 8. Oil prices during 1990-91 Gulf War



Source: Bloomberg, Nomura Research

*West Texas Intermediate

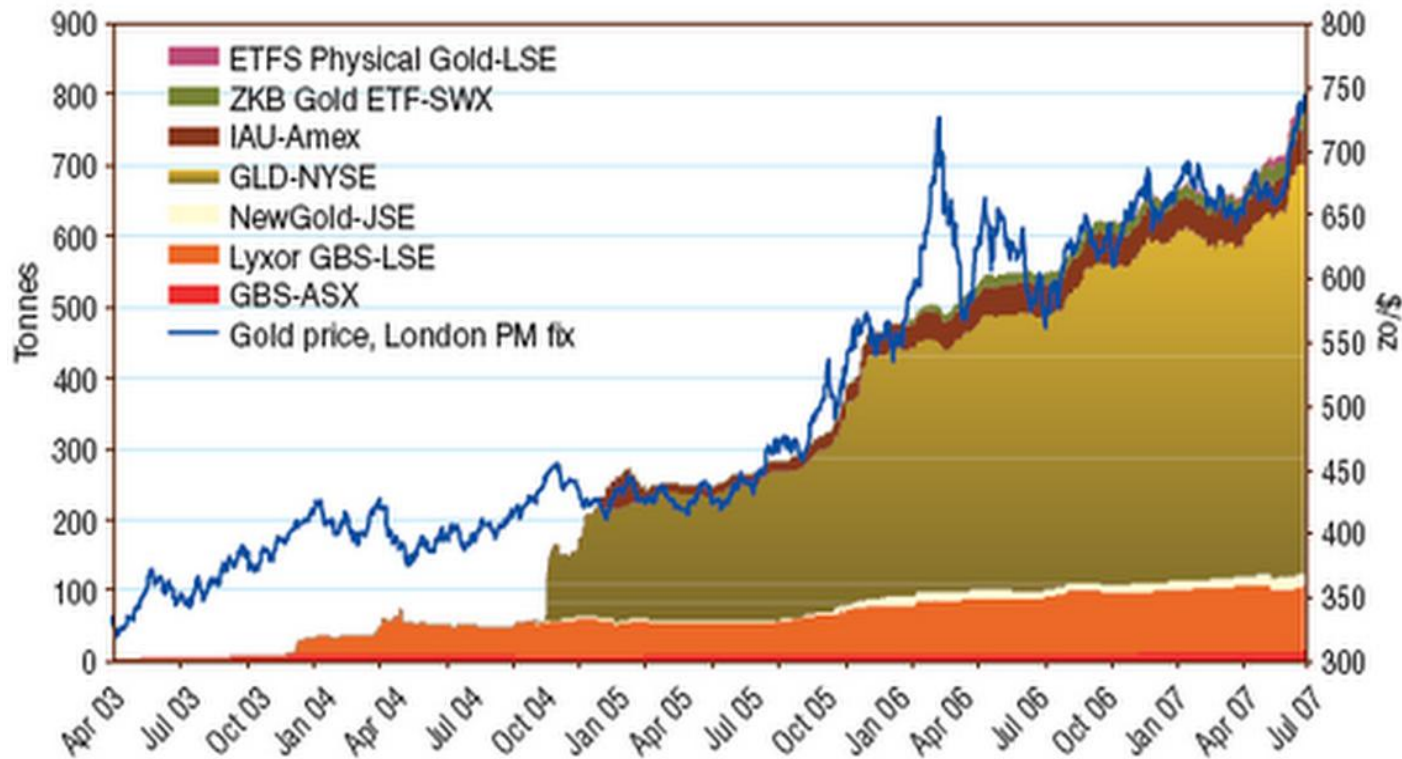
Correlation between copper prices and China



Source: Goldman Sachs, Bloomberg

ETF holdings : gold example

Gold ETF holdings: tonnes

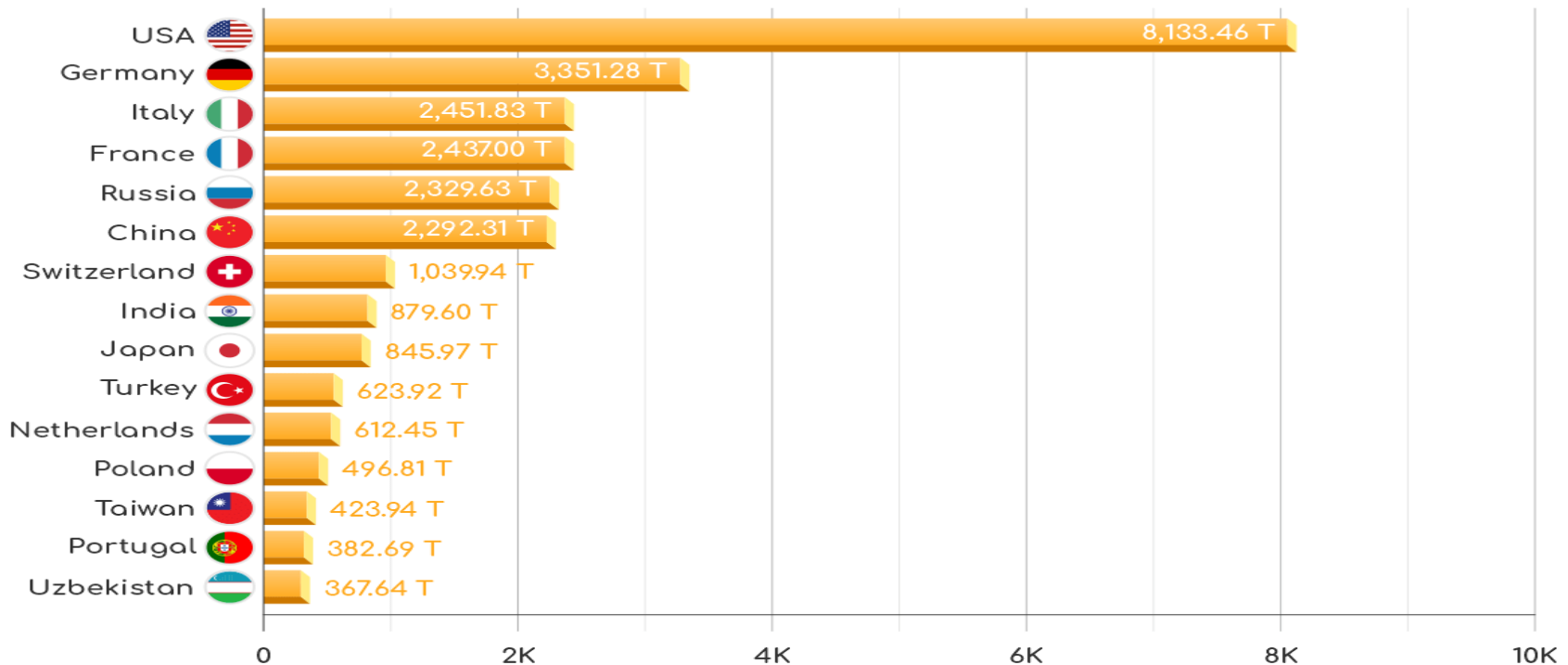


Data: www.ishares.com; www.exchangetradedgold.com; Zurich Kantonalbank; Finans Portföy; Bloomberg; Global Insight Chart: World Gold Council, www.gold.org

Source: SA

Top Gold holders (countries)

Countries with the Largest Gold Reserves as of March 2025



Data Sources: WGC, IMF International Financial Statistics

 BestBrokers.com

Investing in commodity markets

1.2 Market trends

Commodity performance in 2025

Save Defaults		Restore Defaults		Global Commodity Prices													
Monthly Spreads		Mov Avg		Performance		AVAT vs 10d		%Chg YTD		USD							
Commodity		2Day		Price		Norm Chg		%Chg		Δ AVAT		Time		%YTD		%YTDCur	
1) Energy																	
2) NYM WTI Crude				56.23		+0.96		+1.74%		-56.43%		17:22		-21.60%		-21.60%	
3) ICE Brent Crude				59.92		+1.00		+1.70%		+38.62%		17:22		-19.72%		-19.72%	
4) ICE ARA Gasoil				614.25		+6.00		+0.99%		-4.07%		17:22		-11.65%		-11.65%	
5) NYM NYH Gasoline				170.18		+2.09		+1.24%		-15.10%		17:22		-14.97%		-14.97%	
6) NYM NYH Heating Oil				213.82		+0.96		+0.45%		-29.36%		17:22		-7.86%		-7.86%	
7) NYM HH Nat Gas				3.97		+0.09		+2.26%		-27.47%		17:22		+9.39%		+9.39%	
8) ICE TTF Nat Gas				27.32		+0.55		+2.06%		+10.50%		17:22		-44.13%		-36.59%	
9) Metals																	
10) LME Aluminum				2,876.50		--		--		--		12/16/25		+12.74%		+12.74%	
11) LME Copper				11,592.00		--		--		--		12/16/25		+32.21%		+32.21%	
12) LME Nickel				14,263.00		--		--		--		12/16/25		-6.95%		-6.95%	
13) DCE Iron Ore				778.00		+13.00		+1.70%		-39.25%		16:00		+2.44%		+6.16%	
14) SHF Steel Rebar				3,125.00		+43.00		+1.40%		-3.17%		16:00		-5.09%		-1.64%	
15) Spot Gold				4,325.63		+23.31		+0.54%		--		17:32		+64.82%		+64.82%	
16) Spot Silver				65.74		+1.99		+3.12%		--		17:32		+127.47%		+127.47%	
17) Agriculture																	
18) CME Live Cattle				230.70		+0.00		+0.00%		-74.85%		17:22		+19.23%		+19.23%	
19) ICE Coffee				347.90		-4.20		-1.19%		+143.48%		17:22		+11.70%		+11.70%	
20) CBT Corn				439.25		+2.75		+0.63%		+0.39%		17:22		-4.20%		-4.20%	
21) ICE Cotton				63.45		+0.35		+0.55%		+37.44%		17:22		-7.24%		-7.24%	
22) CBT Soybeans				1,067.50		-4.25		-0.40%		-3.68%		17:22		+5.84%		+5.84%	
23) ICE Sugar				14.80		-0.02		-0.14%		-20.00%		17:22		-23.16%		-23.16%	
24) CBT SRW Wheat				506.75		-2.75		-0.54%		+55.10%		17:22		-8.11%		-8.11%	

Source: BBG

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Explanation – on the way up

-Silver : stronger demand (investors, batteries, photovoltaic panels, etc...)

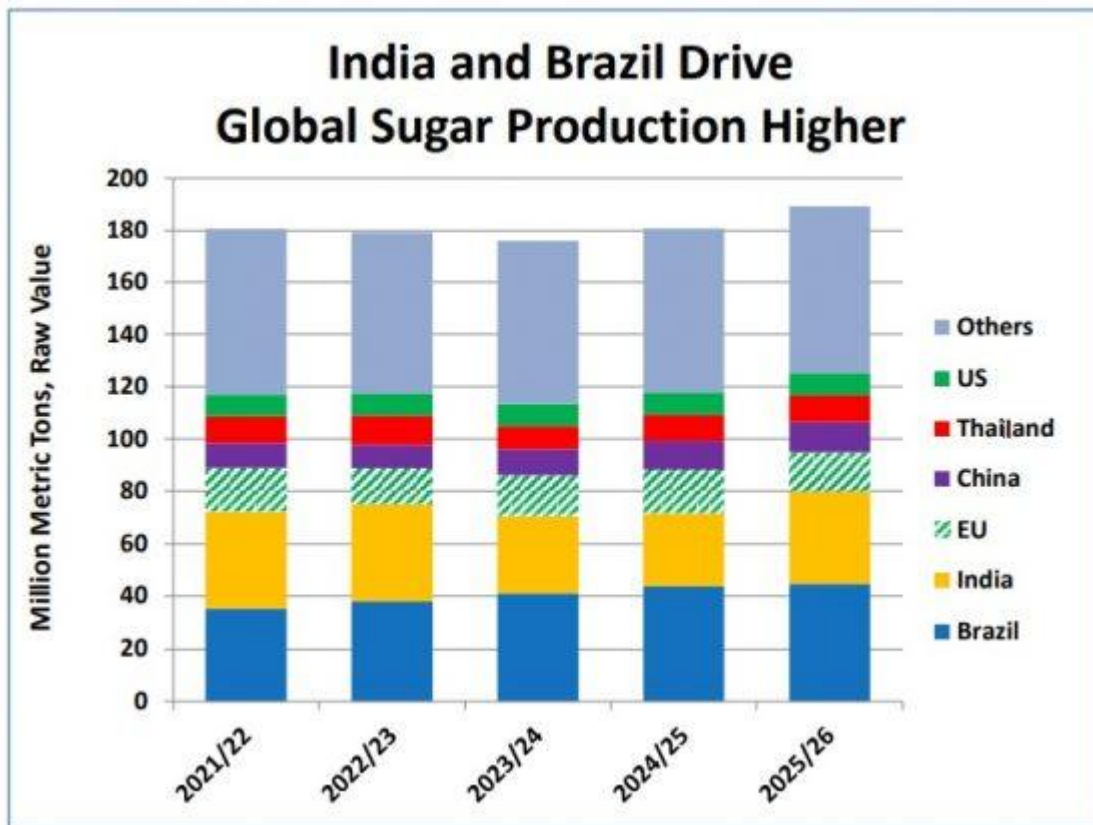


Source: BBG

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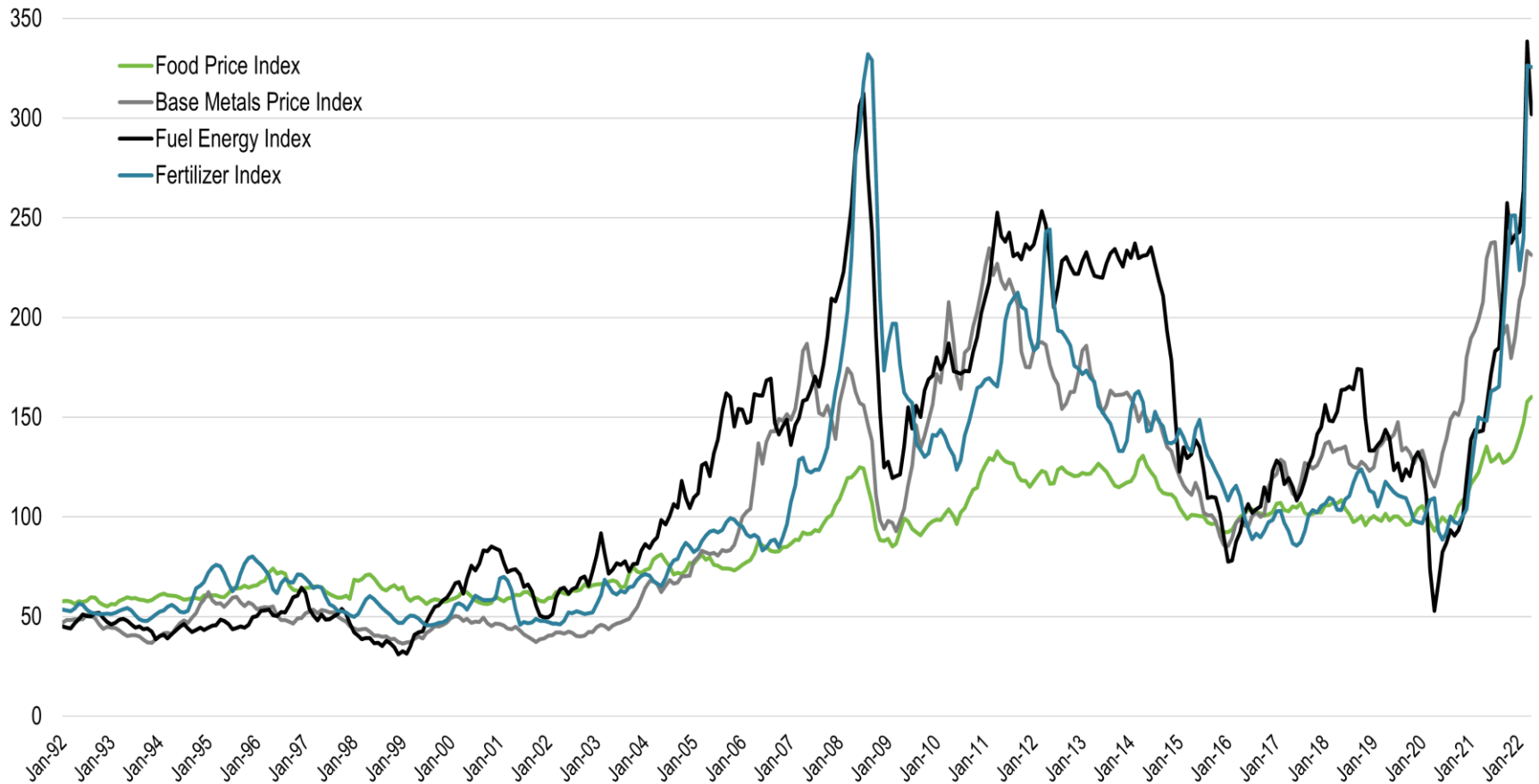
Explanation – on the way down

-Sugar : Oversupply (strong harvests in Brazil and India) and weaker demand



Source: SAM

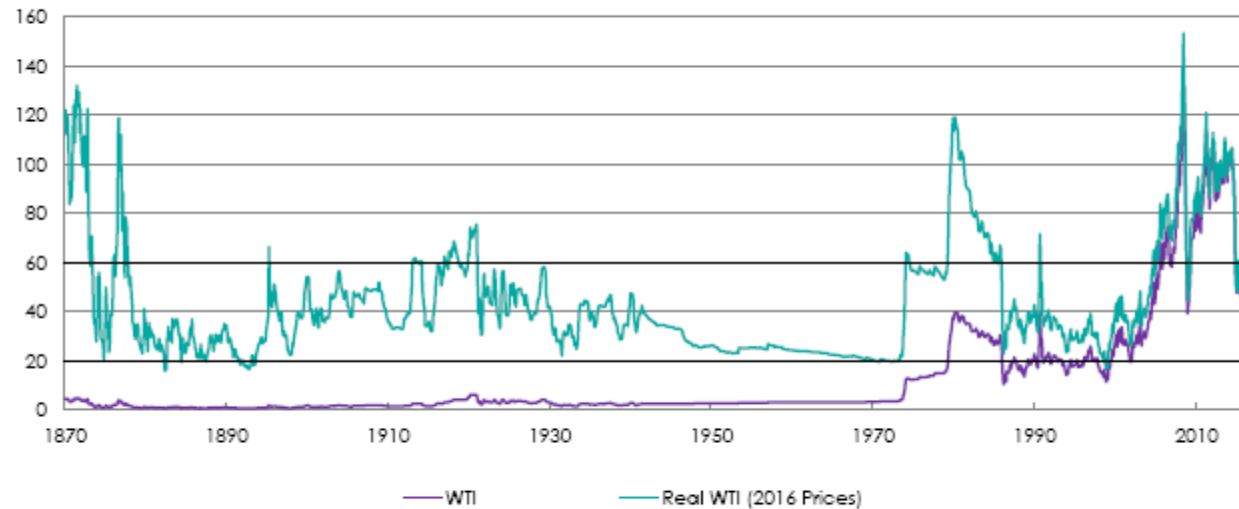
Long term commodity prices



Source : World Bank

A focus on oil

US real oil price since 1870



Source: Global Financial Data, Datastream and Source Research

Seven Sisters

The “Seven Sisters” of oil is a historical nickname given to the seven major oil companies that dominated almost the entire global oil market from the mid-20th century until the 1970s :

Exxon (formerly Standard Oil of New Jersey), Mobil (formerly Standard Oil of New York), Chevron (formerly Standard Oil of California), Gulf Oil, Texaco, BP (British Petroleum, formerly the Anglo-Persian Oil Company), Royal Dutch Shell.

Today, Exxon and Mobil have merged (ExxonMobil), Gulf has disappeared, and Texaco has been absorbed by Chevron.

Seven Sisters – the end

Why were they so powerful?

They set oil prices

They controlled access to oil fields in the Middle East

They carried more weight than many producing states

As a result, producing countries often received only a minimal share of the profits.

Why did it come to an end?

In the 1970s:

Nationalizations (Iran, Saudi Arabia, Venezuela, etc.)

The creation and rise of OPEC

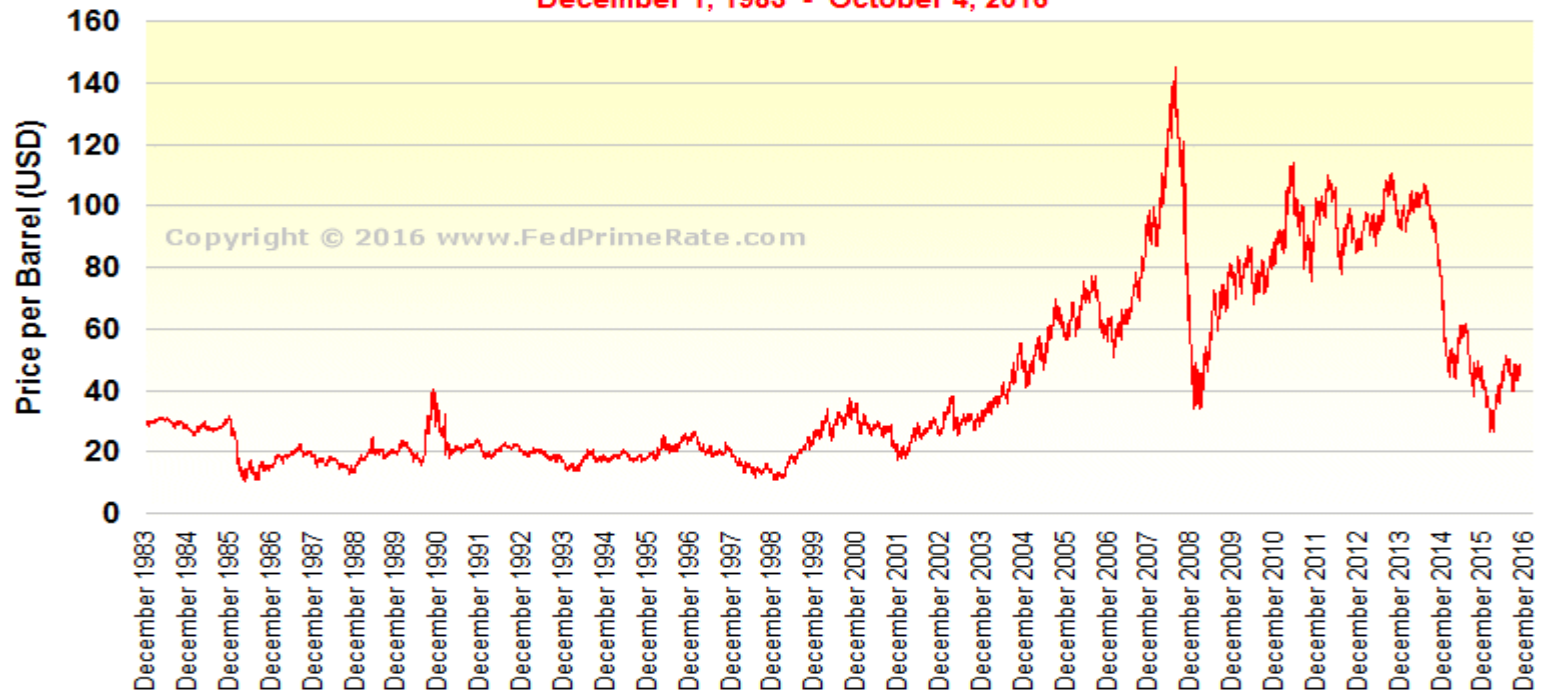
The oil shocks of 1973 and 1979

Recent oil prices (WTI)

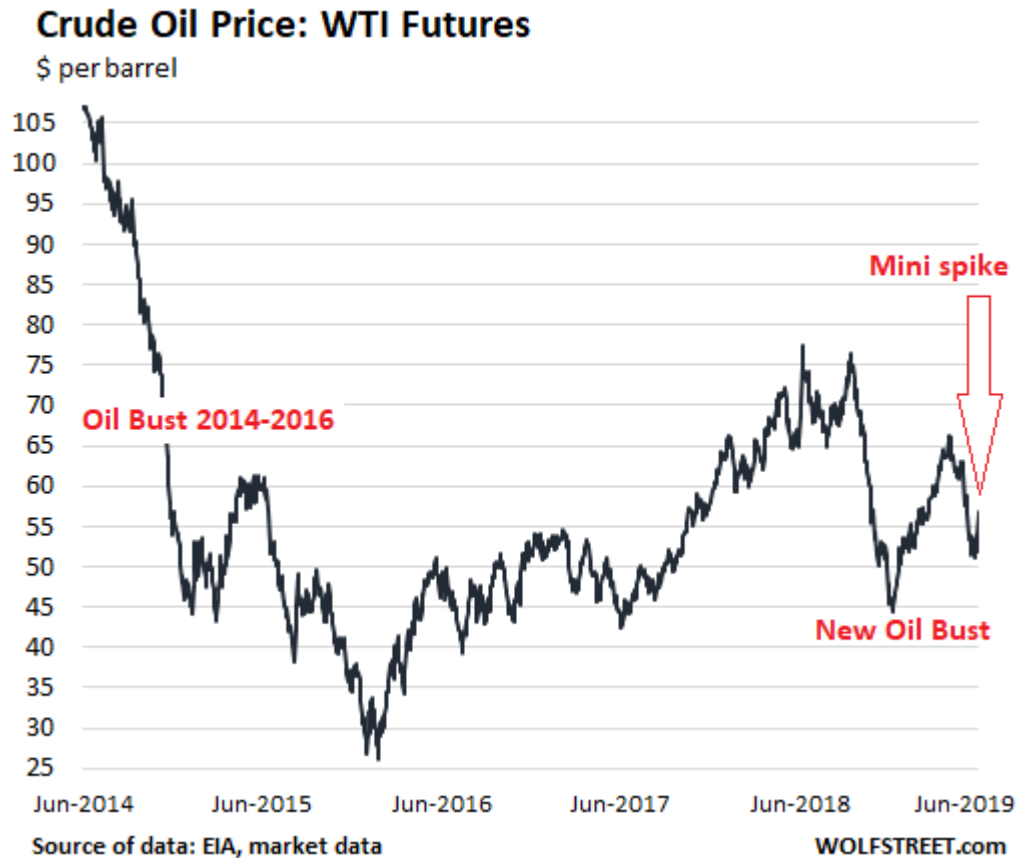
Crude Oil Price History

NYMEX - West Texas Intermediate (WTI)

December 1, 1983 - October 4, 2016

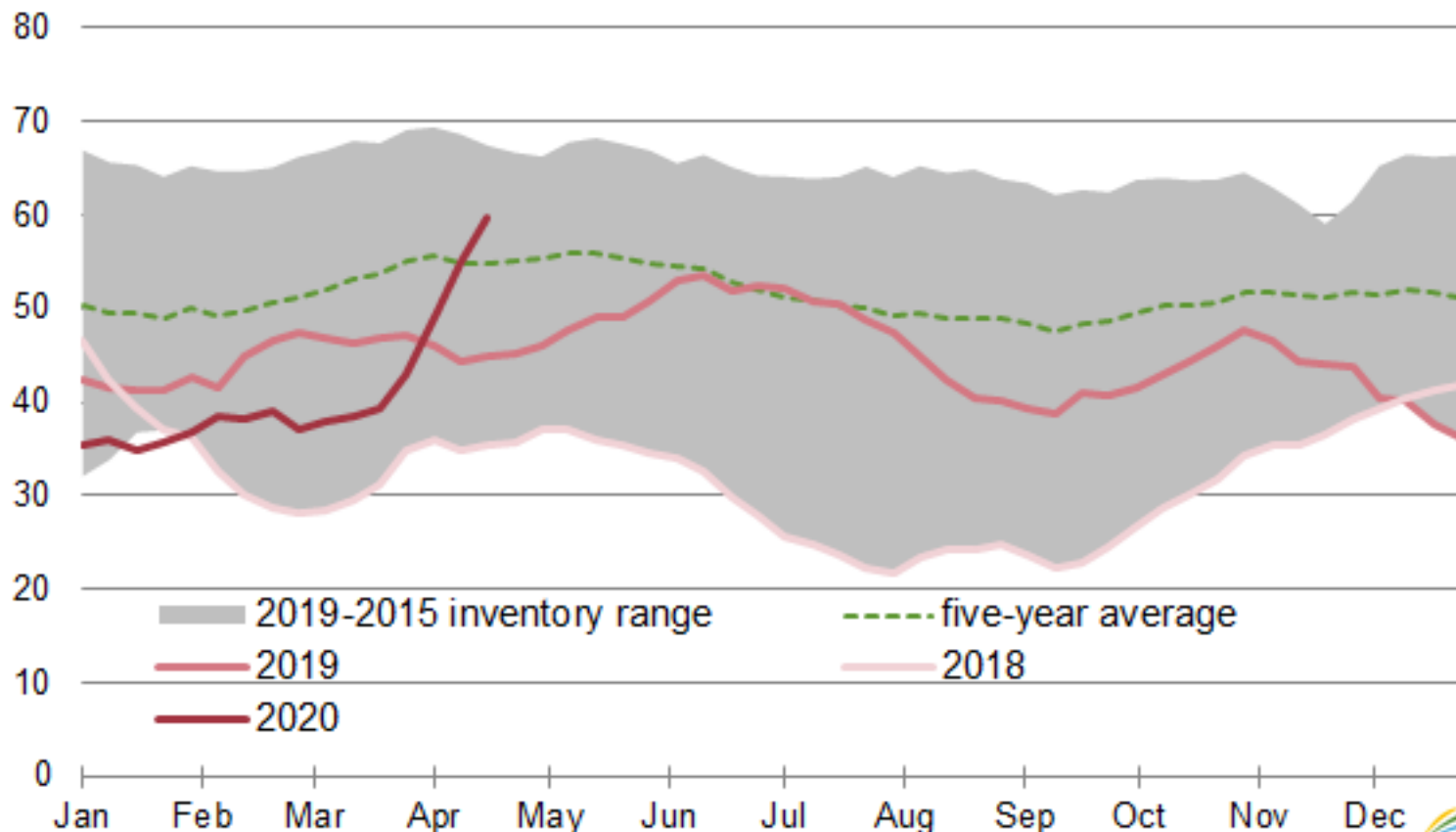


Oil prices (WTI) since 2014



Covid 19 - Too much oil in Q1 2020

Figure 2. Cushing, Oklahoma commercial crude oil inventories
million barrels

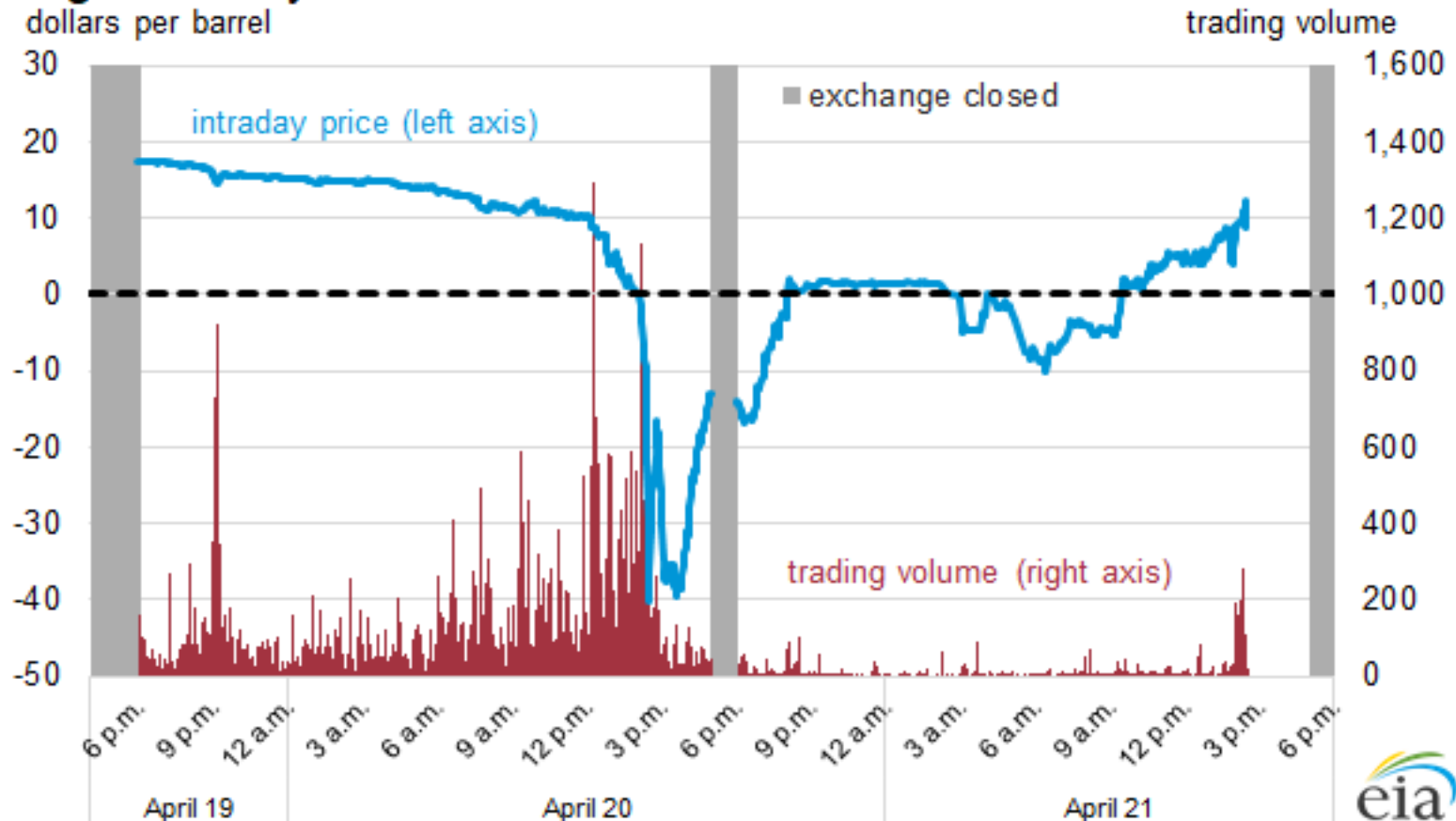


Source: U.S. Energy Information Administration, *Weekly Petroleum Status Report*.

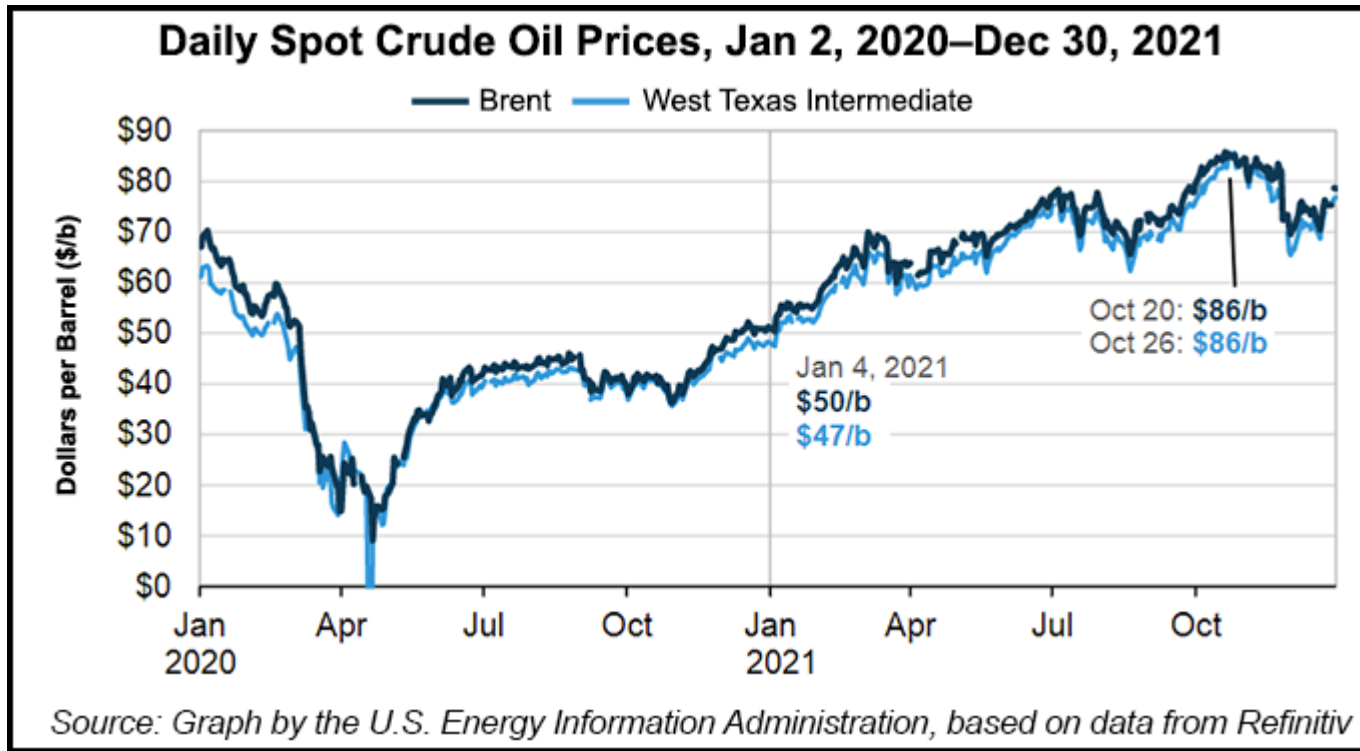


And negative prices

Figure 1. May 2020 West Texas Intermediate futures contract



But under investing → prices go up again



A surge with war in Ukraine

Soaring Crude

Oil futures have surged since Russia invaded Ukraine



Prices should remain high

West Texas Intermediate (WTI) crude oil price and NYMEX confidence intervals
dollars per barrel



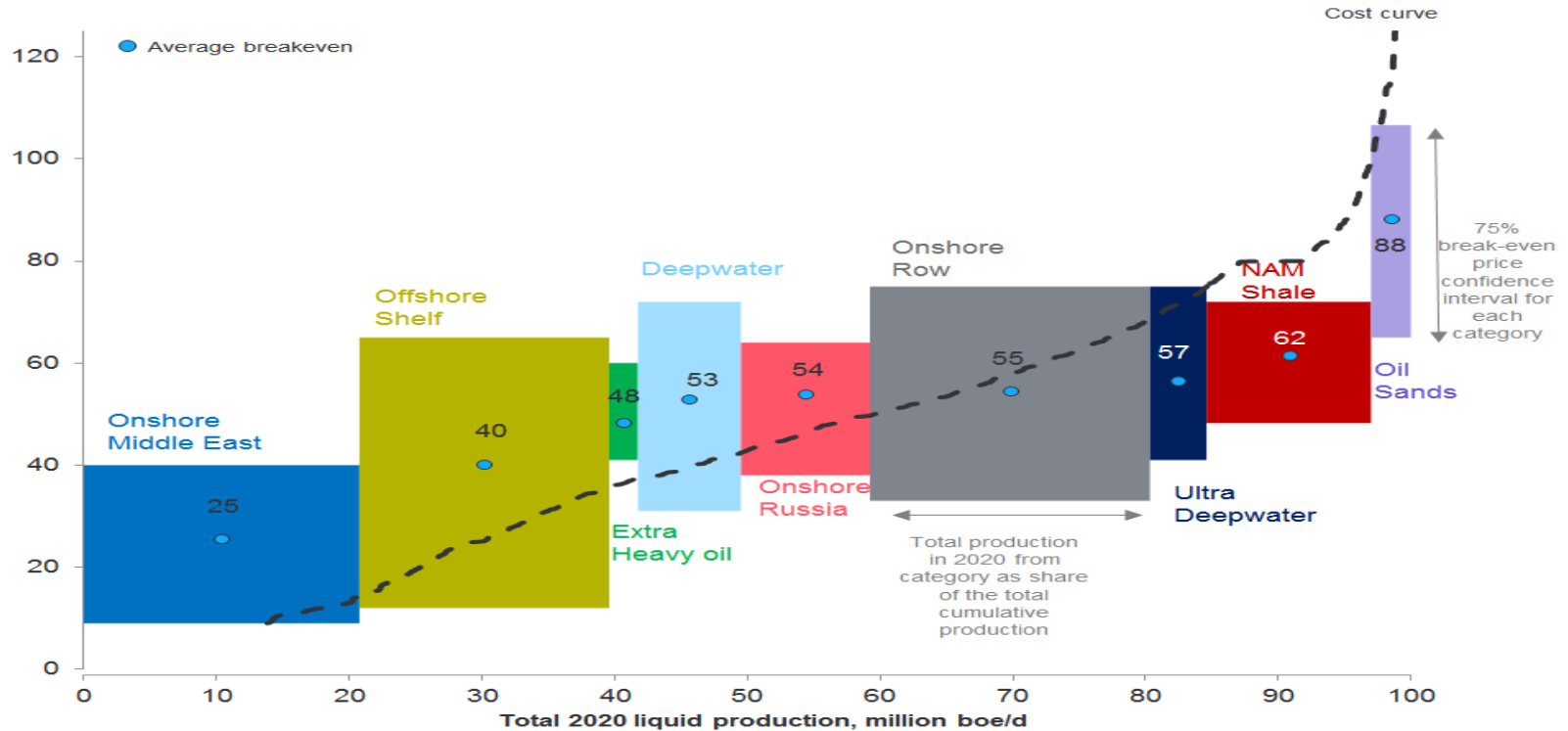
Data source: U.S. Energy Information Administration, Short-Term Energy Outlook, August 2024, CME Group, Bloomberg, L.P., and Refinitiv an LSEG Business

Note: Confidence interval derived from options market information for the five trading days ending August 1, 2024. Intervals not calculated for months with sparse trading in near-the-money options contracts.

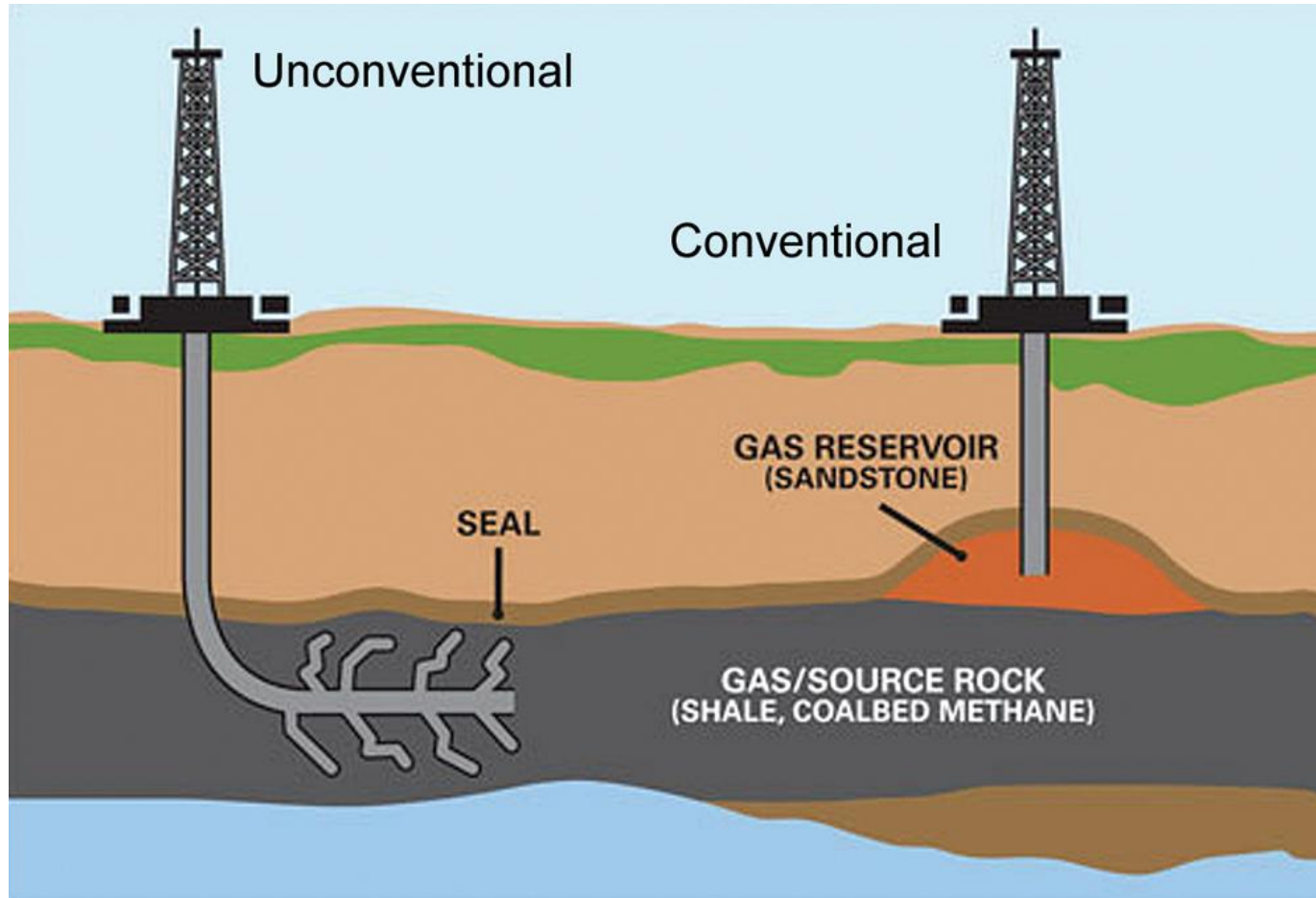


Cost of production

Global liquid supply cost curve
USD/bbl

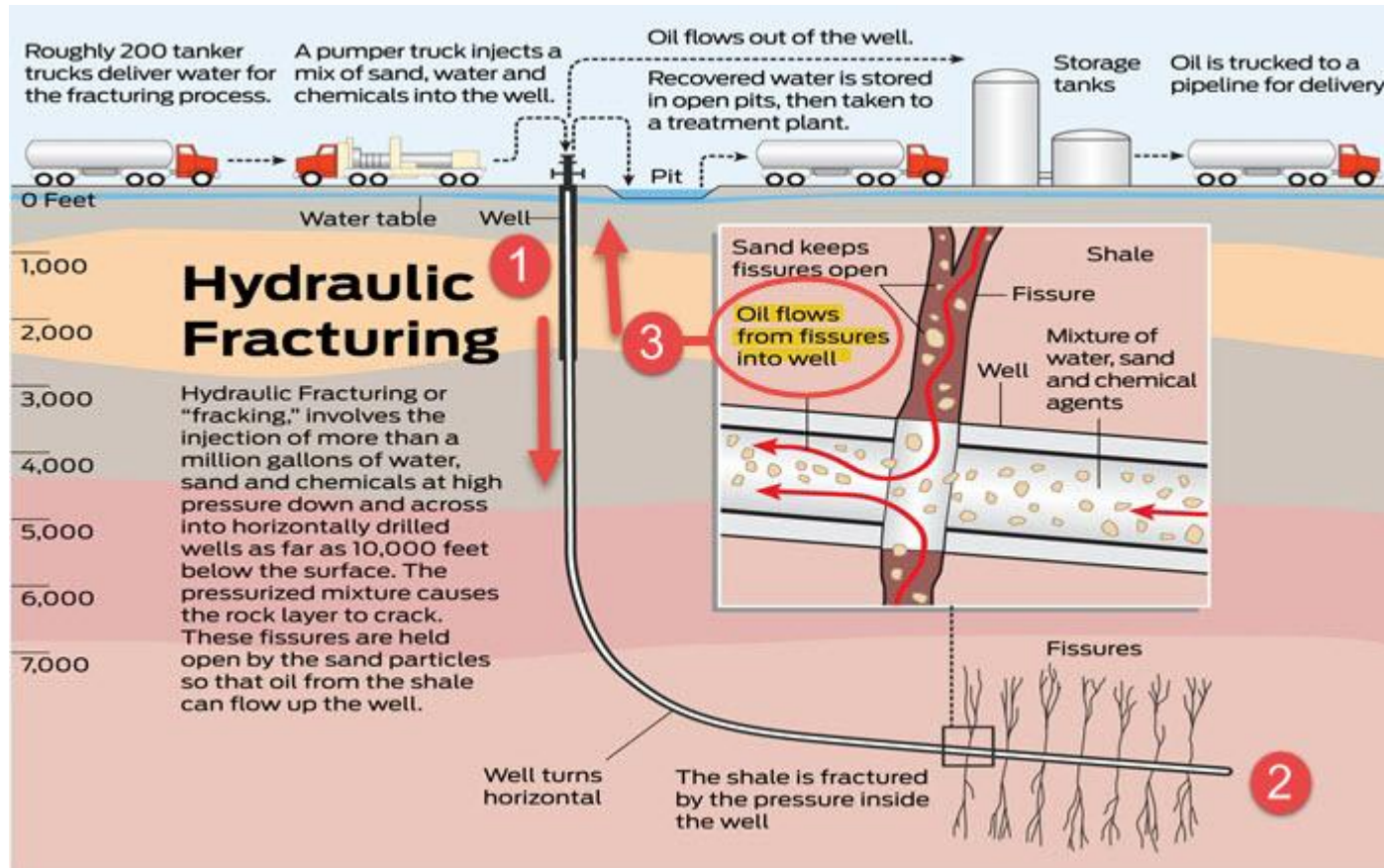


Conventional and unconventional Oil



Source : Shalestuff

Shale oil production : how it works ?



Source : Shalestuff

Oil : everything began in Pennsylvania



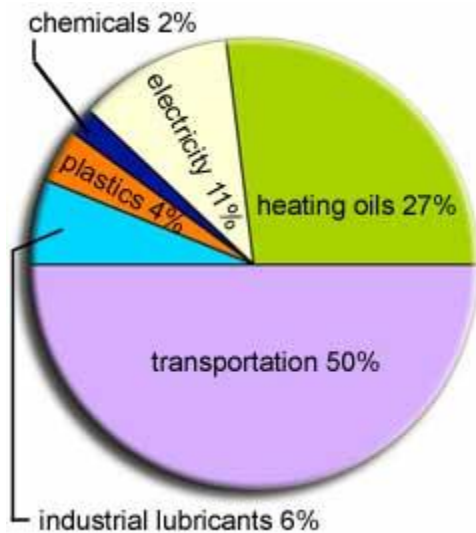
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Uses of oil in history



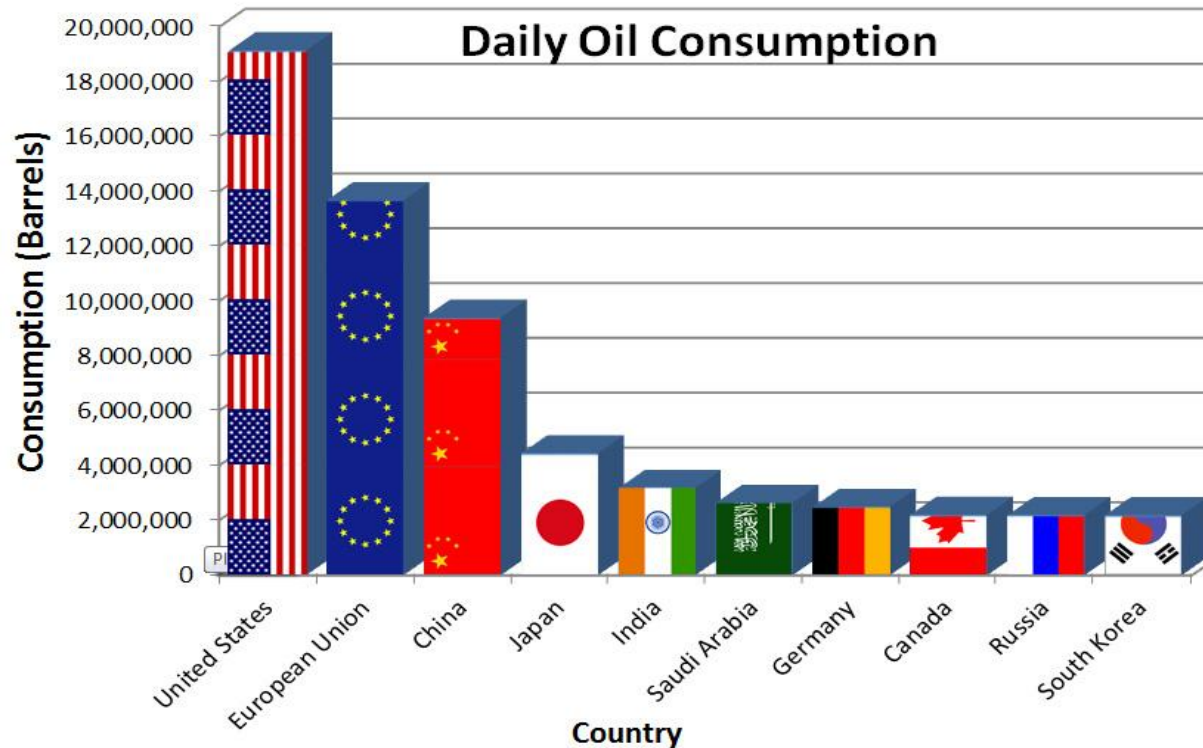
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And now (oil demand by sector)



Source : EIA

US is the world's biggest oil consumer



Source : EIA

Numbers for 2026



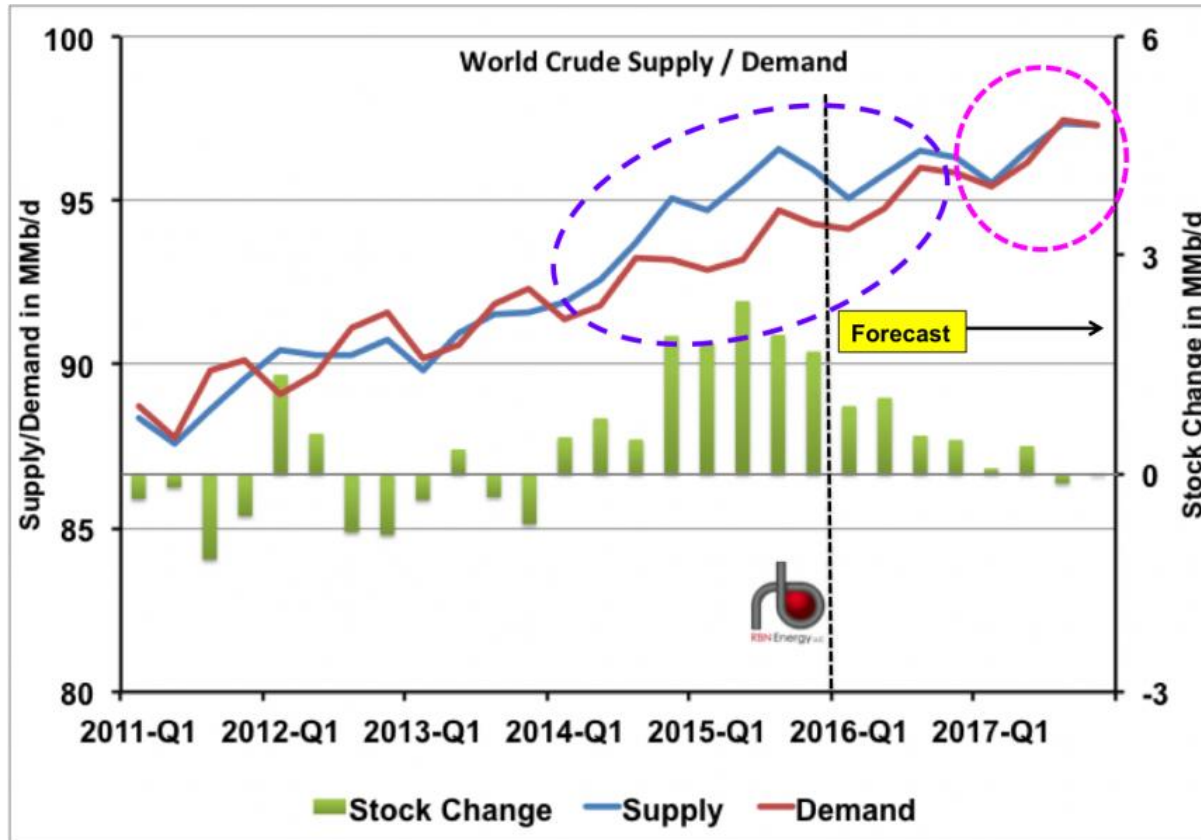
The Quincy agreement (USA/Saudi Arabia)

Roosevelt and King Abdul Aziz met on board the USS Quincy in 1944.

This pact, guaranteeing the Saudi monarchy's military protection in exchange for access to oil (for 60 years). Renewal in 2005.



World supply/demand for oil



Source : EIA

13 OPEC members in 2025

The Organization of the Petroleum Exporting Countries (OPEC) was founded in Baghdad, Iraq, with the signing of an agreement in September **1960** by **five countries** namely Iran, Iraq, Kuwait, Saudi Arabia and Venezuela.

Until September 1, 1965, the headquarters of OPEC was located in Geneva, Switzerland, before being relocated to Vienna, Austria.

In 2025, the Organization has a total of **13 member countries** :

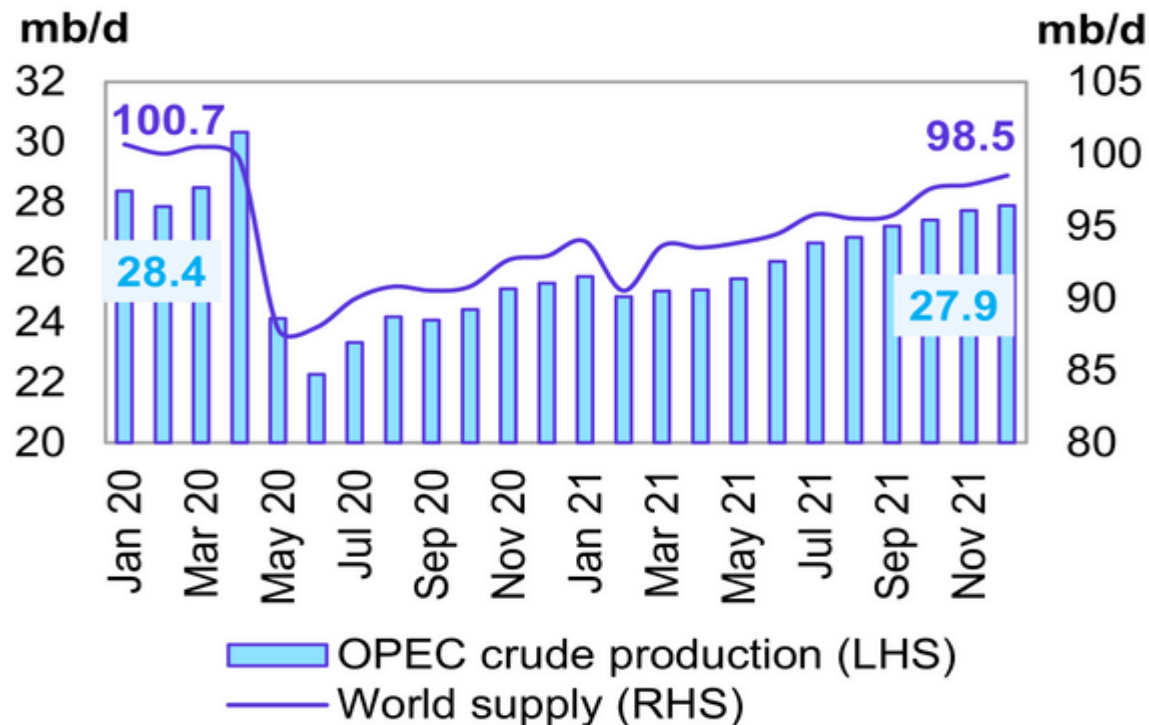
Algeria, Angola, Congo, Equatorial Guinea, Gabon, Iran, Irak, Kuwait, Libya, Nigeria, Saudi Arabia, United Arab Emirates, Venezuela.

OPEC + (since 2016) : Russia and 9 major exporting countries (Mexico, Kazakhstan, Azerbaïdjan, Bahreïn, Brunei, Malaysia, Oman, Sudan, South Sudan)

Source : OPEC

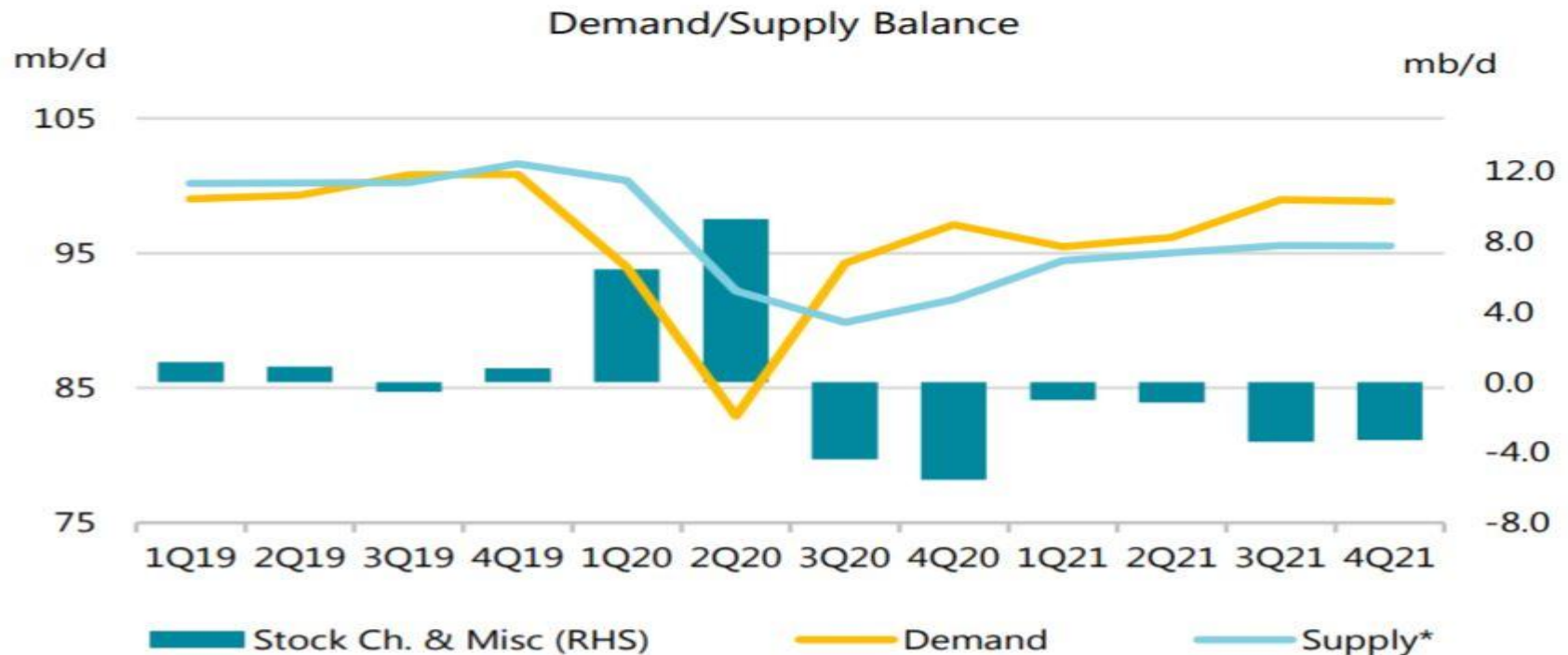
OPEC supply

Graph 5 - 28: OPEC crude production and world oil supply development



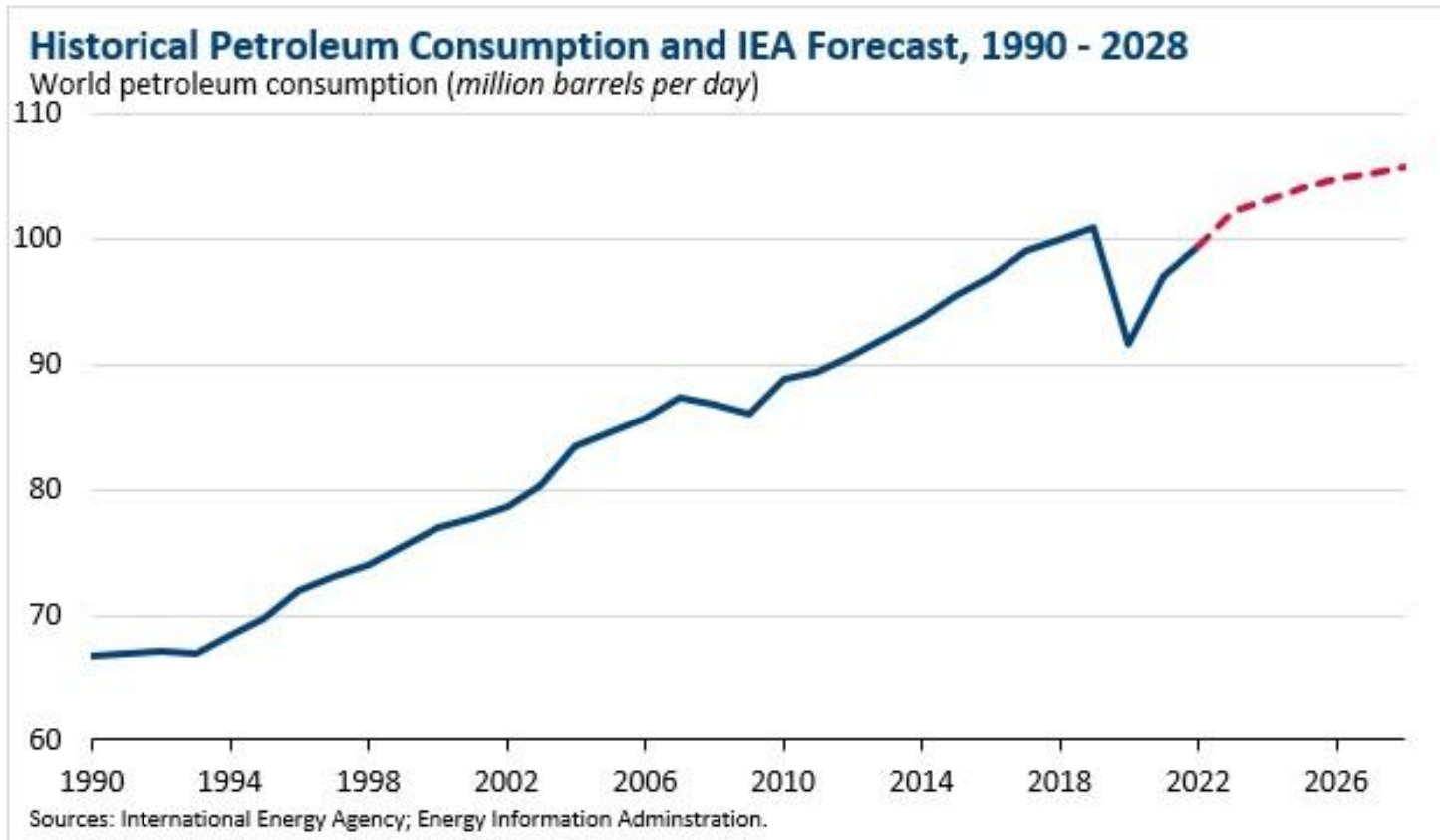
Source: OPEC.

World supply/demand for oil (from 2019)



* Assumes 100% compliance with OPEC+ deal

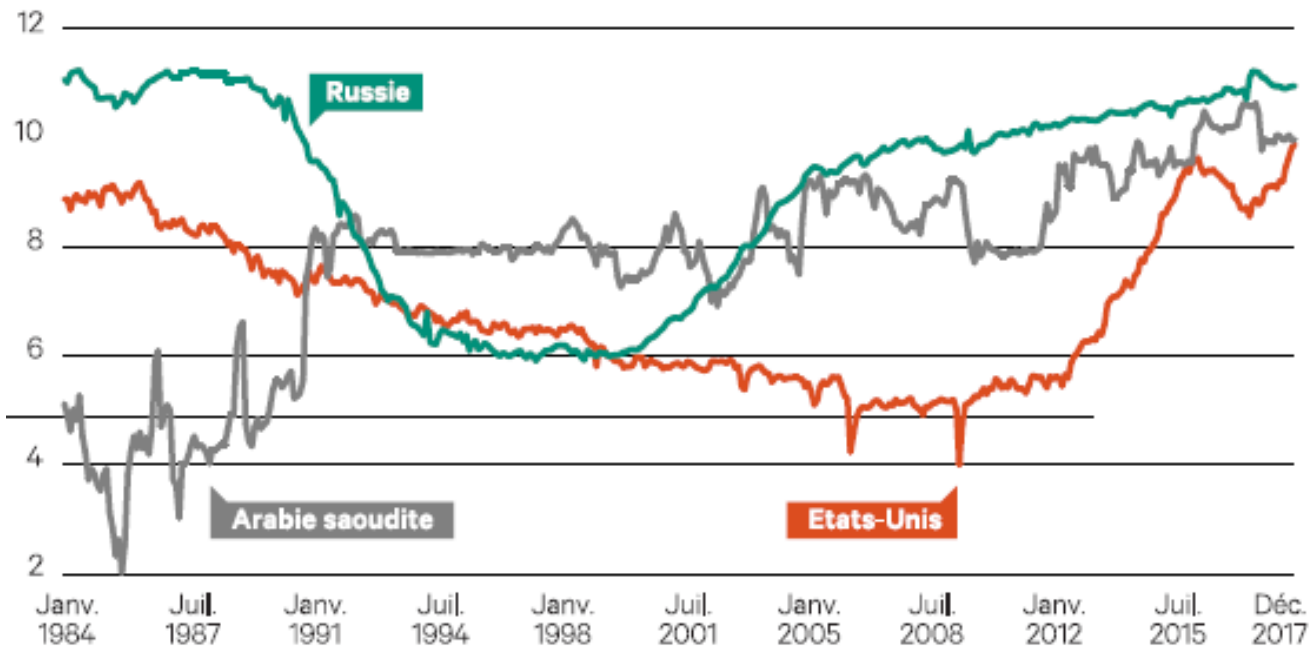
World demand should continue to grow



The biggest producing countries

Le podium mondial des producteurs de pétrole

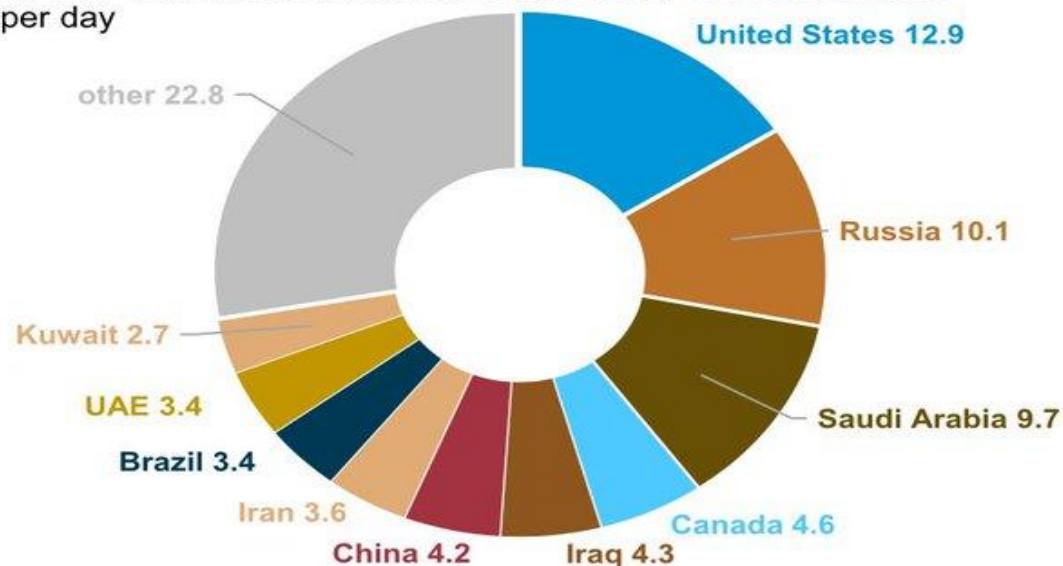
En millions de barils/jour



« LES ÉCHOS » / SOURCE : AGENCE INTERNATIONALE DE L'ÉNERGIE (AIE)

And now

Global crude oil and condensate production in 2023 by select countries
million barrels per day

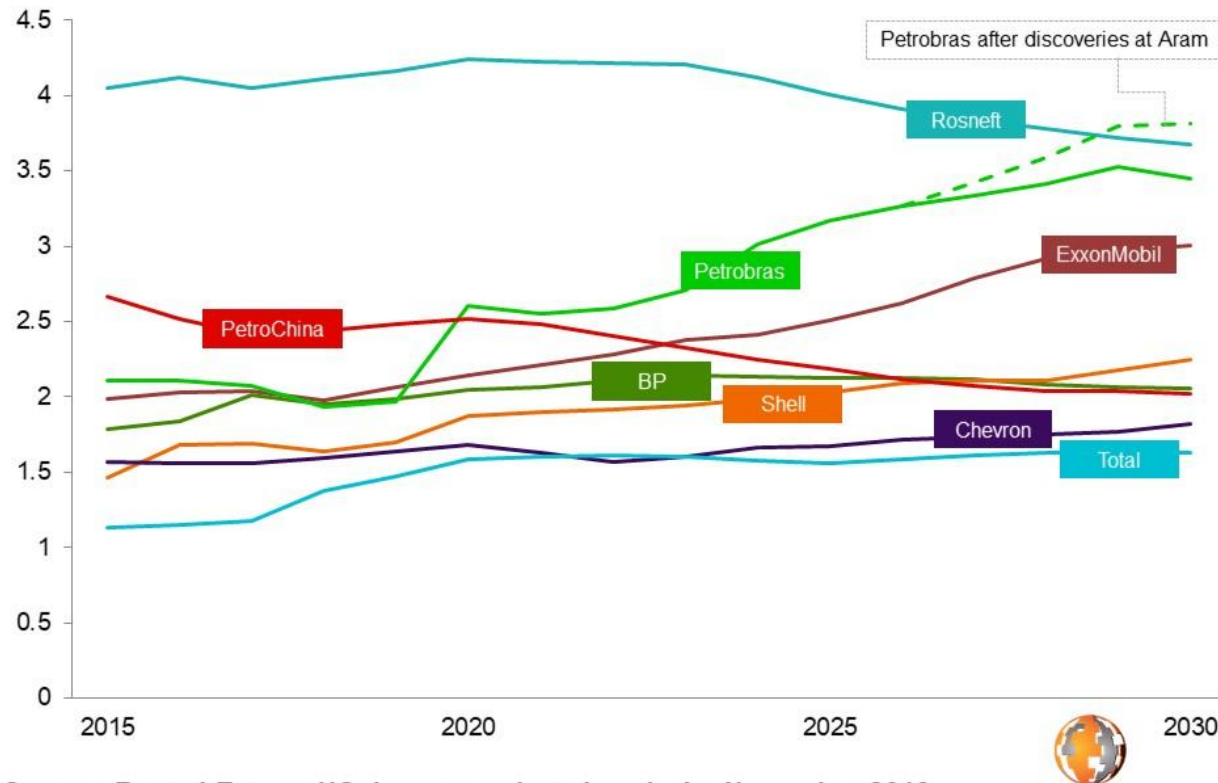


Data source: U.S. Energy Information Administration,
International Energy Statistics

The biggest producing companies (1)

Oil production from top oil producers in the world (net volumes)

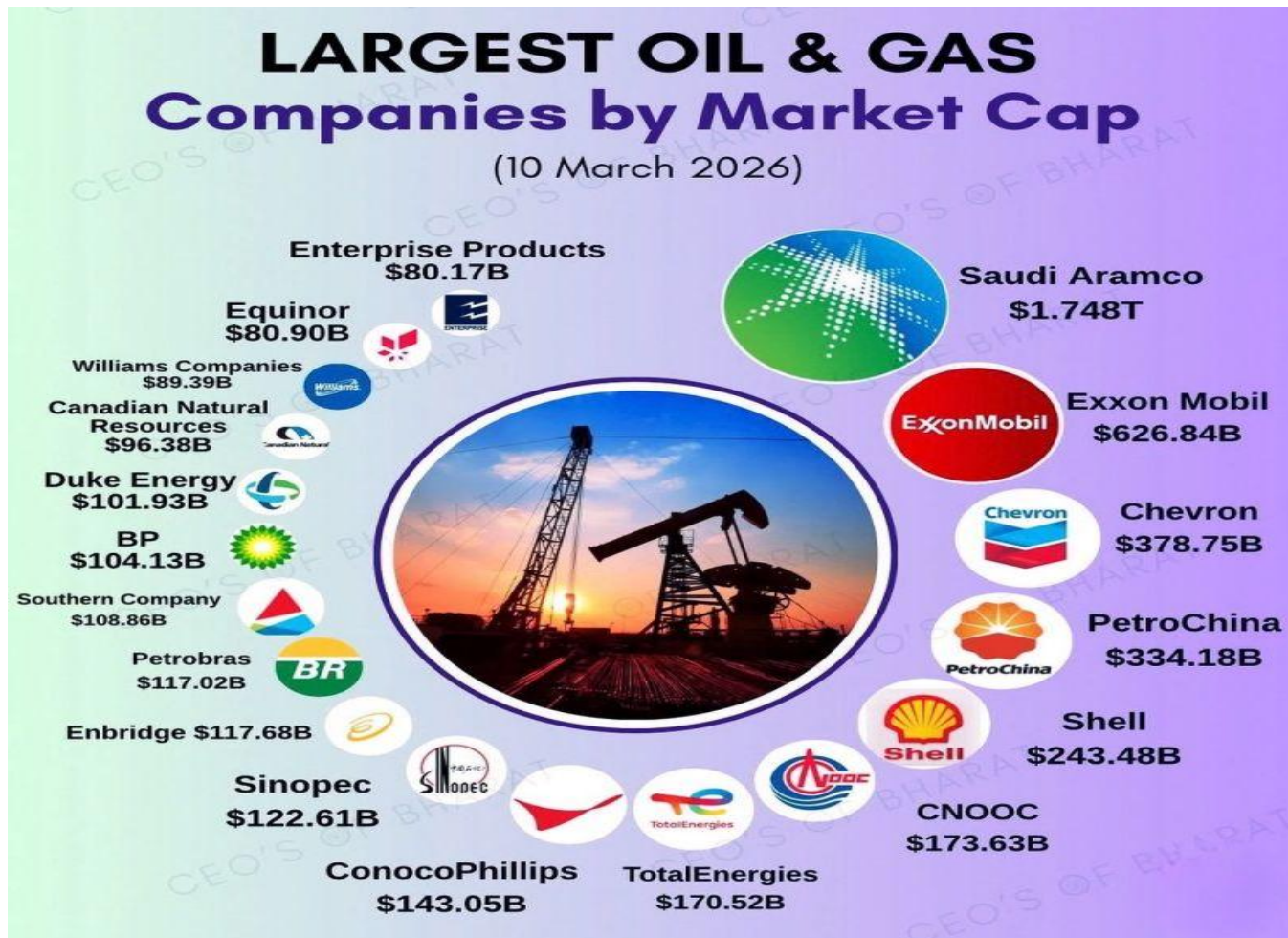
Million barrels per day



Source: Rystad Energy UCube research and analysis, November 2019



The biggest producing companies (2)



Source : BBG

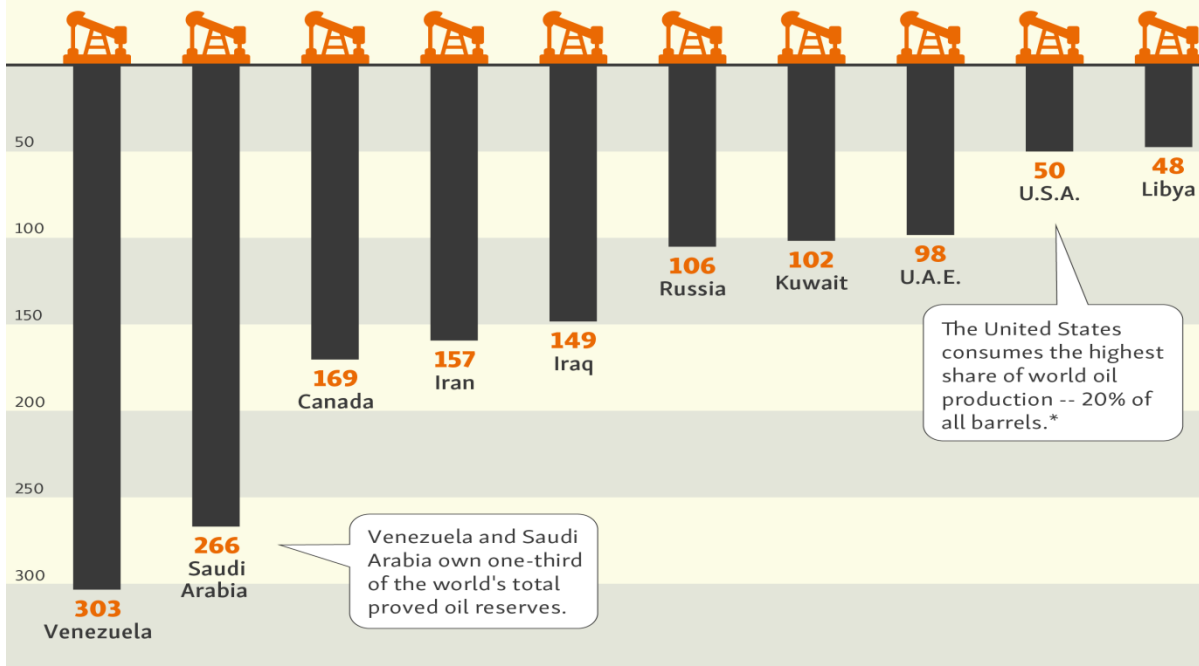
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Total world oil reserves

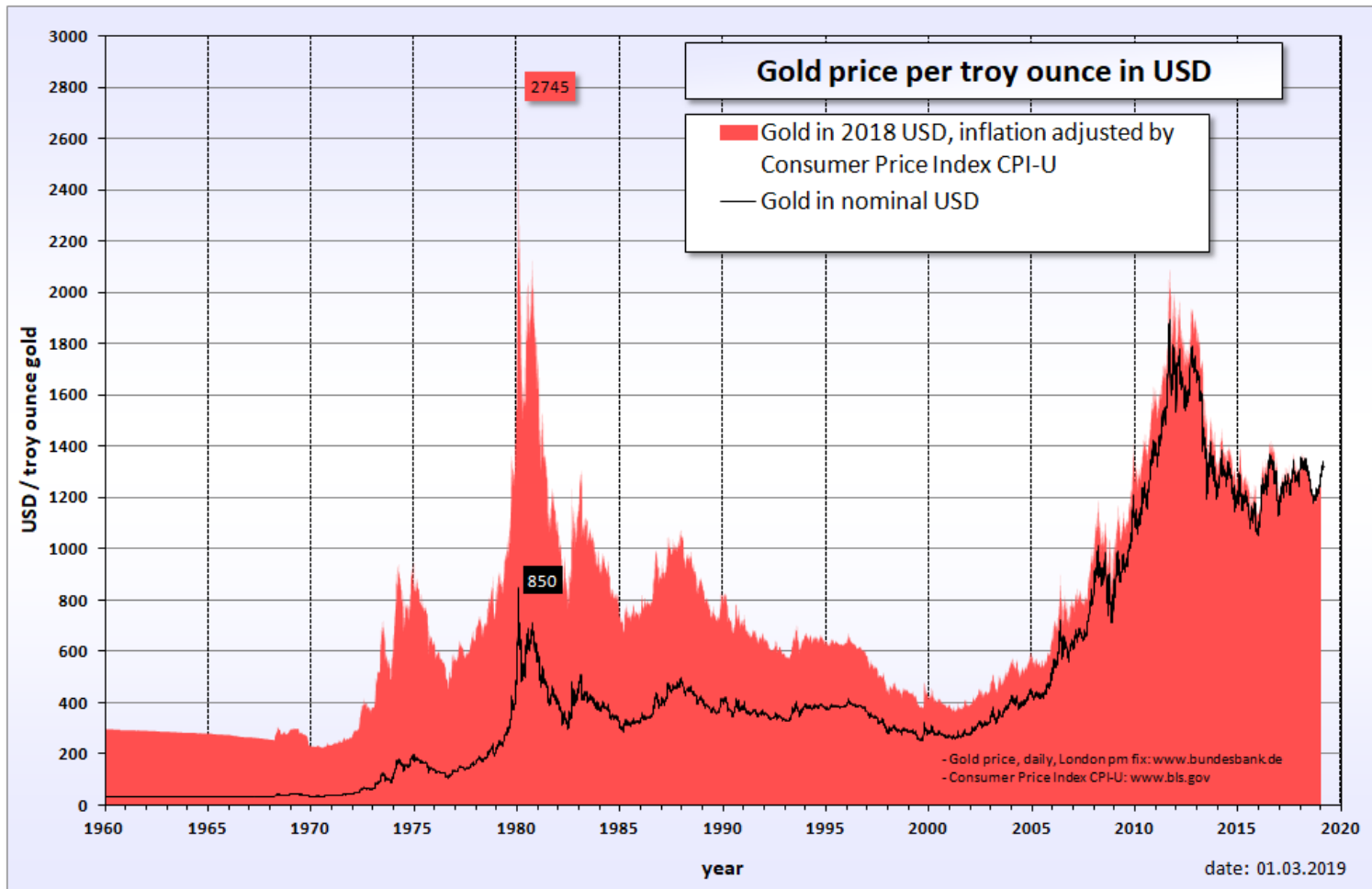
Oil Reserves Around The World

Venezuela, Saudi Arabia, and Canada own the largest proved oil reserves in the world.

Total proved reserves of oil (thousand million barrels)



A focus on gold



Source : Kitco

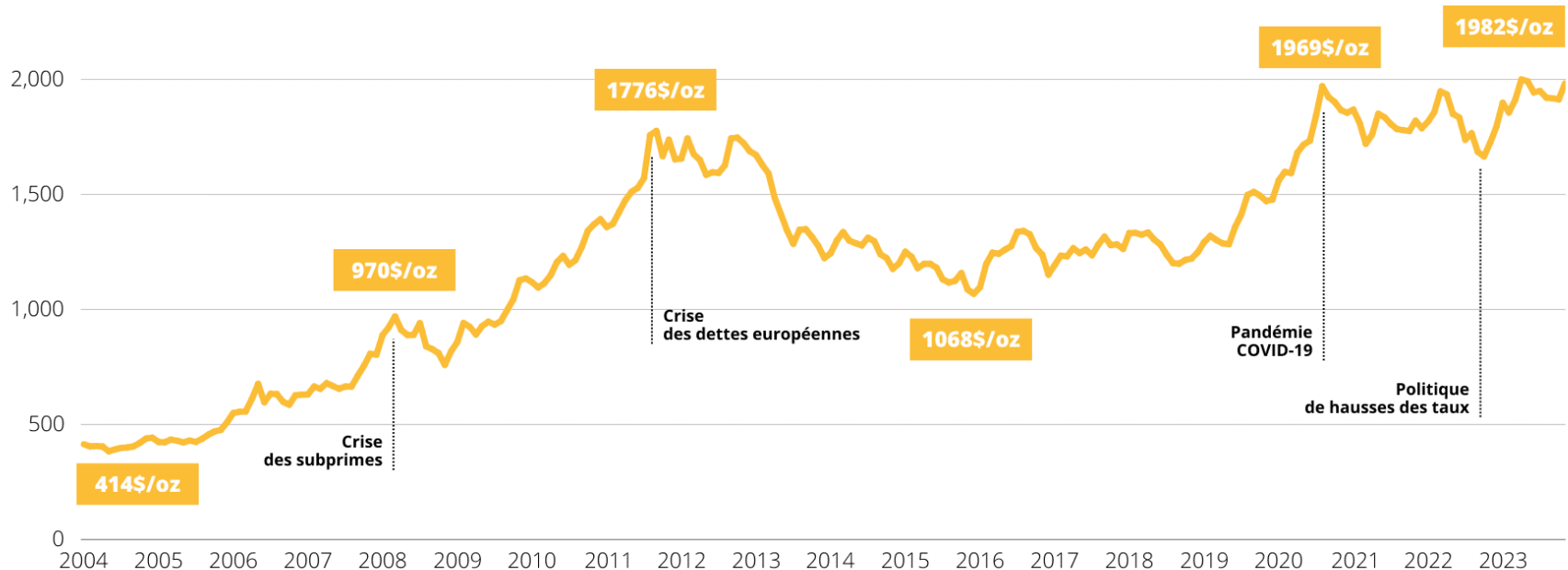
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Recent prices

Évolution du cours de l'or en dollars ces 20 dernières années

moyennes mensuelles de mai 2002 à novembre 2023

© VeraCash - Novembre 2023



Source : goldprice.org

Investing in commodity markets

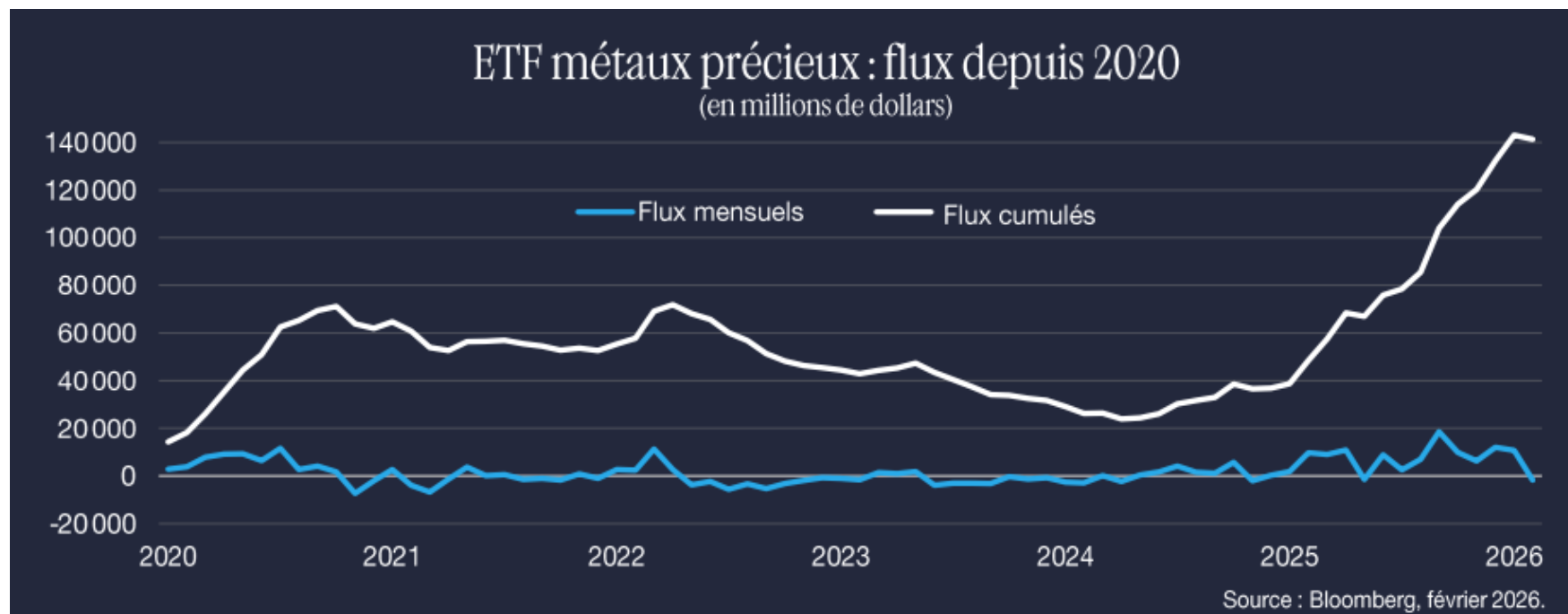
And now 5 000 \$



Source : BBG

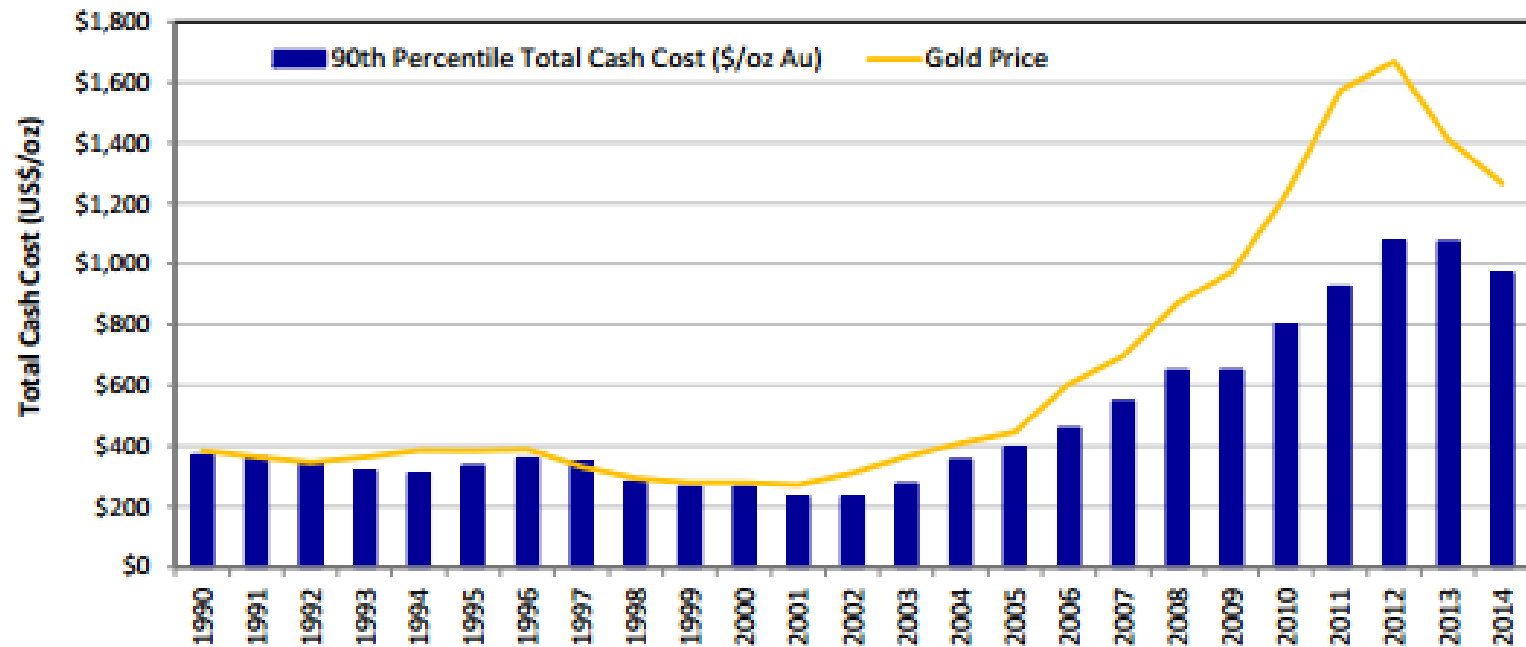
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ETF flows



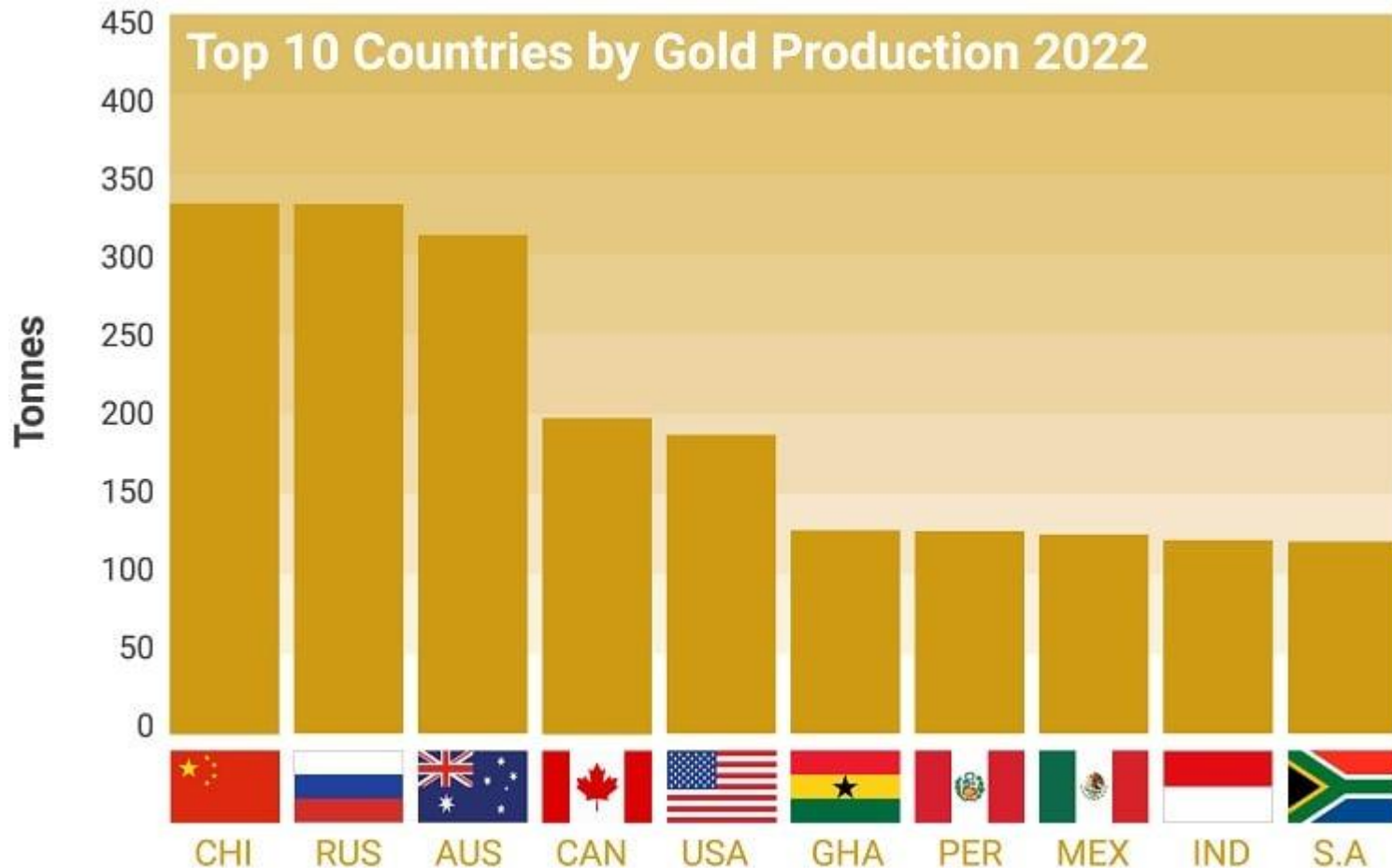
Cost of production

Exhibit 1: Historical 90th Percentile Cash Cost Curve



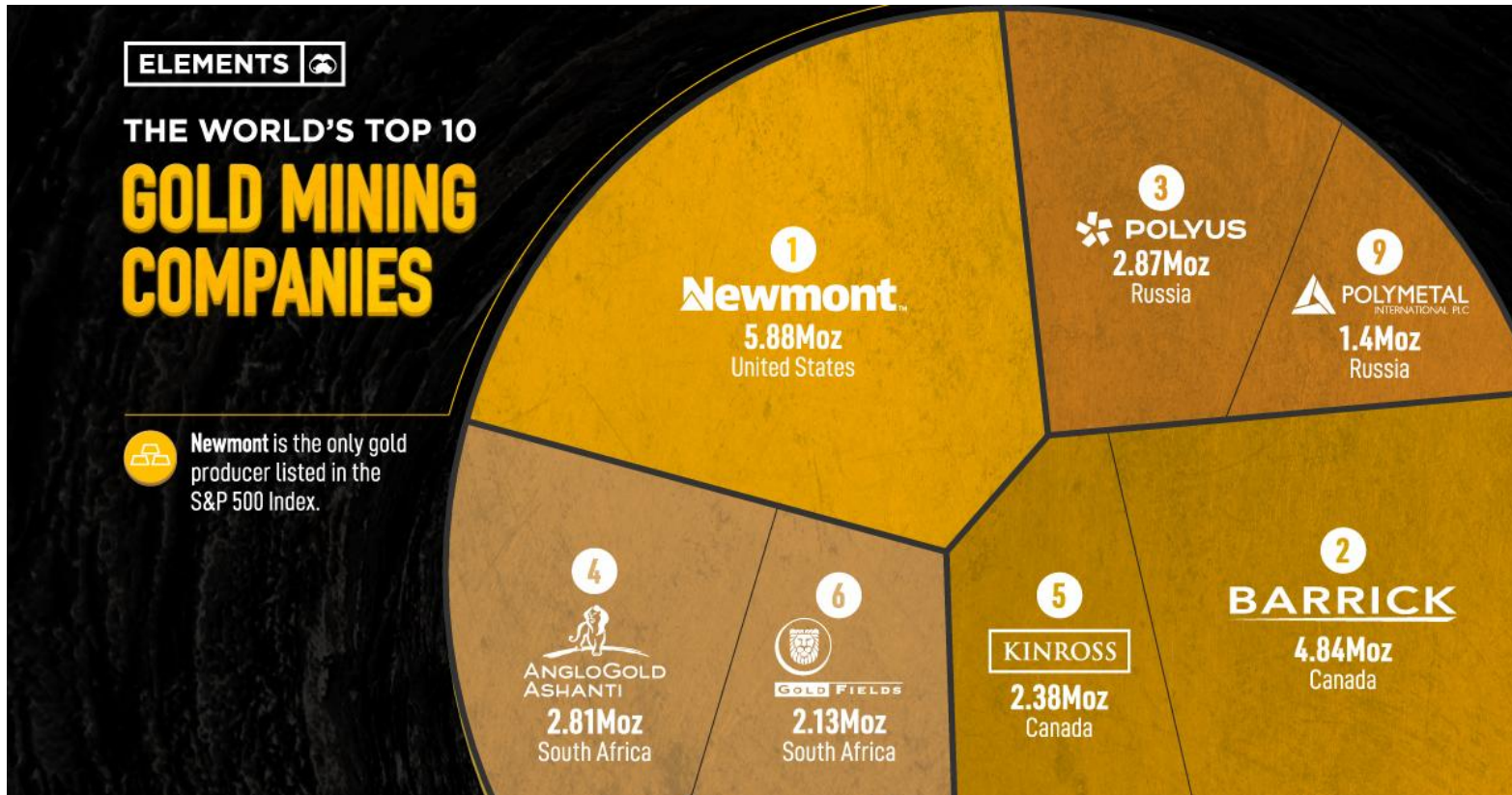
Source: SNL Metals and Mining, Bloomberg RBC Capital Markets

The biggest gold producing countries



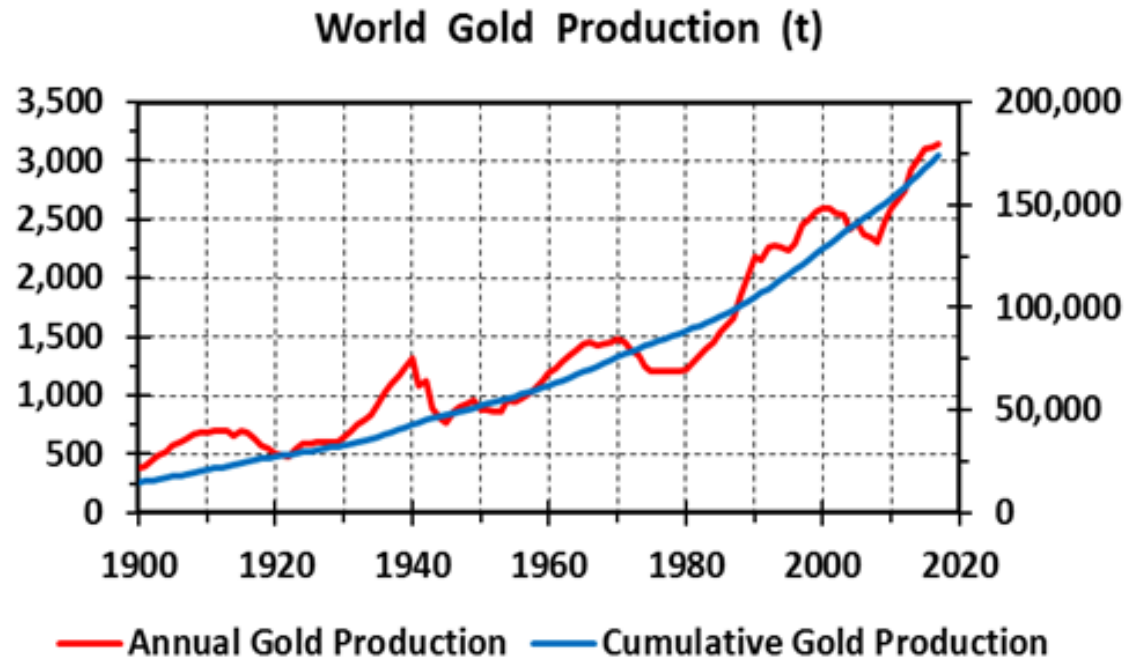
Source: World Gold Council

Biggest gold producing companies



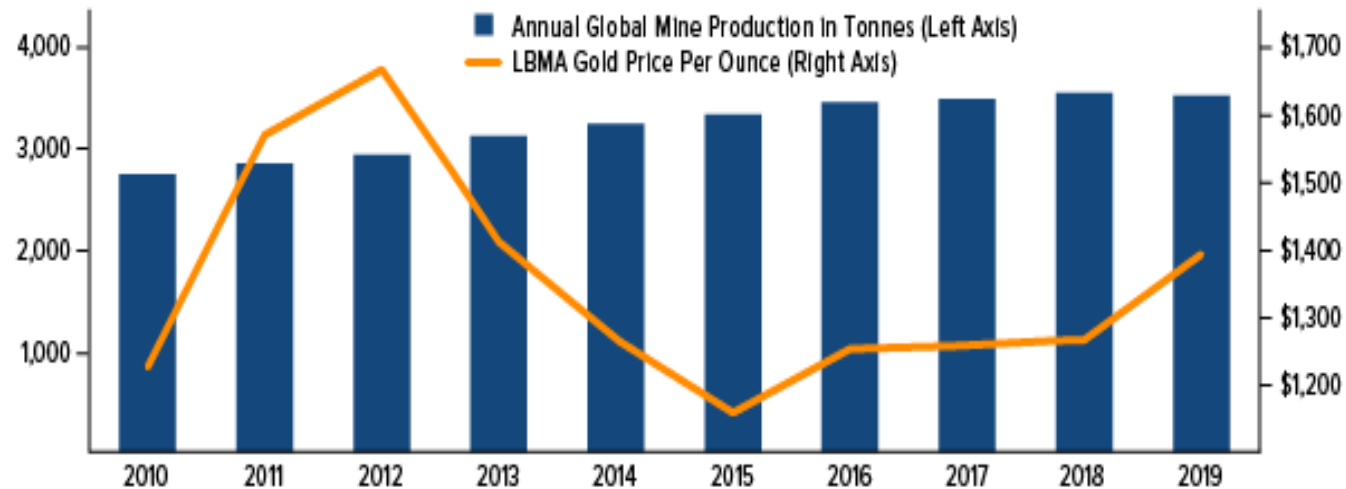
Source : VS

Gold supply over the long term



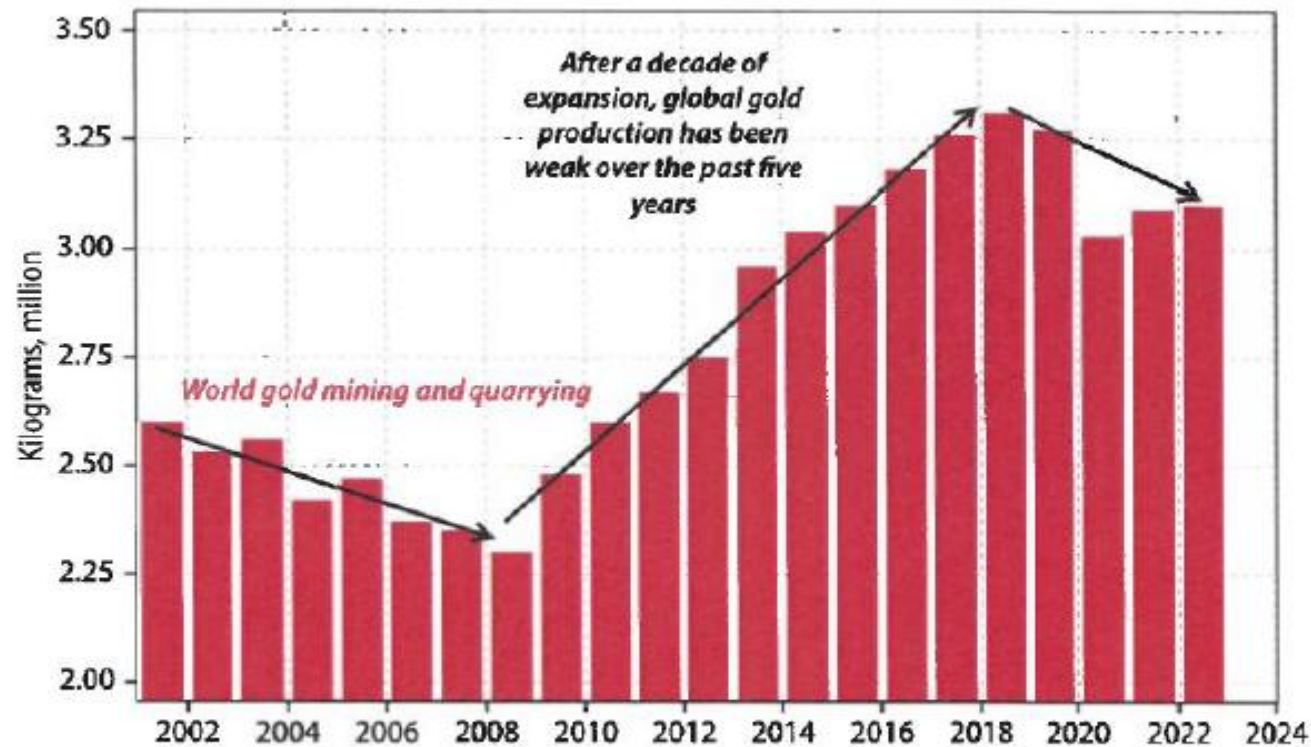
But relatively unchanged now

Gold Supply Relatively Unchanged Over Last Decade



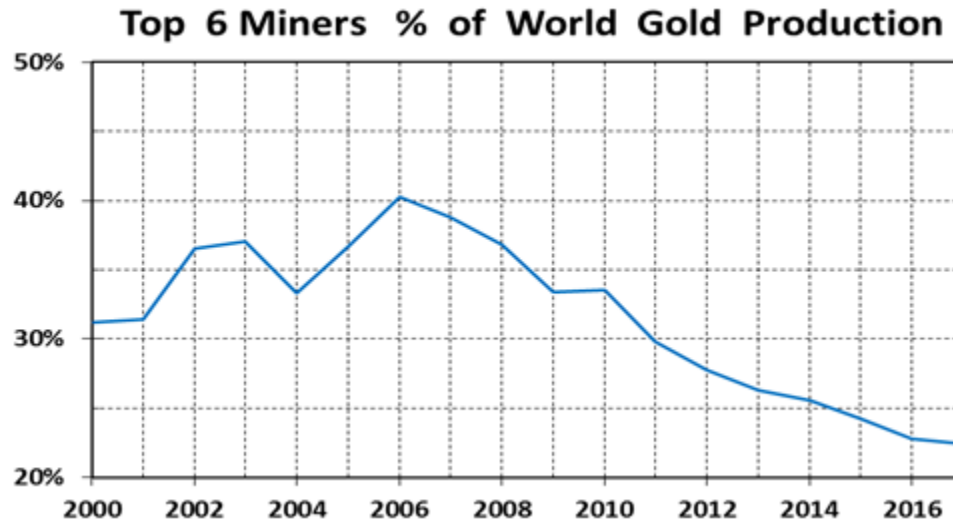
Source: ICE Benchmark Administration, Metals Focus, Refinitiv GFMS, World Gold Council, U.S. Global Investors

With a trend on the way down



Source : Gaveta Research/Macrobond

Top miners market share



Source : Kitco

Open pit or underground mines

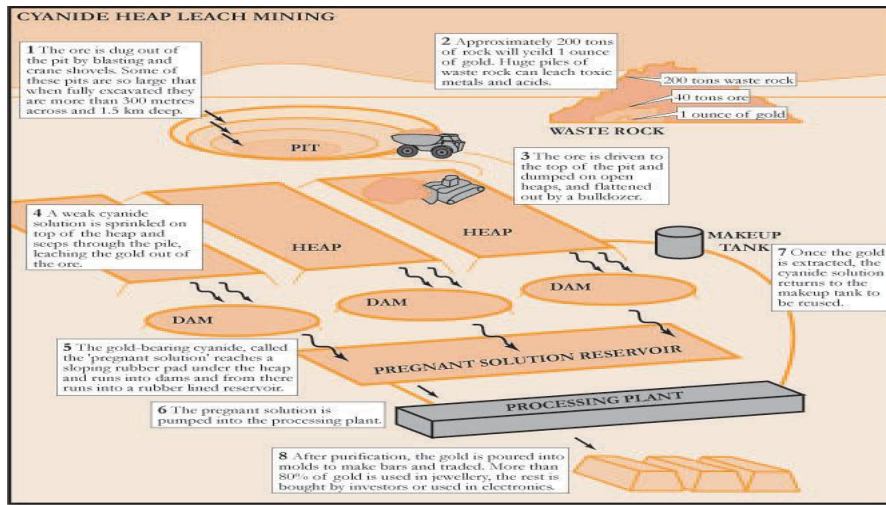
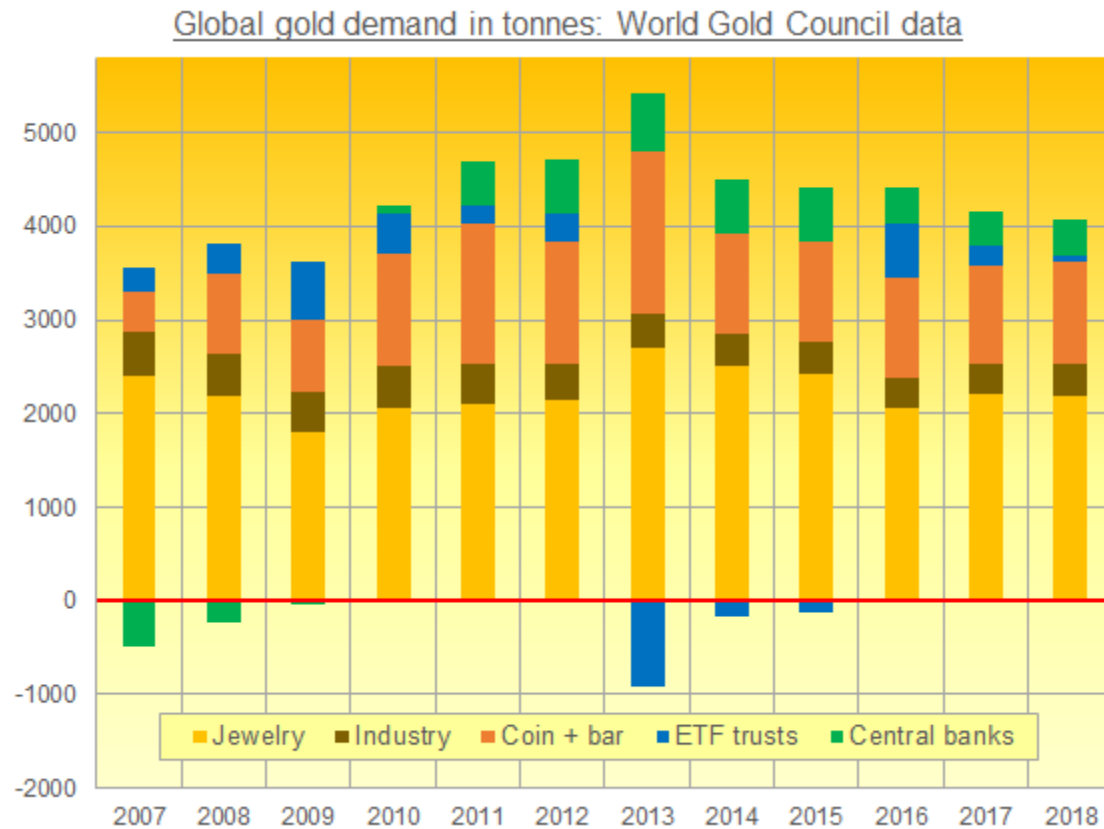


Illustration by Christina Nicholas

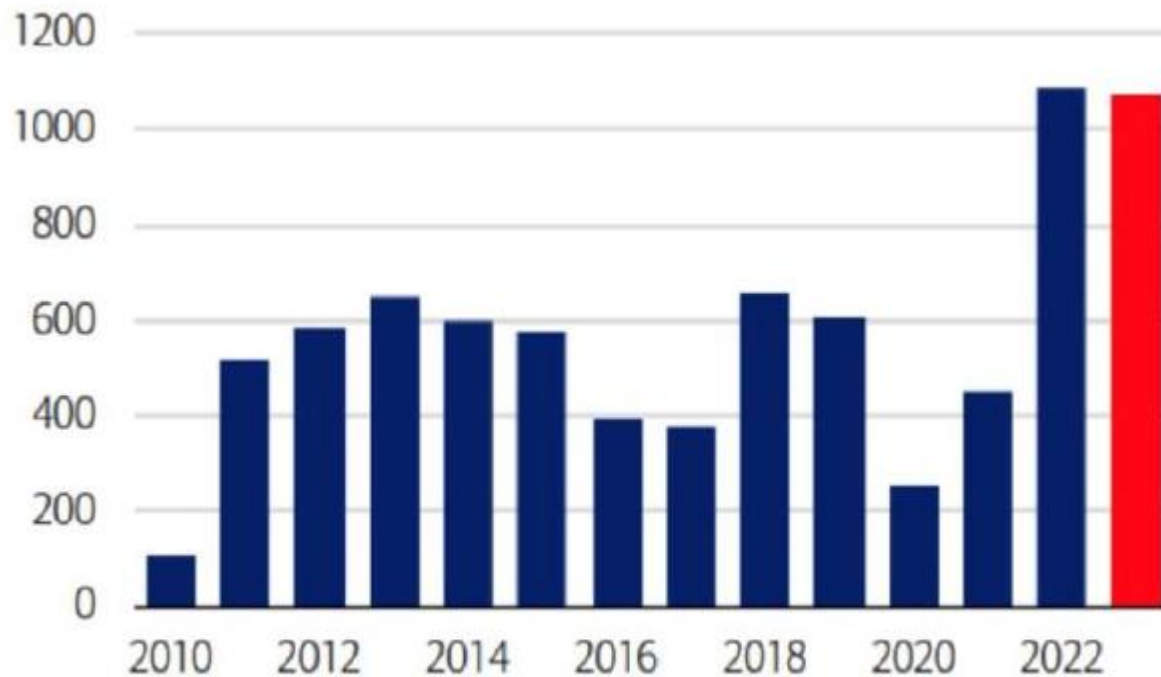


Gold demand



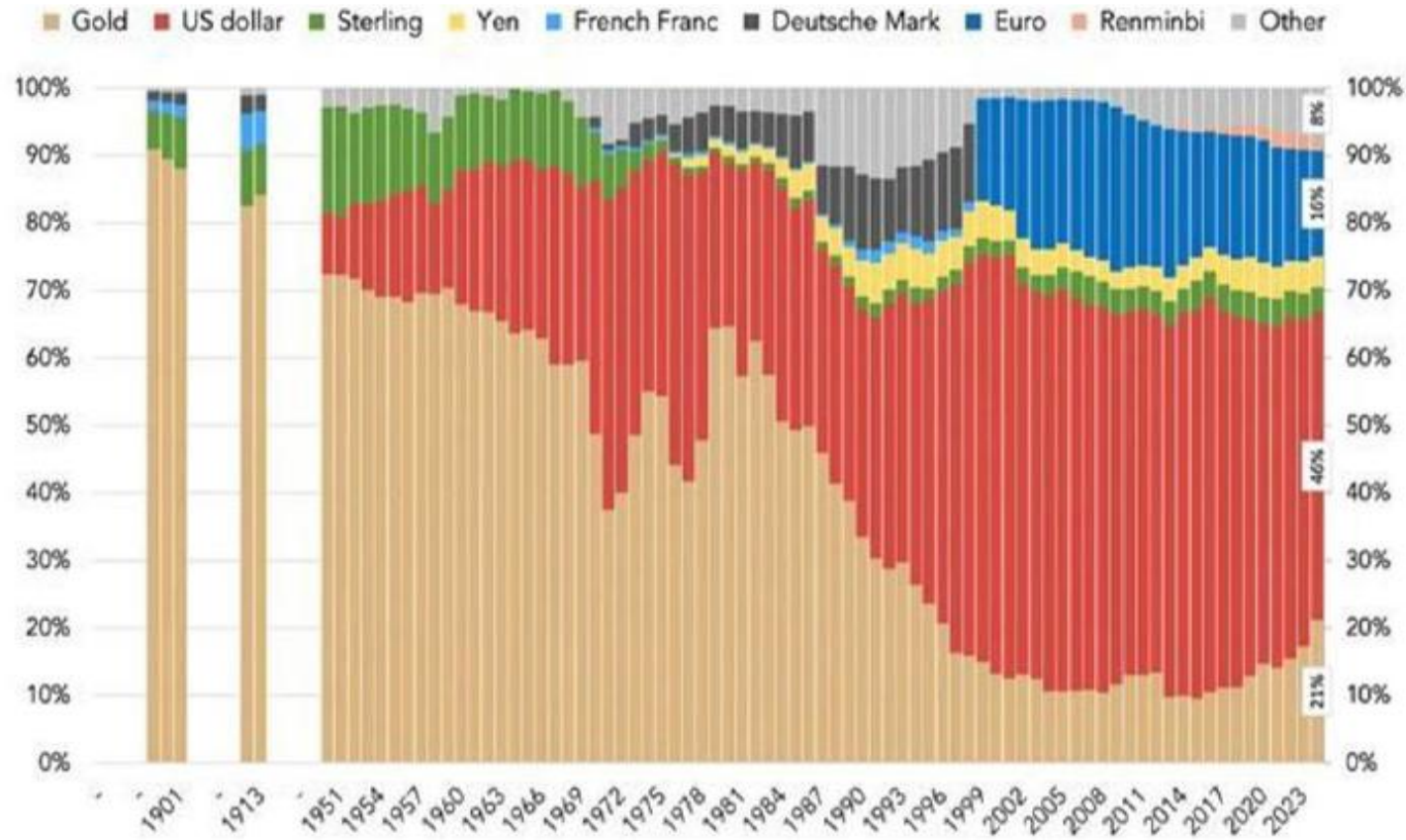
Central bank impact

Annual central bank gold purchases, metric tons



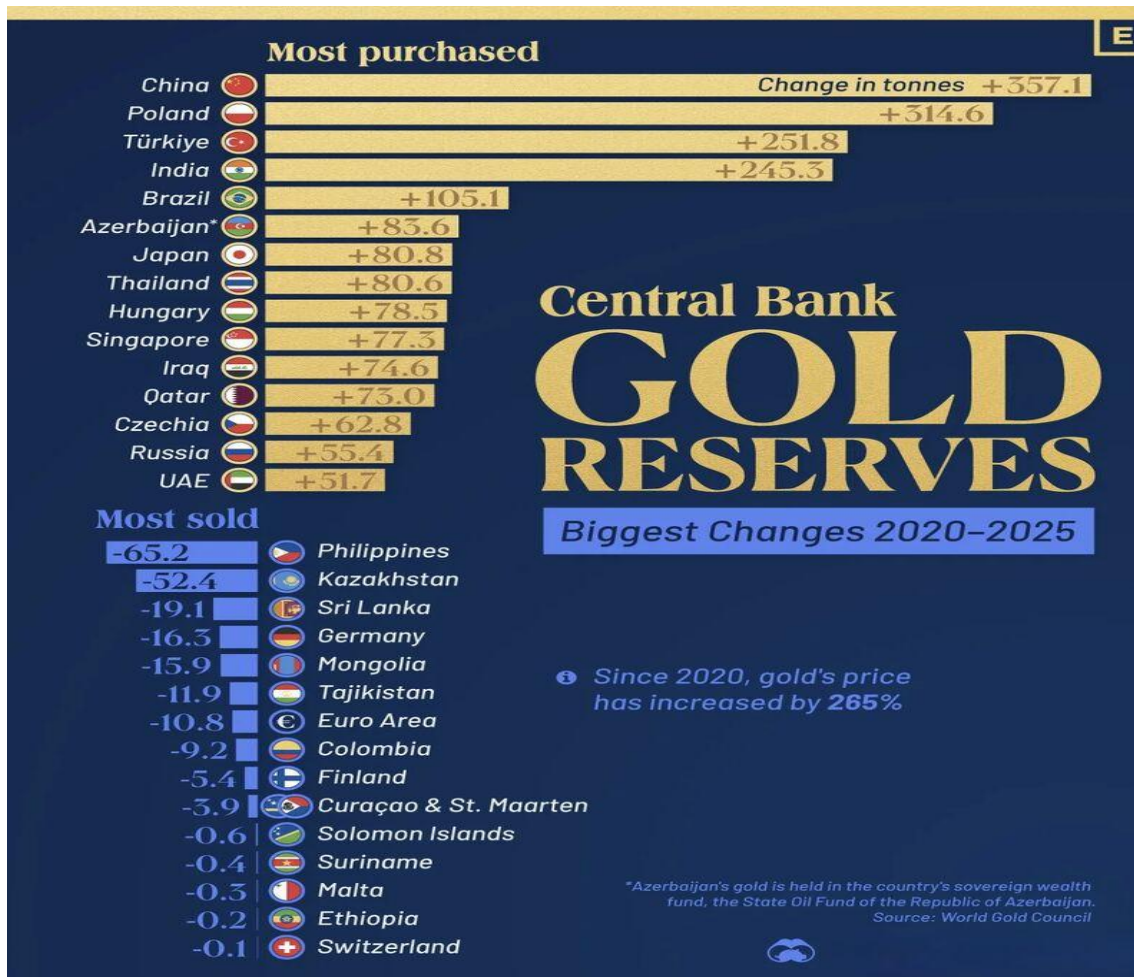
Source : BofA Research Investment Committee, Bloomberg, Metal Focus Data Ltd.
Note: 2023 data (red bar) is annualized

Central bank reserves



Source : CPR AM

Central bank – biggest changes (2020/2025)



Source : World Gold Council

Investing in commodity markets

1.3 Derivative markets

Reference prices are commonly futures market prices

A little bit of history

Futures commodity markets have started in the US, in the city of Chicago.

The CBOT (Chicago Board of Trade) has been set-up in 1848 and the CME (Chicago Mercantile Exchange) in 1874.



Examples of commodity futures



Basics of commodity futures

A commodity futures contract is an agreement to buy or sell a particular commodity at a future date

The price and the amount of the commodity are fixed at the time of the agreement

Most contracts contemplate that the agreement will be fulfilled by actual delivery of the commodity

Some contracts allow cash settlement instead of physical delivery

Most contracts are liquidated before the delivery date

Vocabulary

A lot = A futures contract.

Open interest : total number of outstanding futures contracts at a given time.

Tick : minimum amount that the price of a futures contract can change.

For example on WTI crude oil futures 1 tick = 0,01\$ per barrel.

Commodity futures - month codes

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Code	F	G	H	J	K	M	N	Q	U	V	X	Z

Commodity futures specifications

	Copper	Sugar No 11	Corn	Soybean	Light sweet crude (WTI)	Brent crude oil	Gas oil	Carbon emissions (EUA)	Gold	Silver
Exchange Code	HG./HG	SB./SB	ZC./ZC	ZS./ZS	CL.	B./B	G./G	EUA.	GC.	SI.
Type	Future	Future	Future	Future	Future	Future	Future	Future	Future	Future
Country	US	EU	US	US	US	EU	EU	EU	US	US
Currency	USD	USD	USD	USD	USD	USD	USD	EUR	USD	USD
Exchange	CME Comex (Base)	ICE US	CME GLOBEX	CME GLOBEX	Nymex	ICE EUROPE	ICE EUROPE	ICE EUROPE	CME Comex (Precious)	CME Comex (Precious)
Spread from	50 (10 increments)	8 (8 increments)	1.5 (6 increments)	2.25 (9 increments)	3 (3 increments)	5 (5 increments)	2 (8 increments)	7 (7 increments)	0.3 (3 Increments)	0.02 (4 Increments)
Increment	0.0005	0.01	0.25	0.25	0.01	0.01	0.25	0.01	0.1	0.005
Increment value	USD12.50	USD11.20	USD12.50	USD12.50	USD10.00	USD10.00	USD25.00	EUR10.00	USD10.00	USD25.00
Margin	USD2,000	USD1,000	USD1,000	USD2,000	USD1,000	USD1,000	USD2,000	EUR1,000	USD1,000	USD1,000
Contract unit	25,000 lbs.	112,000 lbs.	5,000 bushels	5,000 bushels	1,000 barrels	1,000 barrels	100 metric tonnes	1,000 tonnes CO2	100 troy oz.	5,000 troy oz.
Duration	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Quarterly	Monthly	Monthly
Trading hours	01:00–00:15	10:30–20:00	03:00–21:15	03:00–21:15	01:00–00:15	03:00–01:00	03:00–01:00	09:00–19:00	01:00–00:15	01:00–00:15
Quoted months	All (not all months are liquid)	Mar (H), May (K), Jul (N), Oct (V)	Mar (H), May (K), Jul (N), Sep (U), Dec (Z)	Jan (F), Mar (H), May (K), Jul (N), Aug (Q), Sep (U), Nov (X)	All	All	All	Most liquid contract – Dec	All (not all months are liquid)	All (not all months are liquid)

Source: LTE

Investing in commodity markets

From open outcry to electronic markets



Market Watch: 12:53:45

Symbol	Bid	Ask	!
EURUSD	1.16430	1.16434	4
GBPUSD	1.32490	1.32501	11
USDCHF	0.98858	0.98898	40
USDJPY	109.499	109.529	30
USDCAD	1.33087	1.33127	40
AUDUSD	0.74190	0.74220	30
EURCHF	1.15111	1.15141	30
EURJPY	127.496	127.526	30
EURGBP	0.87862	0.87892	30
AUDJPY	81.209	81.319	110

+ click to add... 10 / 2295

Symbols | Details | Trading | Ticks |

Market Watch: 12:53:58

Symbol	Bid	Ask	!
EURUSD	1.16440	1.16444	4
GBPUSD	1.32498	1.32505	7
USDCHF	0.98854	0.98894	40
USDJPY	109.498	109.528	30
USDCAD	1.33086	1.33126	40
AUDUSD	0.74190	0.74220	30
EURCHF	1.15116	1.15146	30
EURJPY	127.505	127.535	30
EURGBP	0.87862	0.87892	30
AUDJPY	81.209	81.319	110

+ GB00|

GBPJPY, Great Britain Pound vs Japanese Yen
GBPUSD, Great Britain Pound vs US Dollar

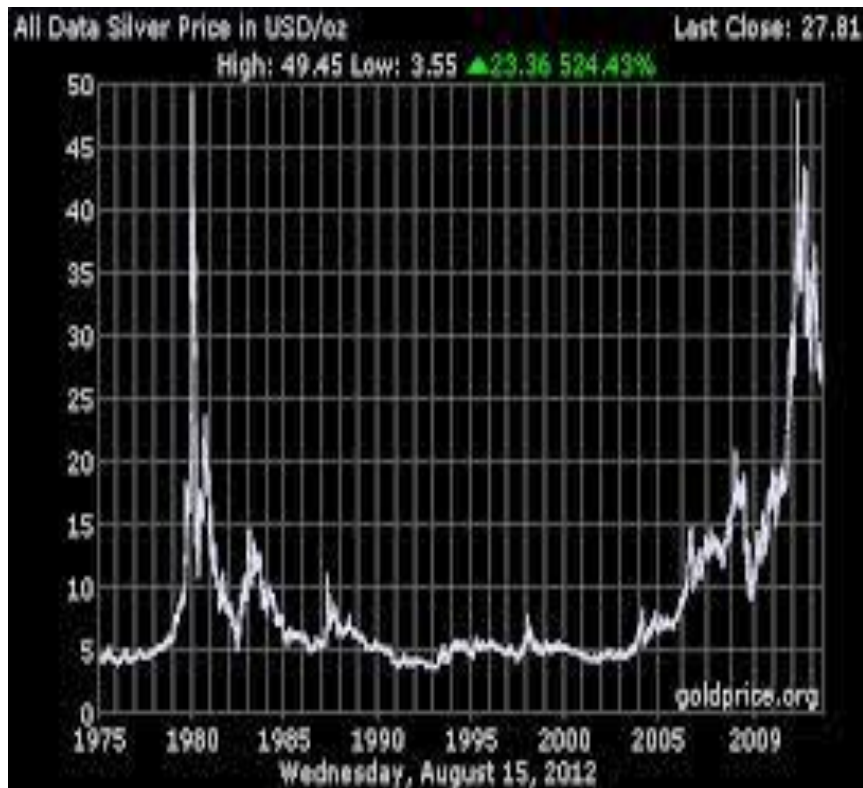
Symbols | Details | Trading | Ticks |

Investing in commodity markets

Futures markets are regulated

For example, In the United States, the Commodity Futures Trading Commission (CFTC) regulates the nation's futures and options markets.

Silver squeeze («corner») / the Hunt brothers



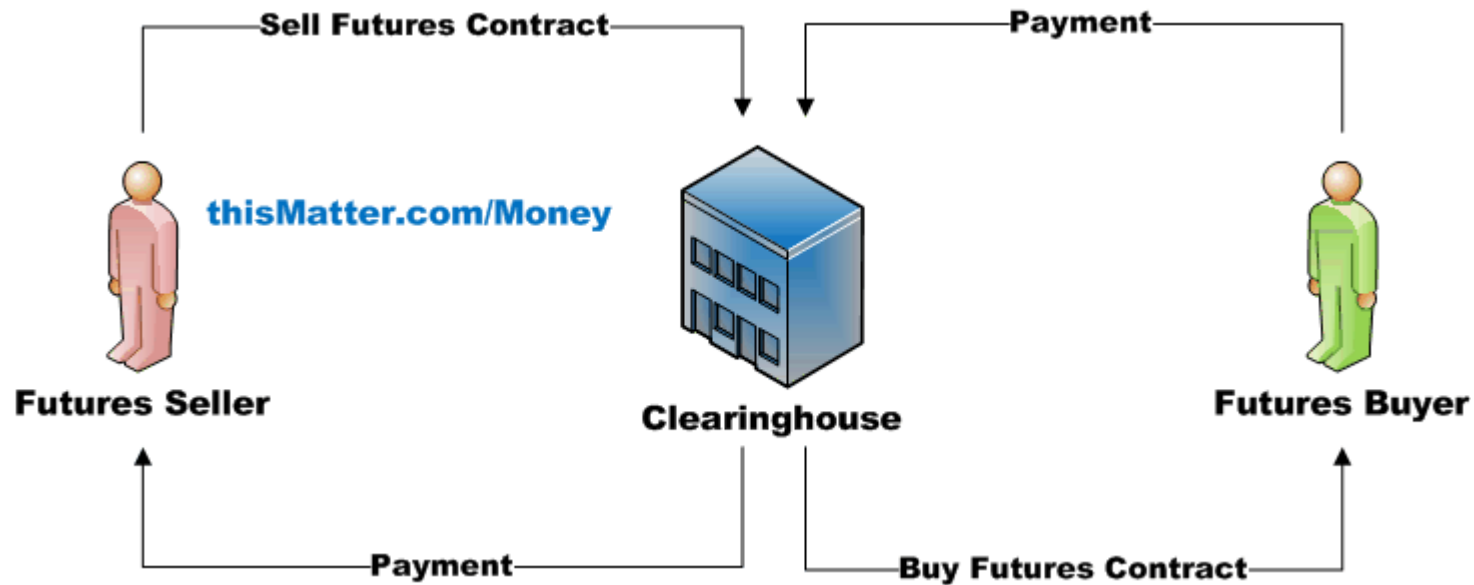
Bernard Madoff and the hot chocolate

Bernard Madoff, who ran a nearly \$65 billion Ponzi scheme, is now in jail.

He cornered the hot chocolate market last year.

He bought up every package of Swiss Miss and sold it for a profit in the prison yard. He monopolized hot chocolate!

The clearing house



How it works ?

Once a trade is done, the clearing house becomes the central counterparty for the buyer and the seller, guaranteeing financial performance of the contract.

In order to open a position, market participants have to pay a **deposit** also called an **initial margin**. The amount of this margin is calculated to cover the largest theoretical loss could incur in a specified, market-specific risk horizon based on historical market prices and their volatility.

All positions are marked-to-market on a daily basis in order to maintain the amount of this initial margin through margin calls. End-of-day variation payments are typically due the following business day.

Relationship between spot and futures prices

Futures price = Spot Price + The Cost of Carry

The Cost of Carry = Financing cost + Storage Cost + Insurance cost

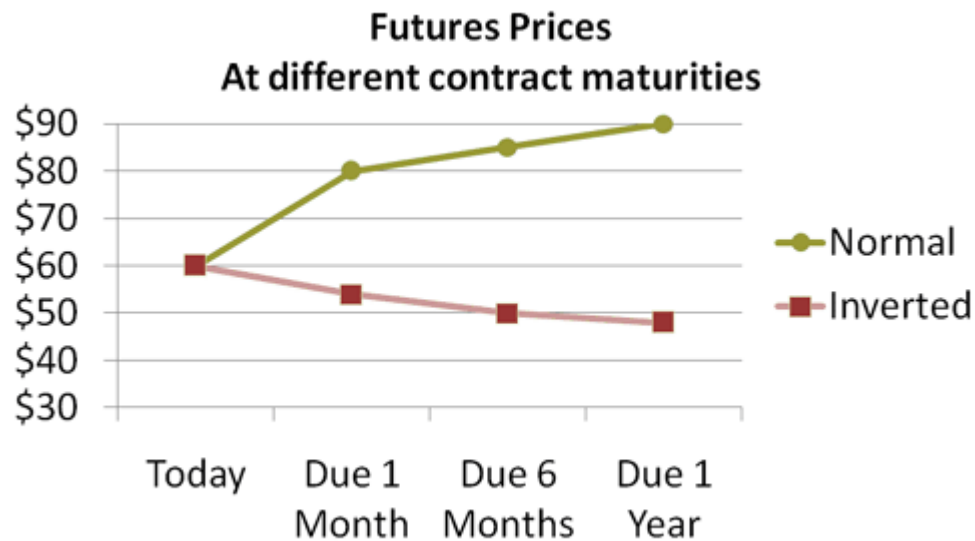
The Cost of Carry is also called « Roll yield »

To sum up

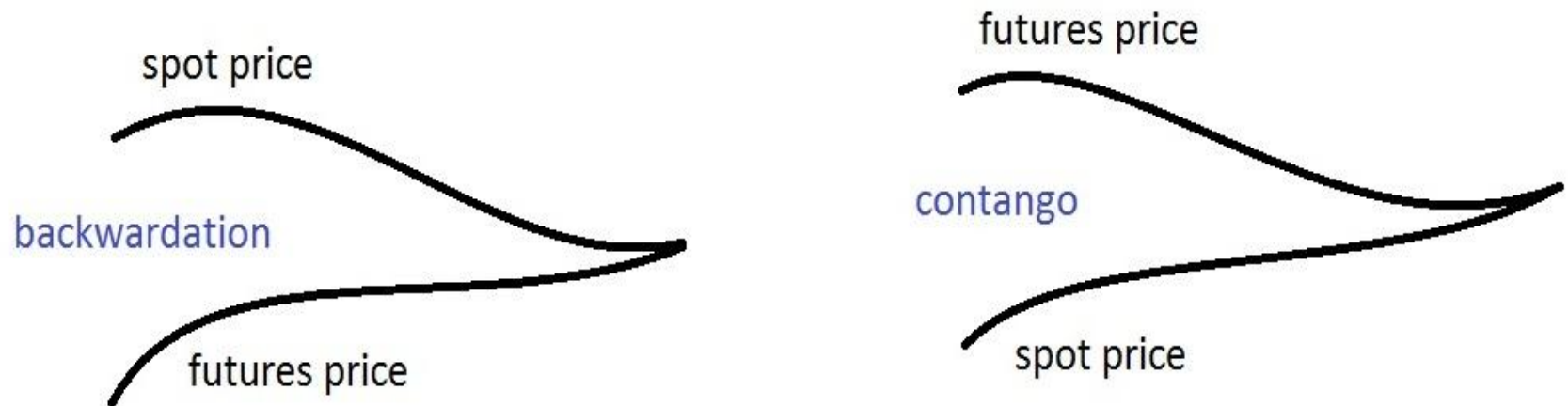


Source: Kotak sec

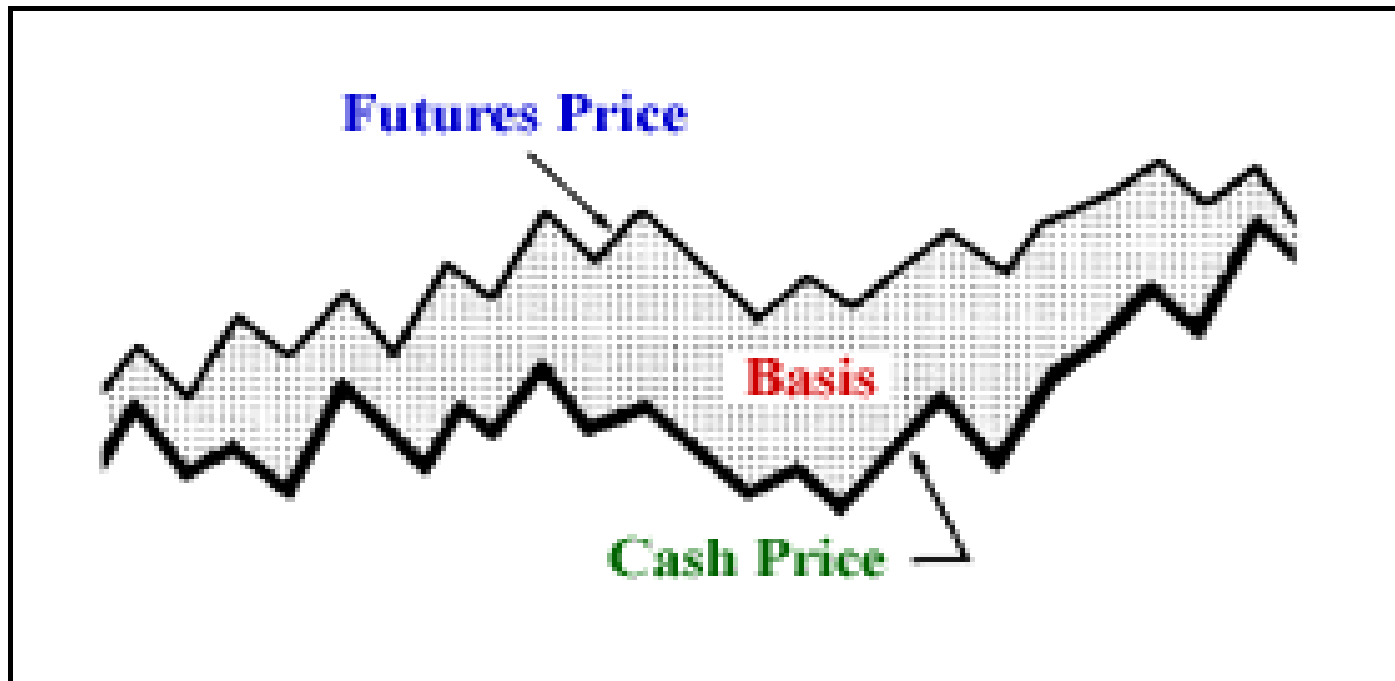
1.4 Commodity market configuration



Backwardation (déport)/Contango (report)



Basis = Cash price – Futures price



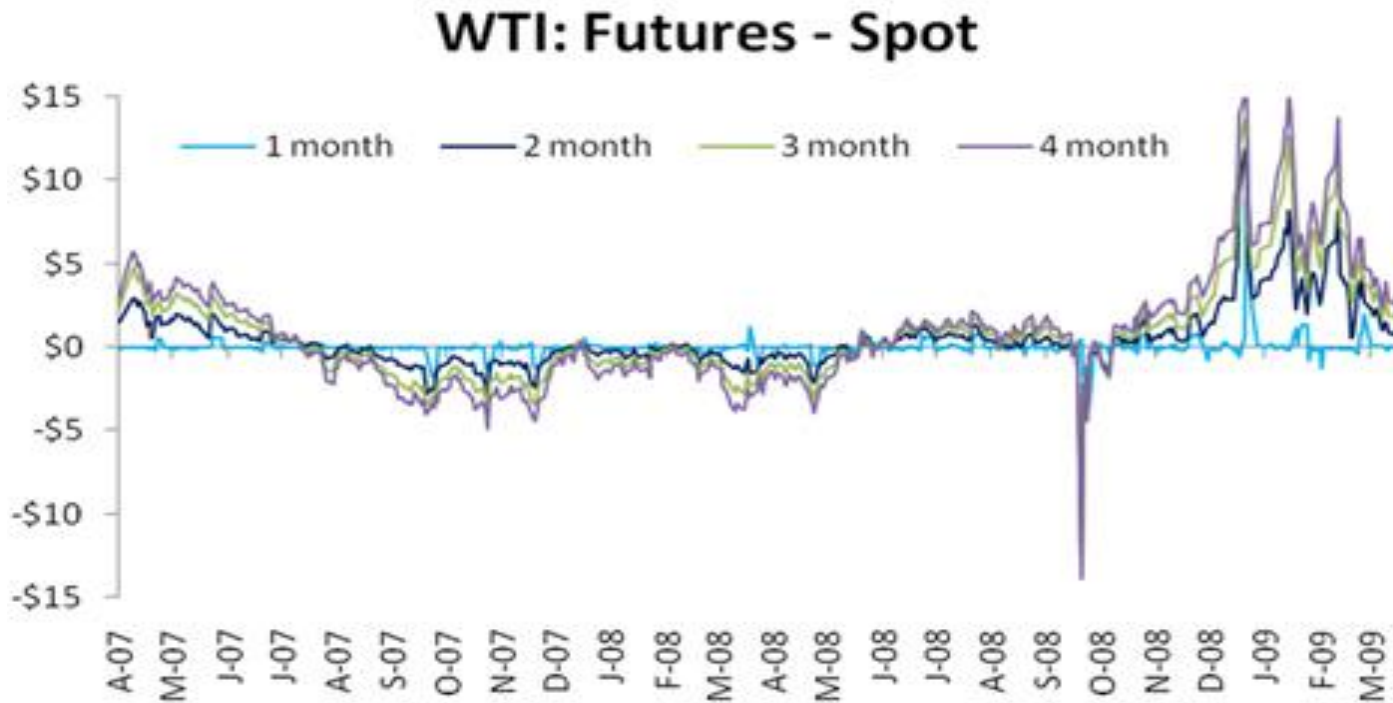
Gold Futures curve example



Source: BBG

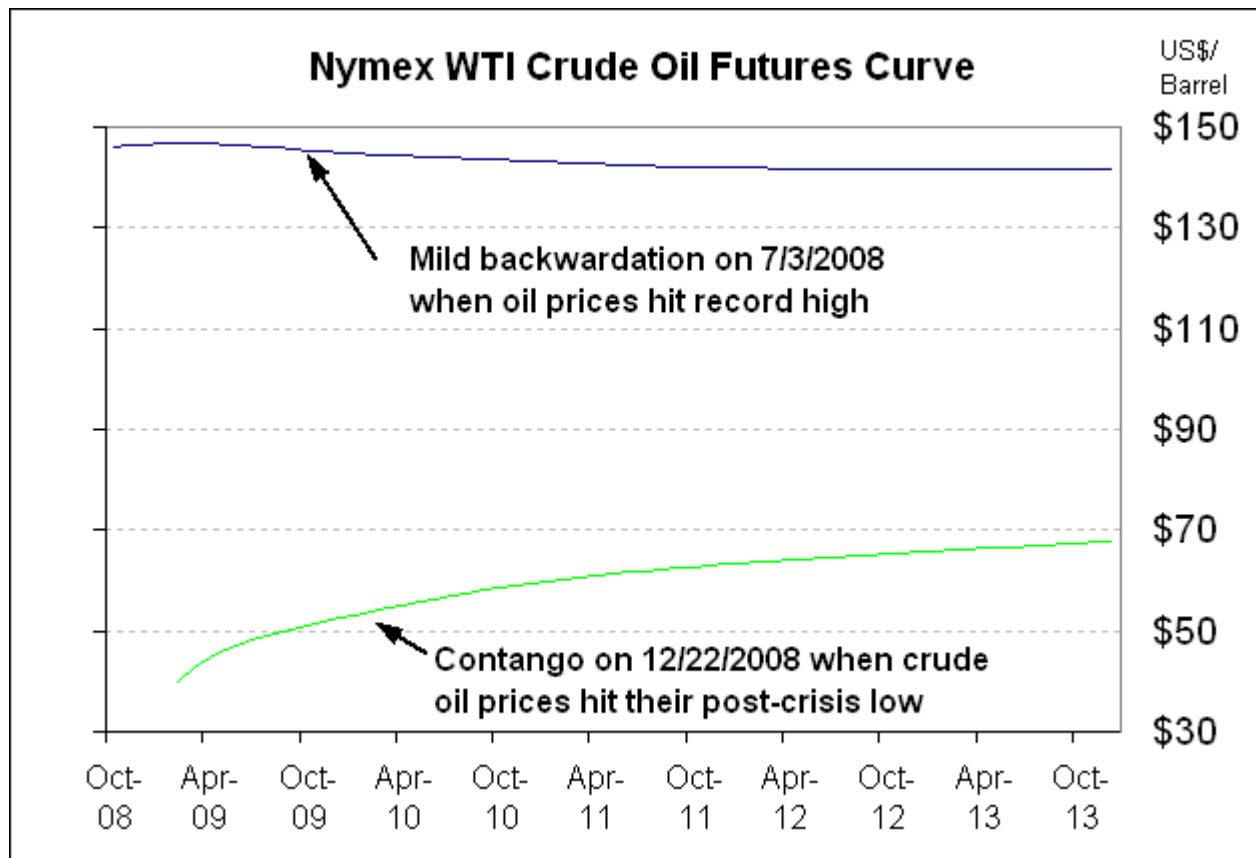
Investing in commodity markets

WTI oil backwardation example



Source: Goldman Sachs, Bloomberg

WTI oil backwardation example



Source: Goldman Sachs, Bloomberg

How do futures contracts roll over?

To avoid getting physical delivery of commodities upon expiry of the futures contract, investors maintain their exposure by rolling their futures position to a contract with later maturity.

This process can incur a carry return because futures prices may converge to spot prices over time, i.e., there is a gain or loss in carrying the futures contract up or down the curve.

When futures curves are in contango, i.e., prices are upward sloping, roll yield is typically negative and when curves are in backwardation, i.e., prices are downwards sloping, roll yield is typically positive

Less than 5 % of contracts go to maturity - the roll over process -



Source: Commodity.com

Contango impact for oil ETF

GESTION D'ACTIFS

Un tracker spécialisé a pris 6,77 %, contre près de 47 % pour le WTI.

Etienne Goetz

@etiennegoetz

A la chasse, on appellerait cela rater un éléphant dans un couloir. Le fonds indicier coté US Oil (USO), spécialisé dans le pétrole le plus populaire au monde, est passé à côté de la remontée des cours de l'or noir. En effet, cet ETF (« exchange traded fund ») n'a gagné que 6,77 % depuis le début de l'année. Dans le même temps, son indice de référence, le baril de pétrole texan, le WTI, a bondi de 46,84 %.

Comment expliquer un tel décalage alors que les ETF sont censés coller au plus près à l'évolution des cours des indices ou des matières premières qu'ils suivent ? La faute reviendrait au phénomène technique du « contango » (lire ci-contre).

une situation anormale de marché entraînant des pertes lors des renouvellements de contrats sur les marchés dérivés. Le contango « a pesé sur les rendements des investisseurs, de telle sorte que, même si la valeur nominale d'un contrat à terme a augmenté, maintenir sa position sur le pétrole avait un coût élevé », explique à Bloomberg John Love, président de la société gérant USO, tout en rappelant que les investisseurs sur le marché physique de l'or noir doivent eux aussi faire face à des coûts de stockage, d'assurance, etc.

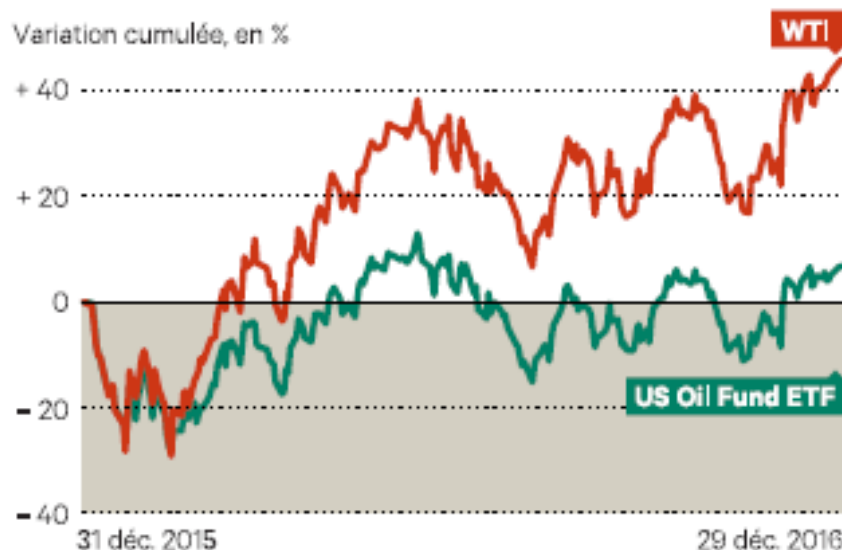
La courbe devrait s'aplanir

Le nuage est cependant en train de se dissiper. Après le fort rebond de 2016, la courbe des prix à terme devrait s'aplanir courant 2017, en raison d'un marché plus tendu. Citigroup relevait déjà début décembre que la différence entre un baril WTI livrable dans un mois ou dans un an était tombée à 2,95 dollars, contre 5,32 dollars en novembre.

Les fonds indiciers traquant les cours du pétrole sont des instru-

L'ETF US Oil Fund a raté le rebond du baril de WTI américain

Variation cumulée, en %



« LES ÉCHOS » / SOURCE : BLOOMBERG

ments financiers permettant de parier sur l'or noir sans avoir besoin d'un capital important, comme cela est nécessaire quand on investit directement dans des matières premières. D'où une popularité crois-

sante. La taille d'USO est passée de 437 millions de dollars en mai 2014 à 3,2 milliards fin décembre. Les investisseurs espéraient alors jouer par ce biais un rebond des cours. Pas sûr que ce fut le meilleur choix. ■

Source: Les Echos

2. Understanding commodity indices

-Direct investment in physical commodities is not easy : investors need to face logistical difficulties such as transportation, storage and insurance of the stocks

-it can be more practical and cost efficient to use derivatives, especially commodity futures

Commodity related equity is not a pure play

Commodities prices could impact a producing or processing company's stock price

BUT

Capacity utilisation, debt-equity ratio, valuation, government policies and corporate governance issues for example could also have a big impact on stock price

2.1 Major commodity indices

S&P GSCI – inception date: 31-12-1969

BCOM* – inception date: 31-12-1990

*Bloomberg Commodity Index ex DJ-AIG Index

Methodology

Weighting : world production value & market liquidity

Futures contract used : front month contract

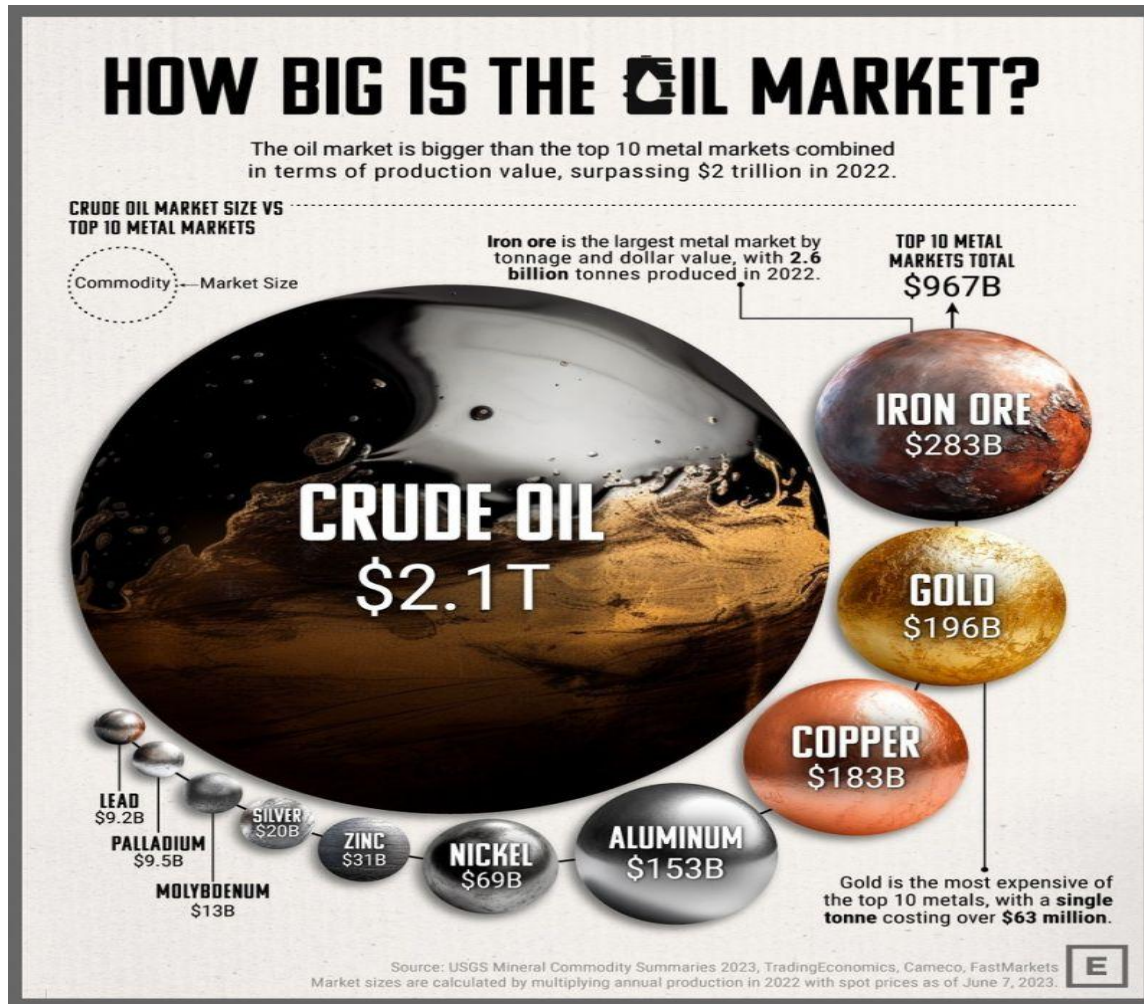
Number of markets : S&P GSCI (24)+BCOM (22)

Rebalancing frequency : annual

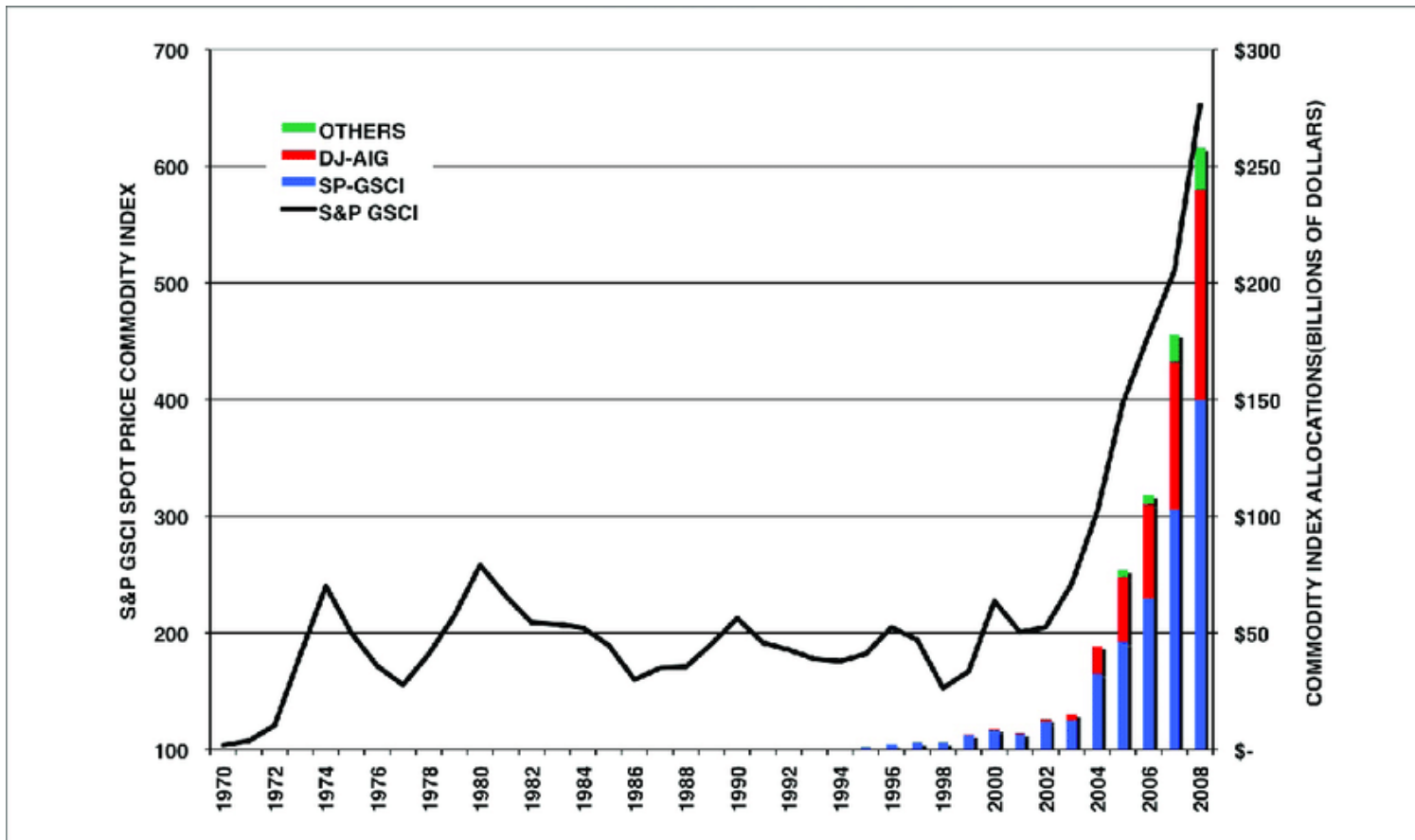
Sector weighting constraints : max 33 % for BCOM

Roll period : five business days (fifth to the ninth business day of the month)

Examples of production value



S&P GSCI is the reference



Source: Research Gate

Investing in commodity markets

S&P GSCI – 2025 weights

2025 COMMODITY REFERENCE PERCENTAGE DOLLAR WEIGHT COMPARISON - PRO FORMA

Symbol	Commodity	2024 CPW	2025 CPW	2025 ACRP	2025 TDW	2024 RPDW	2025 RPDW	RPDW BP Diff
W	Chicago Wheat	19982.69	18940.32	5.824583333	110319.4722	3.175%	2.502%	-67
KW	Kansas Wheat	7751.352	8978.328	6.180833333	55493.54898	1.449%	1.258%	-19
C	Corn	44787.06	45400.45	4.427916667	201029.4092	6.001%	4.558%	-144
S	Soybeans	12724.26	12999.81	11.89604167	154646.2814	3.966%	3.507%	-46
KC	Coffee	22077.66	22909.76	1.993958333	45681.10687	0.854%	1.036%	18
SB	Sugar	393391.2	397190.2	0.220716667	87666.49698	1.888%	1.988%	10
CC	Cocoa	5.303011	5.530011	6537.750000	36153.82942	0.335%	0.820%	48
CT	Cotton	55831.68	55873.44	0.809833333	45248.17416	1.015%	1.026%	1
LH	Lean Hogs	111483.7	130487.7	0.835916667	109076.8432	2.105%	2.473%	37
LC	Live Cattle	110944.0	114369.8	1.801854167	206077.7007	4.001%	4.673%	67
FC	Feeder Cattle	35136.87	38776.70	2.453145833	95124.90004	1.624%	2.157%	53
CL	WTI Crude Oil	11252.59	11300.89	79.05166667	893354.1893	19.31%	20.26%	95
HO	Heating Oil	74629.31	70482.35	2.642750000	186267.2305	4.812%	4.224%	-59
RB	RBOB Gasoline	78382.22	72252.20	2.381458333	172065.6038	4.385%	3.902%	-48
LCO	Brent Crude Oil	11345.07	11026.17	82.89916667	914060.3045	20.73%	20.73%	0
LGO	Gas Oil	311.2954	327.9326	794.6250000	260583.4423	5.729%	5.909%	18
NG	Natural Gas	42118.17	43851.52	2.407083333	105554.2629	3.466%	2.393%	-107
MAL	Aluminum	69.00000	70.18000	2364.555000	165944.4699	3.499%	3.763%	26
MCU	Copper	24.28000	24.50000	8917.437500	218477.2188	4.487%	4.954%	47
MNI	Nickel	2.162800	2.281200	17449.48500	39805.76518	1.130%	0.903%	-23
MPB	Lead	11.64000	11.84000	2126.120000	25173.26080	0.540%	0.571%	3
MZN	Zinc	13.58000	13.54000	2655.546667	35956.10187	0.827%	0.815%	-1
GC	Gold	103.2039	102.5609	2194.900000	225110.9194	4.252%	5.104%	85
SI	Silver	841.7066	818.5581	25.90441667	21204.27009	0.422%	0.481%	6

Source: S&P

Investing in commodity markets

S&P GSCI – 2025 weights

2025 S&P GSCI RPDW SECTOR WEIGHTS - PRO FORMA

Sector	2024 Weight	2025 Weight	Sector RPDW BP Diff
Energy	58.43%	57.41%	-102
Agriculture	18.68%	16.69%	-199
Livestock	7.730%	9.303%	157
Industrial Metals	10.48%	11.01%	52
Precious Metals	4.674%	5.585%	91

Source: S&P

BCOM – 2025 weights

Group	Commodity	Ticker	2025 Target Weight	2024 Target Weight
Energy	Brent Crude Oil	CO	8.0339950%	7.6379950%
	Natural Gas	NG	7.7783690%	7.9841690%
	WTI Crude Oil	CL	6.9660050%	7.3620050%
	Low Sulphur Gas Oil	QS	2.8583140%	2.7798210%
	ULS Diesel	HO	2.2039330%	2.1604180%
	RBOB Gasoline	XB	2.1702560%	2.2073490%
			30.0108720%	30.1317570%
Grains	Soybeans	S	5.9660320%	5.9068480%
	Corn	C	5.6241090%	5.6623490%
	Soybean Meal	SM	3.5290020%	3.5401900%
	Soybean Oil	BO	3.3642880%	3.3491640%
	Wheat	W	2.8250830%	2.8184120%
	HRW Wheat	KW	1.8422120%	1.8189100%
			23.1507260%	23.0958730%
Industrial Metals	Copper	HG	5.3747820%	5.2978220%
	Aluminum	LA	4.1595000%	4.1056260%
	Zinc	LX	2.2654760%	2.4945660%
	Nickel	LN	2.3993070%	2.5842820%
	Lead	LL	0.9219700%	0.8661400%
			15.1210350%	15.3484360%
Precious Metals	Gold	GC	14.2907530%	14.3468000%
	Silver	SI	4.4913890%	4.4770770%
			18.7821420%	18.8238770%
Softs	Coffee	KC	3.0069210%	2.9741780%
	Sugar	SB	2.9956340%	2.8076370%
	Cotton	CT	1.6071050%	1.5703150%
			7.6096600%	7.3521300%
Livestock	Live Cattle	LC	3.5918600%	3.4650870%
	Lean Hogs	LH	1.7337010%	1.7828380%
			5.3255610%	5.2479250%

Source: BBG

Investing in commodity markets

The S&P GSCI is more focused on energy

S&P GSCI

Energy weighting (2025) = 57,41 %

VS

BCOM

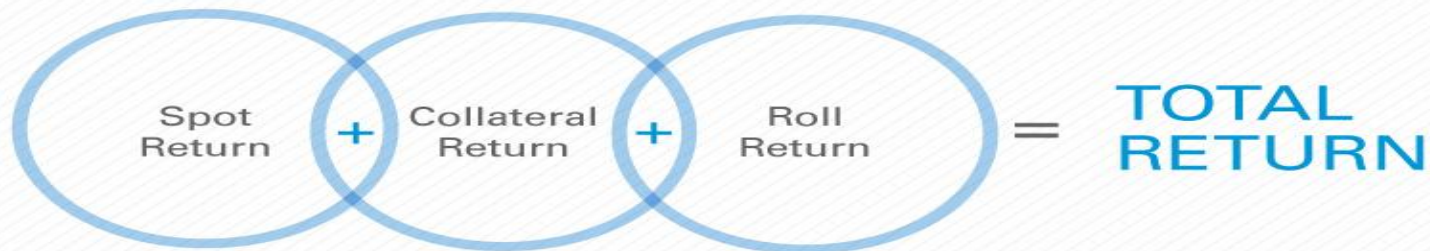
Energy weighting (2025) = 30,01 %

2.2 Returns components

- Spot return = return associated with spot price of commodity
- Excess return = spot return + carry or roll return (return from maintaining futures exposure by rolling positions)
- Total return = excess return + collateral return (return from collateral used to guarantee futures position)

Returns components illustration

THE THREE COMPONENTS THAT MAKE UP THE TOTAL RETURN ON A FUTURES CONTRACT



SPOT RETURN: The return generated from selling a commodity for cash

COLLATERAL RETURN: The return generated from holding US Treasury Bills or other securities which are used to secure a futures contract.

ROLL RETURN: The return, either positive or negative, generated by rolling from a short-term futures contract to a longer term futures contract.

Source: Deutsche Bank

Spot, Excess and Total Return Indices

SPGC
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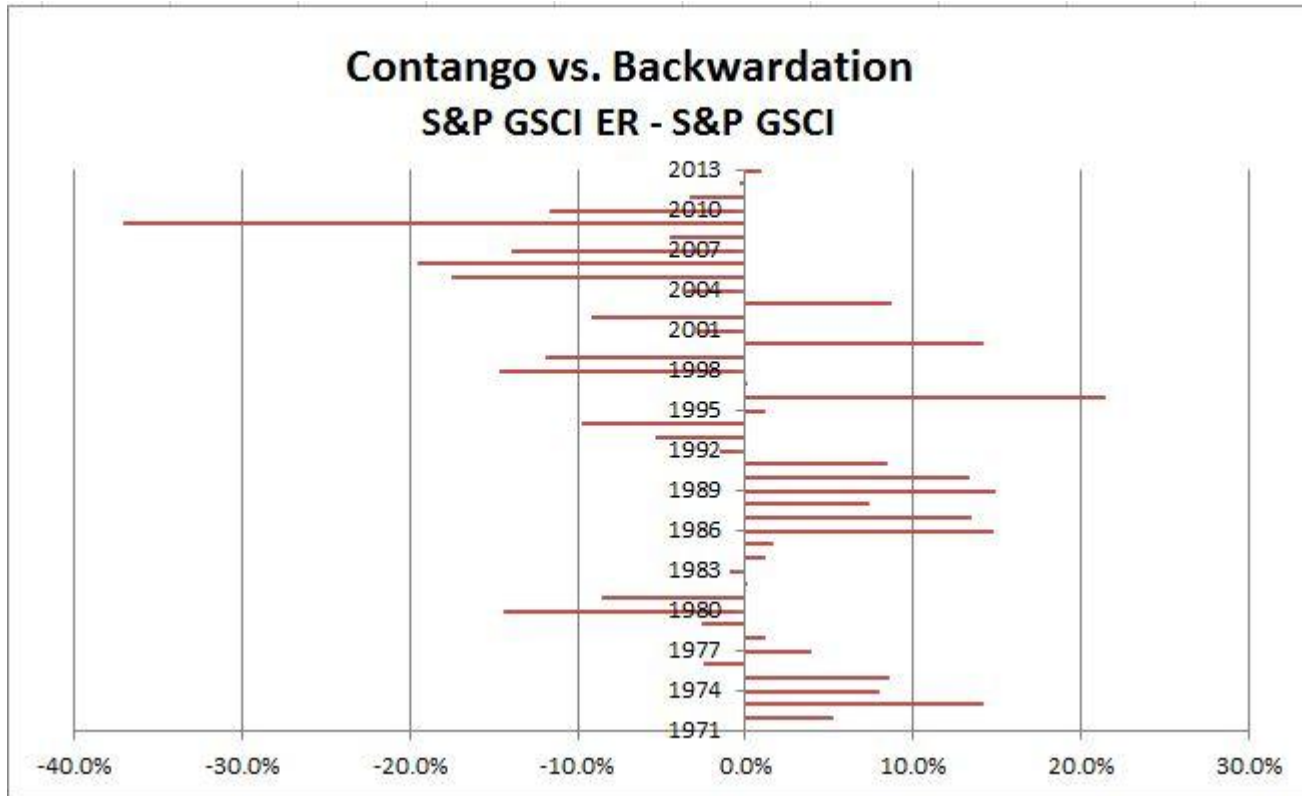
					S&P GSCI Indices	
	Current	Official	Change	MTD	YTD	12 Month
1) Spot Index	Value	Close	On Day	%Change	%Change	%Change
4) S&P GSCI	658.572	658.572300	-4.870	.00	+4.16	+6.91
5) S&P GSCI Energy	349.076	349.076200	-1.862	.00	+3.26	+8.22
6) S&P GSCI Indust. Mtl	358.588	358.587500	+2.743	.00	+2.50	+5.82
7) S&P GSCI Agriculture	349.779	349.779300	-12.734	.00	-.53	-8.12
8) S&P GSCI Livestock	407.325	407.325300	+1.861	.00	+27.37	+30.37
2) Excess Return Index						
9) S&P GSCI ER	494.696	494.696200	-3.658	.00	+5.69	+9.51
10) S&P GSCI Energy ER	320.015	320.015000	-1.707	.00	+5.59	+13.70
11) S&P GSCI Ind. Mtl ER	203.117	203.117300	+1.554	.00	+1.23	+5.31
12) S&P GSCI Agri ER	59.732	59.732280	-2.175	.00	+1.64	-8.41
13) S&P GSCI Livestock ER	242.422	242.421700	+1.107	.00	+20.90	+22.90
3) Total Return Index						
14) S&P GSCI TR	5105.464	5105.464000	-37.745	.00	+5.71	+9.56
15) S&P GSCI Energy TR	1189.290	1189.290000	-6.341	.00	+5.61	+13.75
16) S&P GSCI Ind.Mtl TR	1383.872	1383.872000	+10.588	.00	+1.25	+5.36
17) S&P GSCI Agri TR	616.623	616.622500	-22.447	.00	+1.66	-7.40
18) S&P GSCI Livestock TR	2502.079	2502.079000	+11.433	.00	+20.92	+22.96

Australia 61 2 9777 8600 Brazil 5511 3048 4500 Europe 44 20 7330 7500 Germany 49 69 9204 1210 Hong Kong 852 2977 6000
Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2014 Bloomberg Finance L.P.
SN 977691 EDT GMT-4:00 H191-2632-0 01-Jul-2014 11:11:01

Source: BBG

Investing in commodity markets

S&P GSCI – roll return impact



Source: S&P

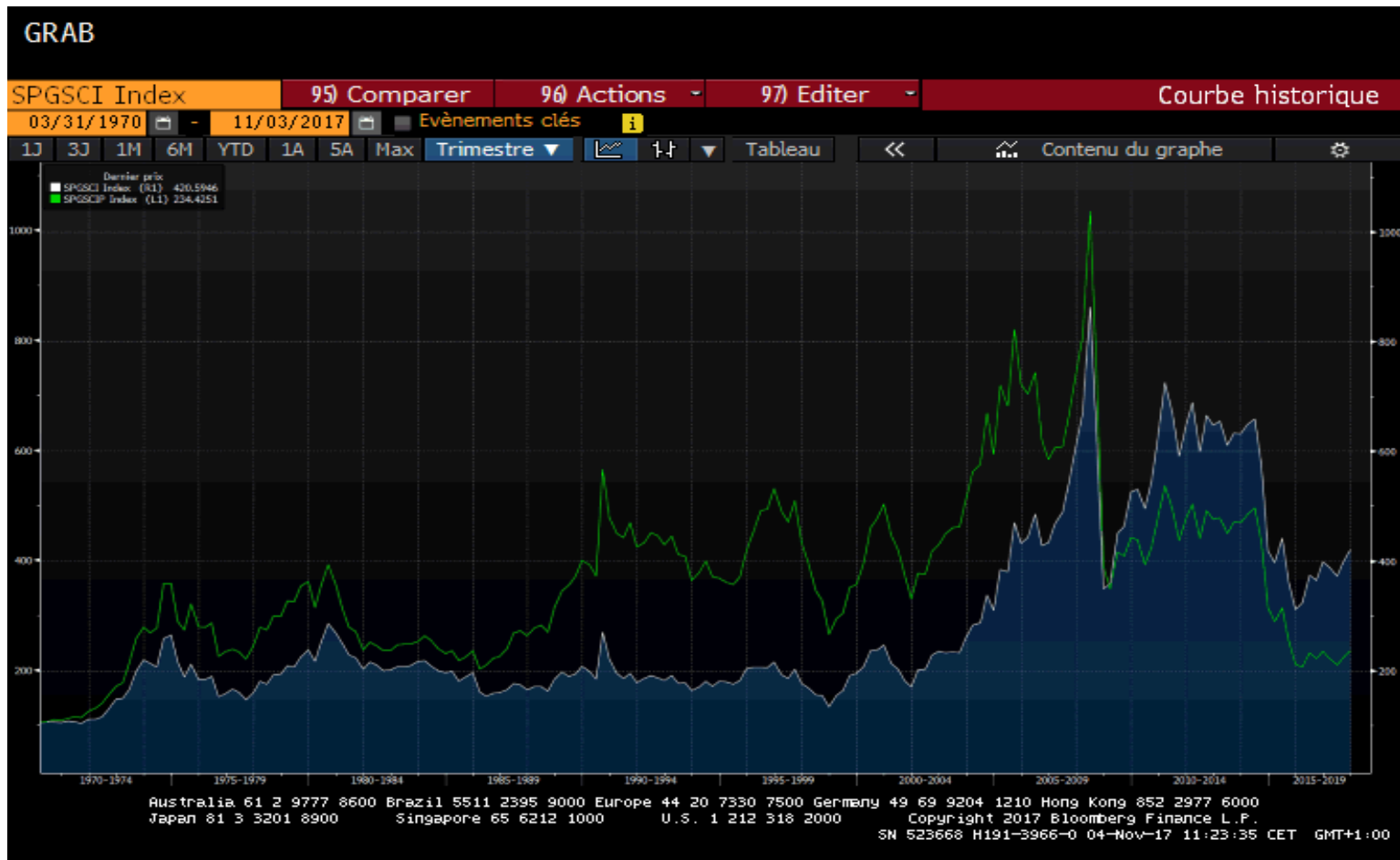
S&P GSCI – roll impact chart

Fig 6. S&P GSCI Spot and Total Return



Source: Bloomberg, S&P, Ecstrat, h/t to Oliver Jacob @petromatrix

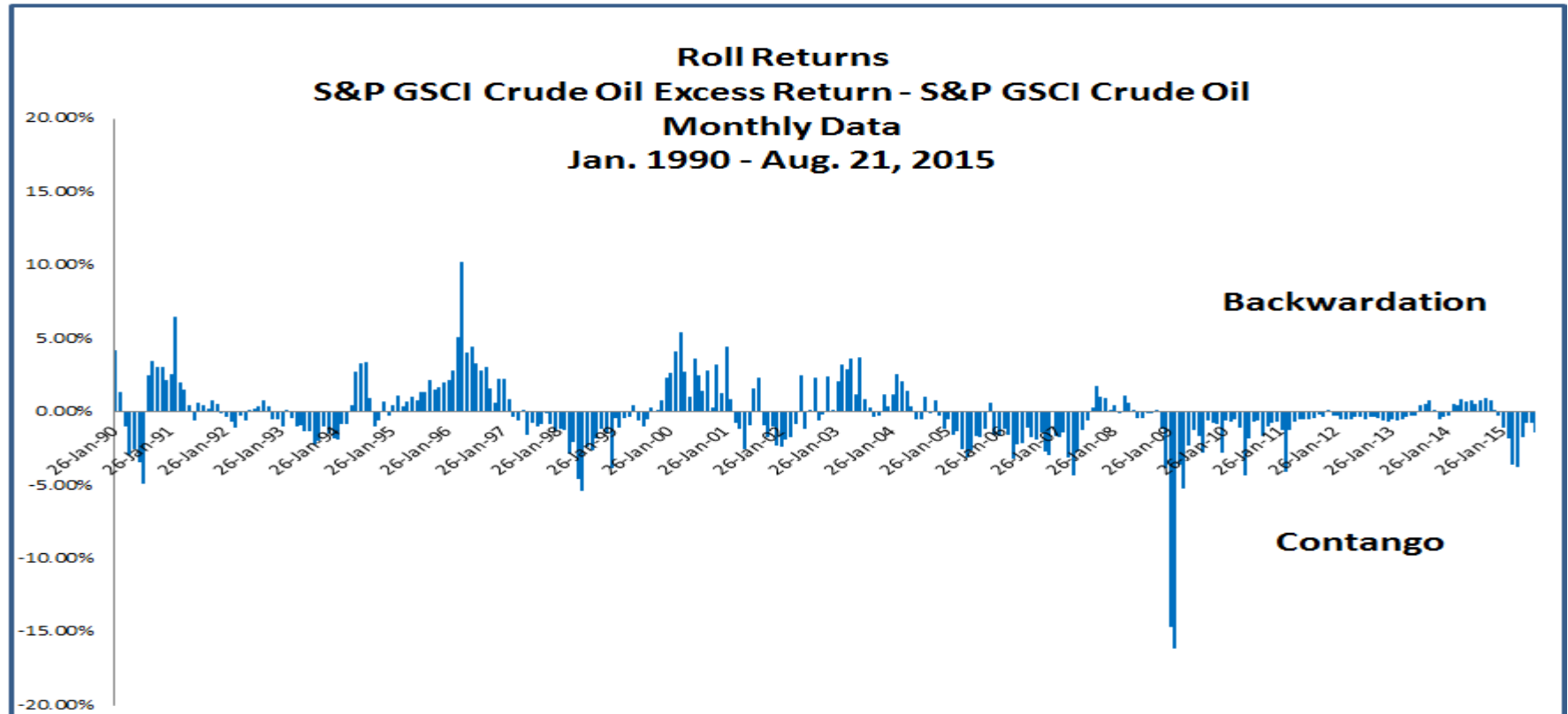
S&P GSCI Index (Spot vs ER)



Source: BBG

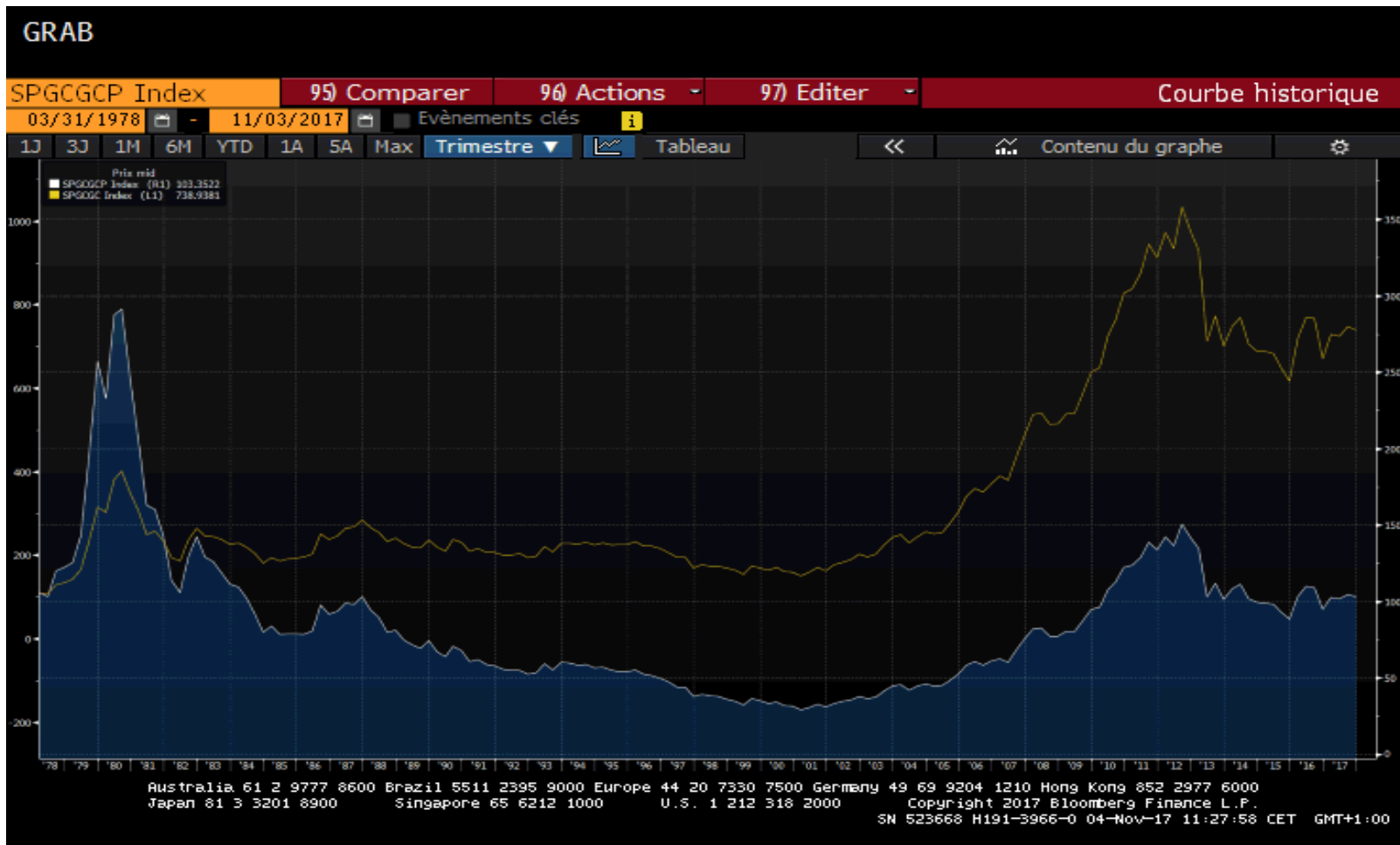
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Example for WTI Crude oil (S&P GSCI)



Source: S&P

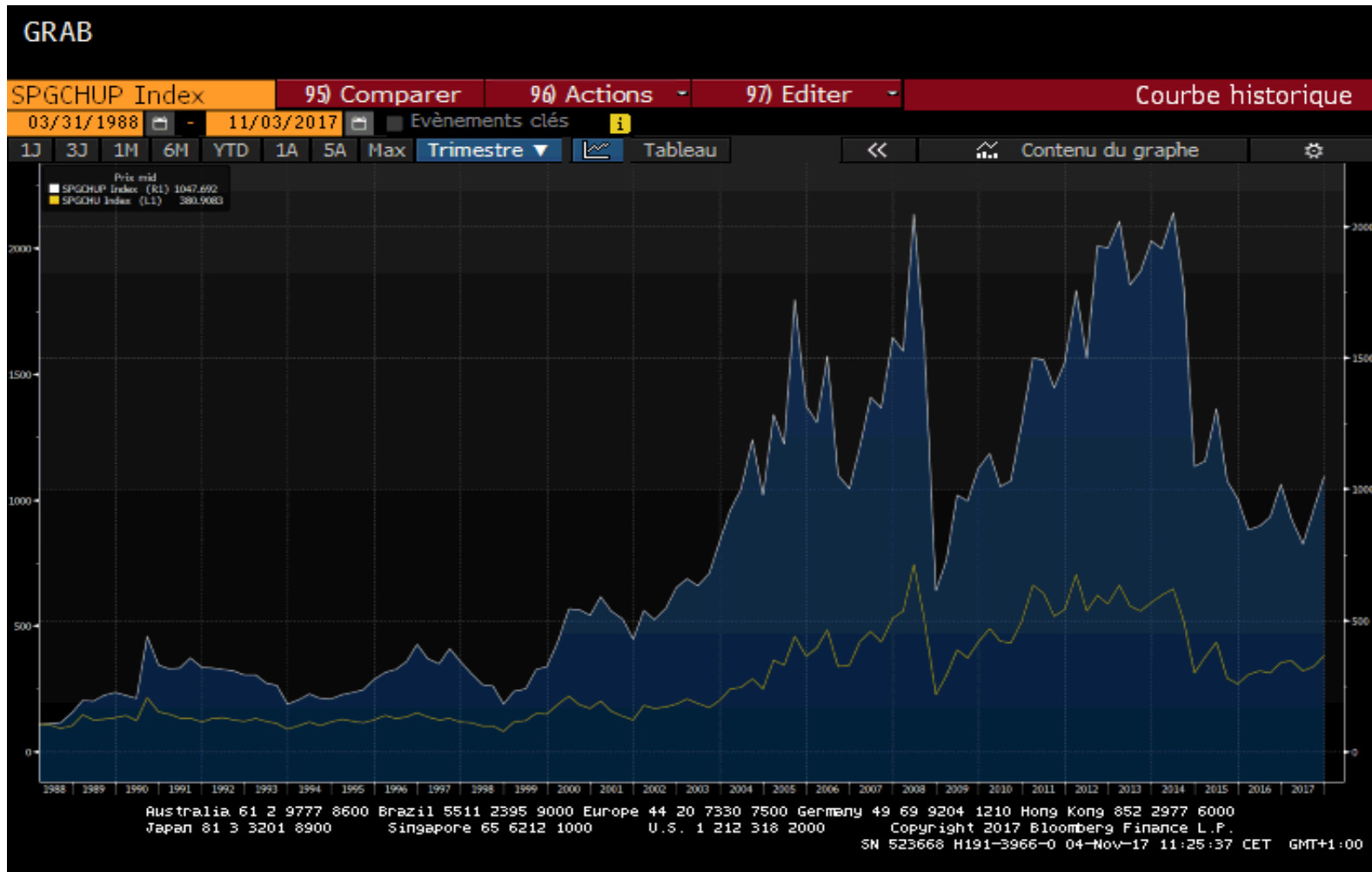
S&P GSCI Gold (Spot vs ER index)



Source: BBG

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S&P GSCI Gasoline (Spot vs ER index)



Source: BBG

Investing in commodity markets

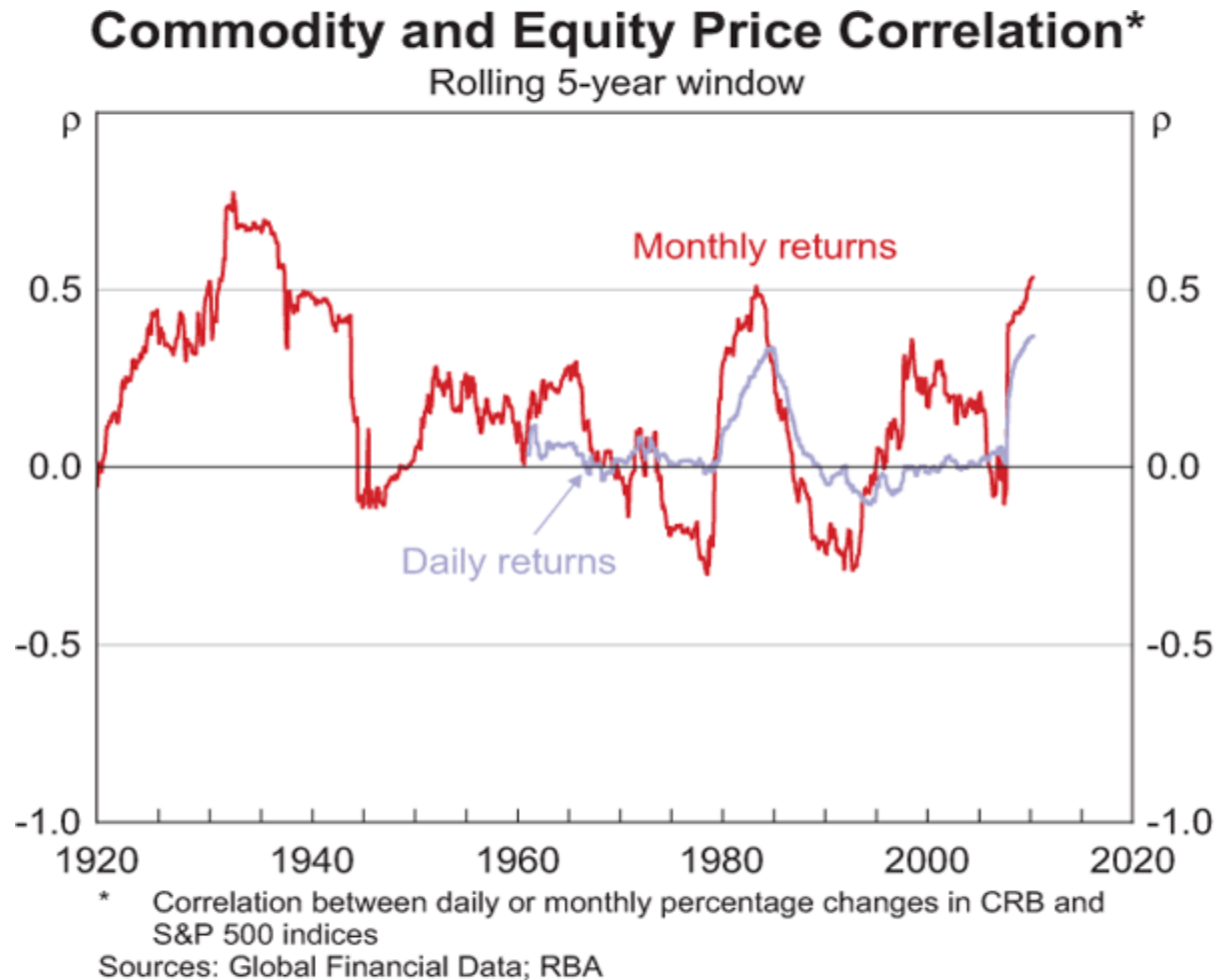
S&P GSCI ER – Historical Volatility



Source: BBG

Investing in commodity markets

2.3 Commodity correlation



S&P GSCI ER – Correlation vs S&P 500

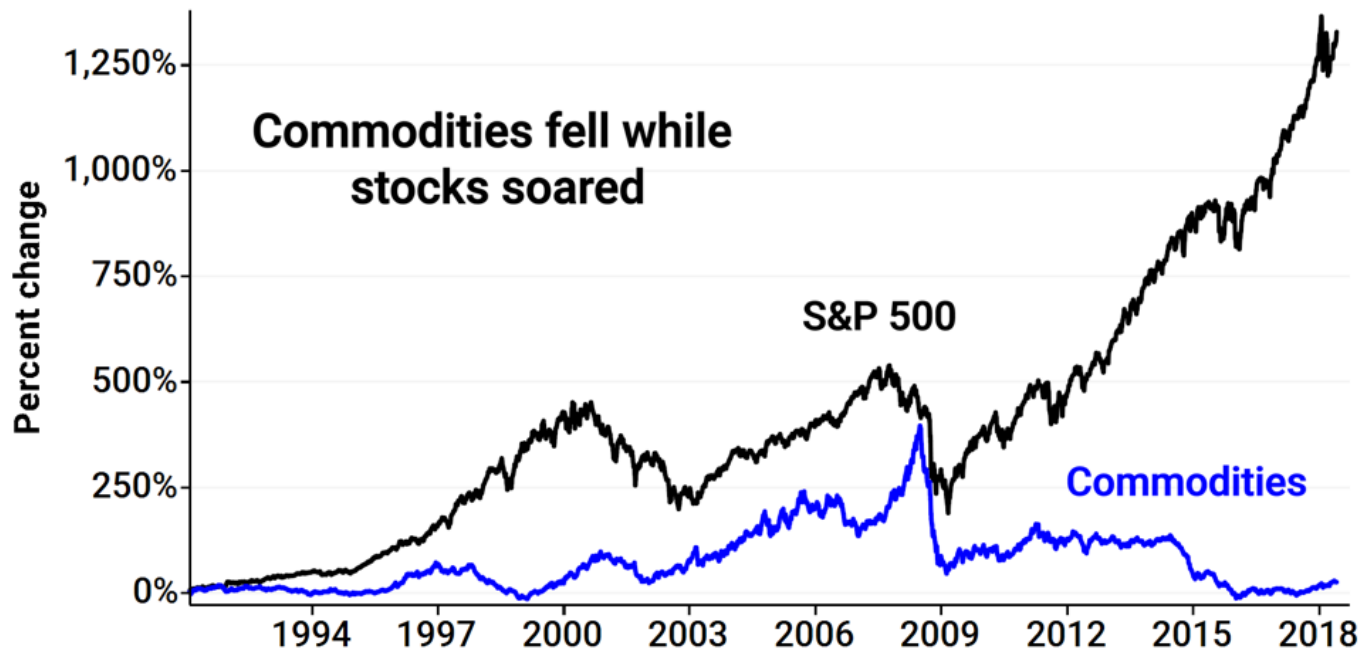


Source: BBG

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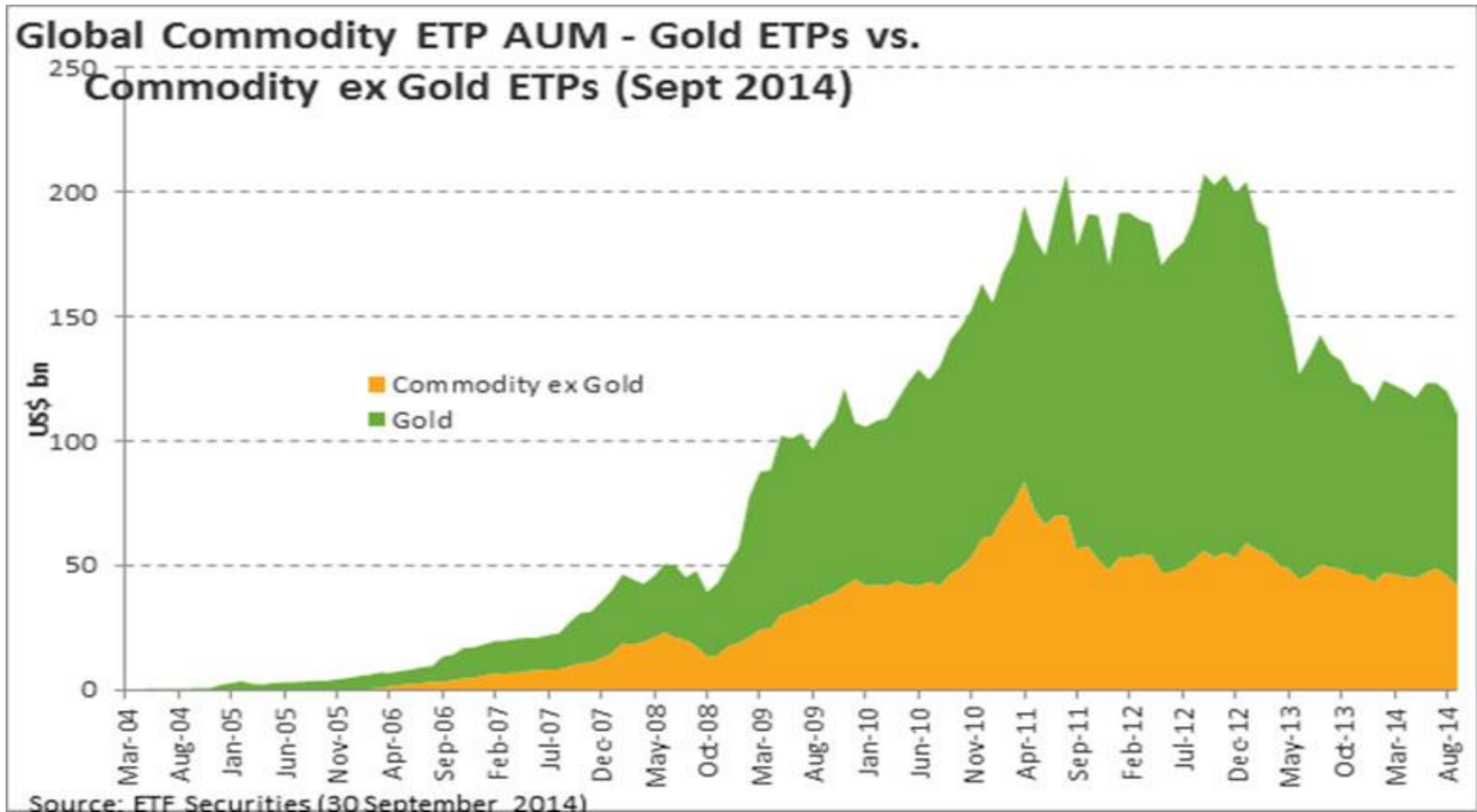
A change after the financial crisis

S&P 500 vs. Commodities

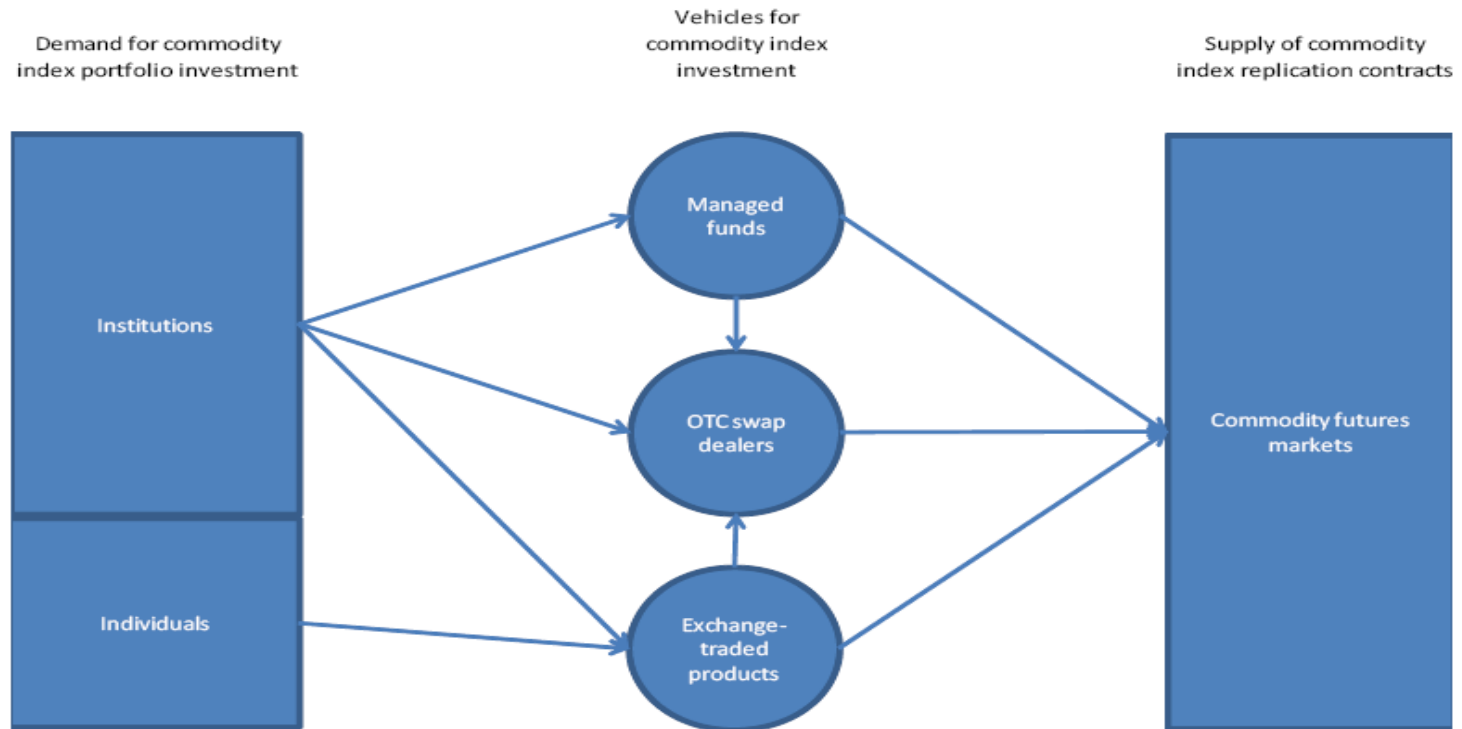


Source: S&P

More and more 'financial players'



3. Different types of investment vehicles



Source: Risks

3.1 ETF – exchange traded funds

-Advantages : low management fee, trade like a stock at real time price, no minimum investment, liquidity

-Disadvantages : bid/ask spread could be large

Top 3 commodity ETF by AUM (31-dec-25)

Name	Sponsor	Assets under management (AUM in B USD)
SPDR Gold Trust	SPDR	69,42
iShares Gold Trust	iShares	30,16
iShares Silver Trust	iShares	13,75

Source: ETFDB

3.2 Commodity index swaps

- Advantages : lowest management fee, trade like a stock at real time price, liquidity
- Disadvantages : high minimum investment (only for institutional investors)

3.3 Managed commodity funds

- Advantages : total return strategies

- Disadvantages : high management fee, illiquidity

Concept = Capturing commodity risk premium

- Hedging pressure

- Scarcity

- Financial flows

Hedging pressure

- The futures markets exist to facilitate hedging, not to forecast prices
- The producers go short to protect against price drops and the consumers go long to protect against price increases

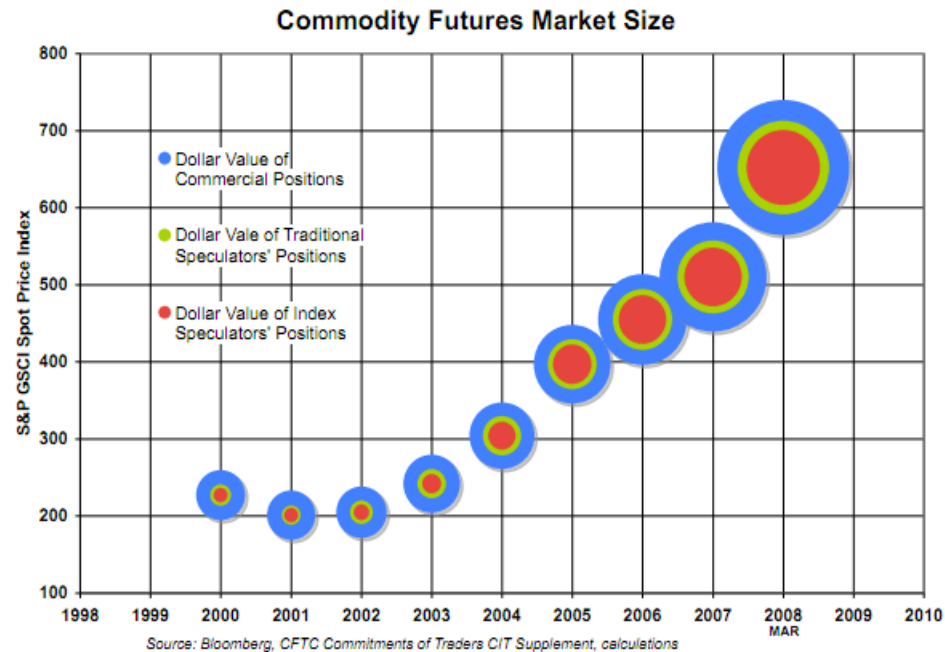
Scarcity

The prices of commodities in physically tighter markets, i.e backwardation, historically tended to outperform those in less physically tight markets

Financial flows

- Investors could have a significant short term impact on directional prices and the shape of forward curve
- It reflects the 'financialization' of commodity markets

The 'financialization' of commodity markets



Strategies based on hedging pressure

Trend following strategies are able to capture trends using momentum indicators, combining moving averages and volatility break-out filters

Trend following is based on moving averages

The commonly used moving averages is the simple moving average which is the simple average of a security over a defined number of time periods.

It gives an indication about price trends.

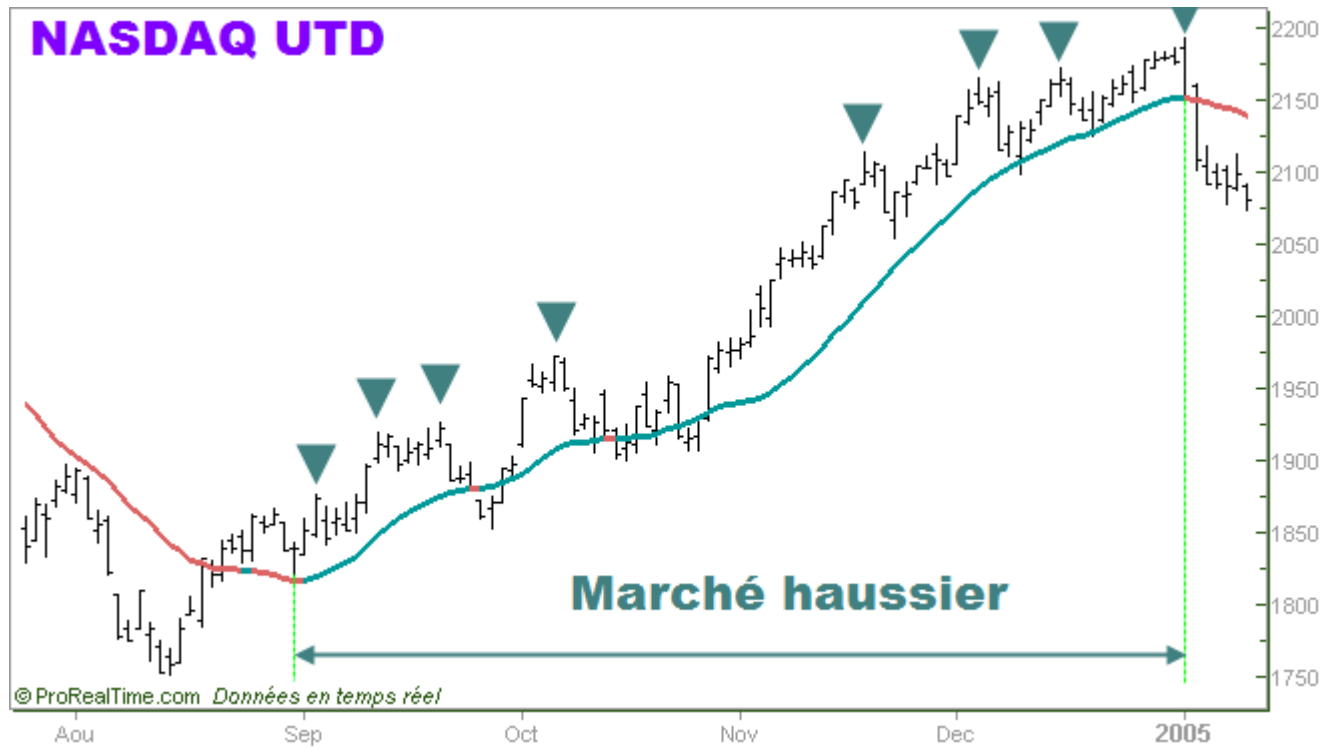
A moving average (MM) is calculated from closing price in general.

We add the last n closing prices divided by n , corresponding to the period.

$$\text{Formule MM} = \frac{\text{somme des } n \text{ clôtures}}{n}$$

n = MM period

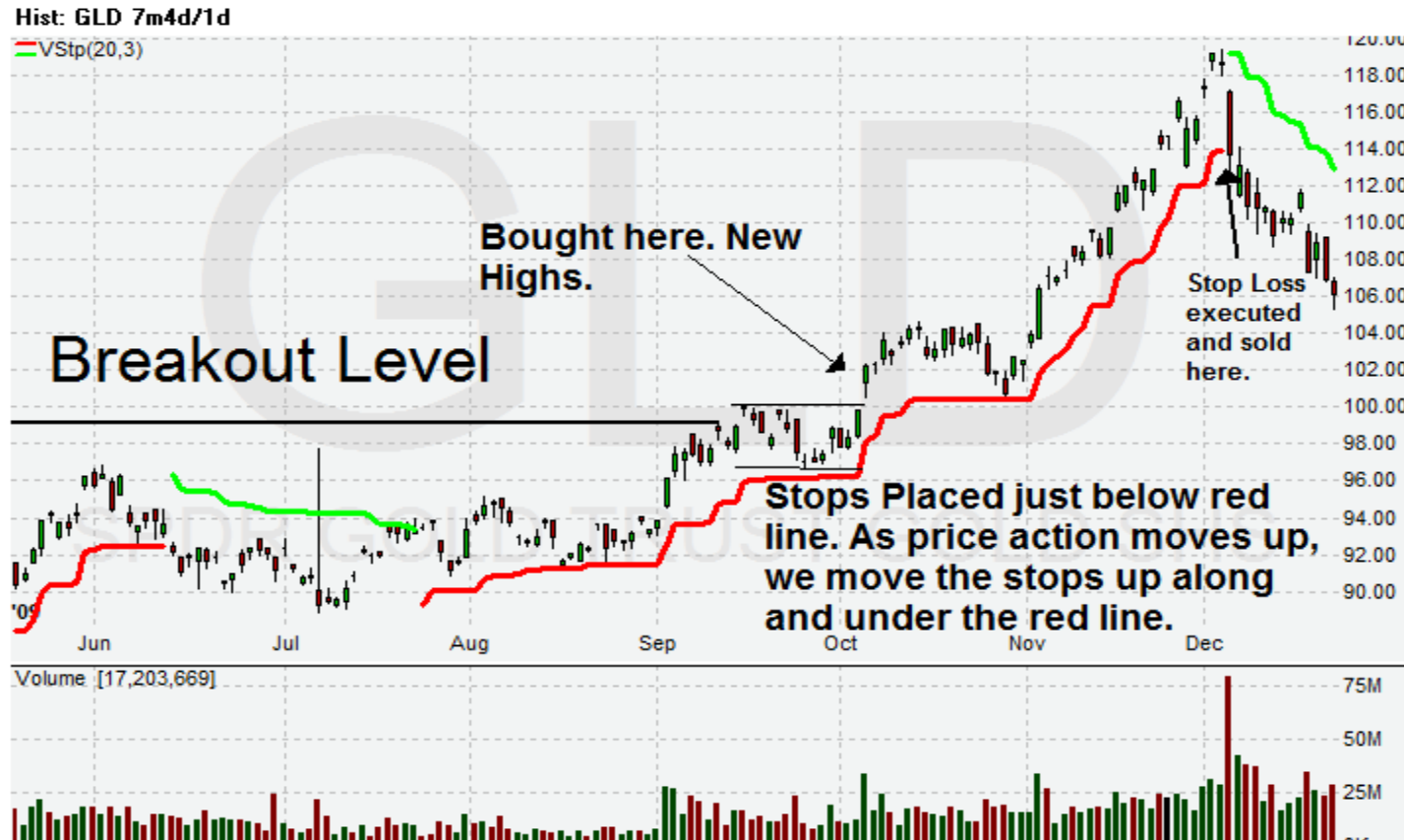
Bull market



Bear market



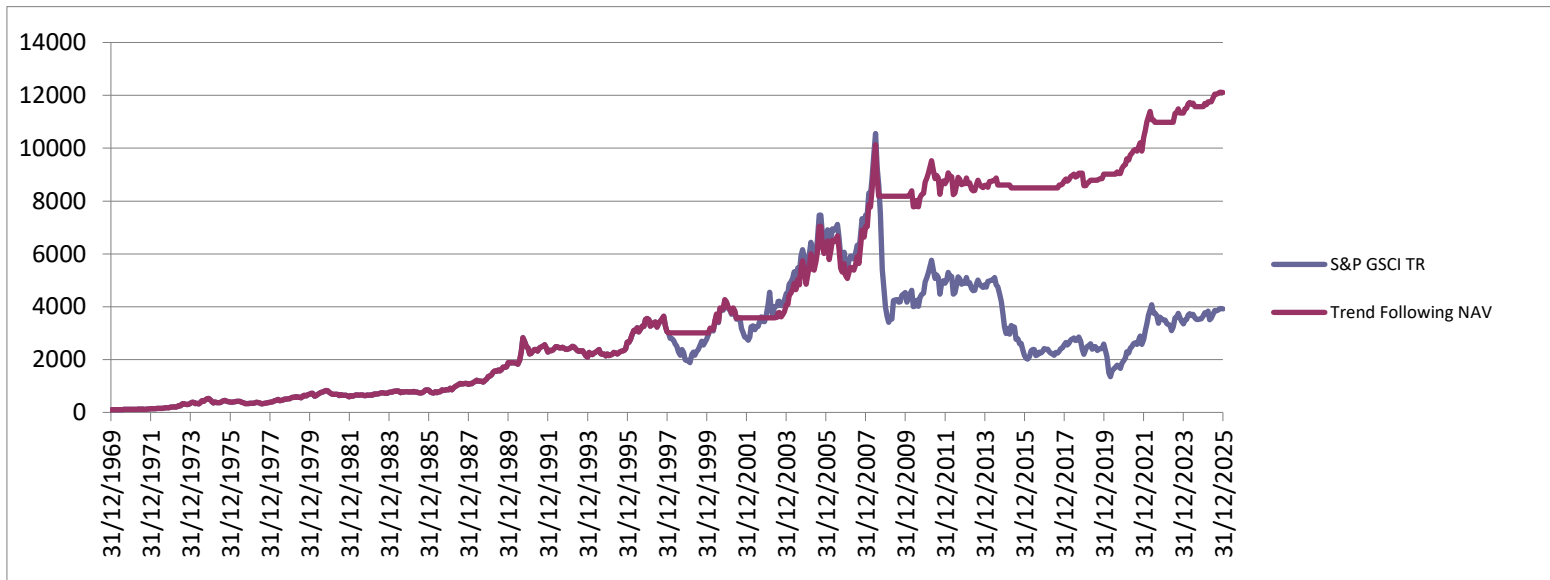
The volatility breakout concept



Source: Turtletrading

Trend following strategies NAV

(From 31-12-1969 to 31-12-2025)



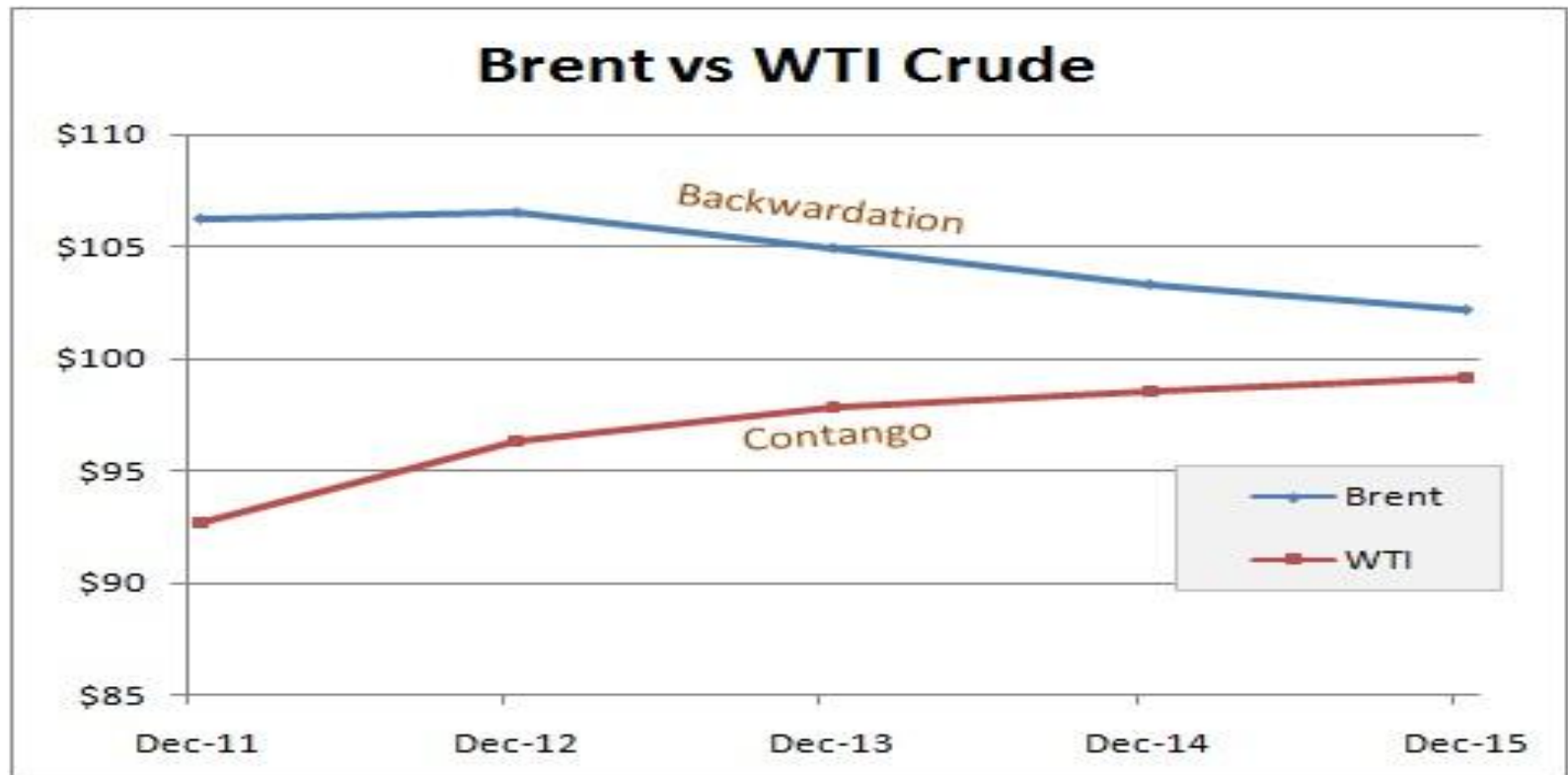
Strategies based on scarcity

Long/Short strategies are able to select the most backwardated and the most contangoed commodities in order to build an optimal portfolio

These strategies are mainly based on inter commodity spreads

Inter commodity Calendar Spread = buying a futures contract and simultaneously selling a futures contract in different commodities.

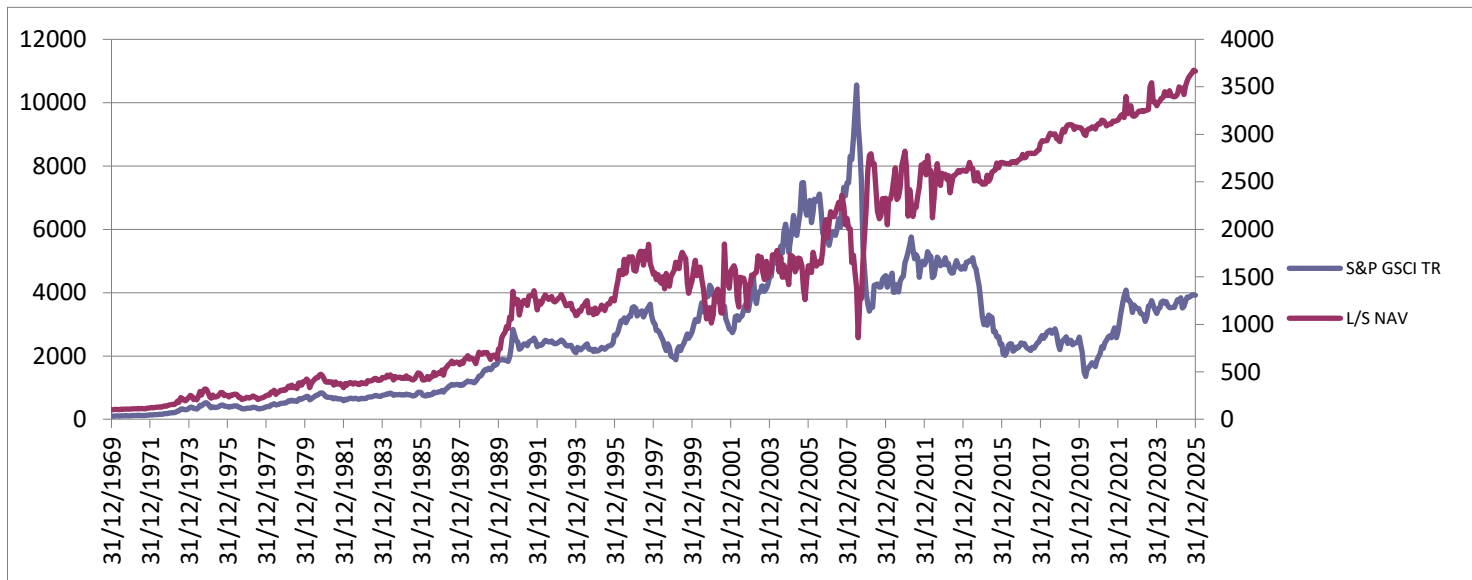
Backwardated vs 'Contangoed' commodity



Source: ICE

L/S strategies track record NAV

(From 31-12-1969 to 31-12-2025)



Strategies based on financial flows

These strategies are mainly based on intra commodity spreads also called futures calendar spreads

How it works

What is a Futures Calendar Spread?

A futures strategy that consists of buying/selling futures contracts that have different expiration months together as one position.

•How do Calendar Spreads Profit?

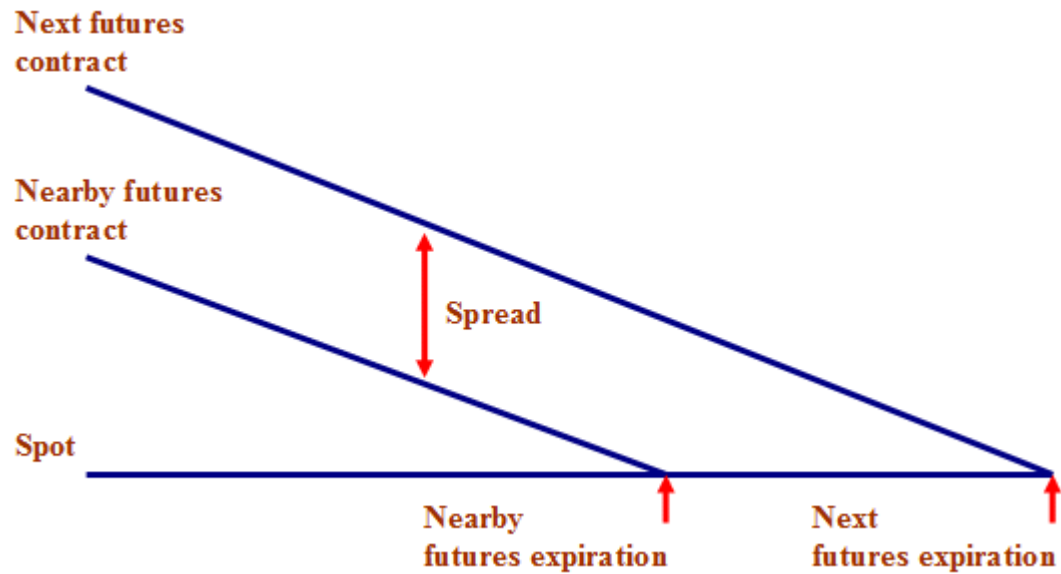
Futures Calendar Spreads Profit from the difference in rate of movement between near term futures contracts and longer term futures contracts.

•Why use a futures calendar spread?

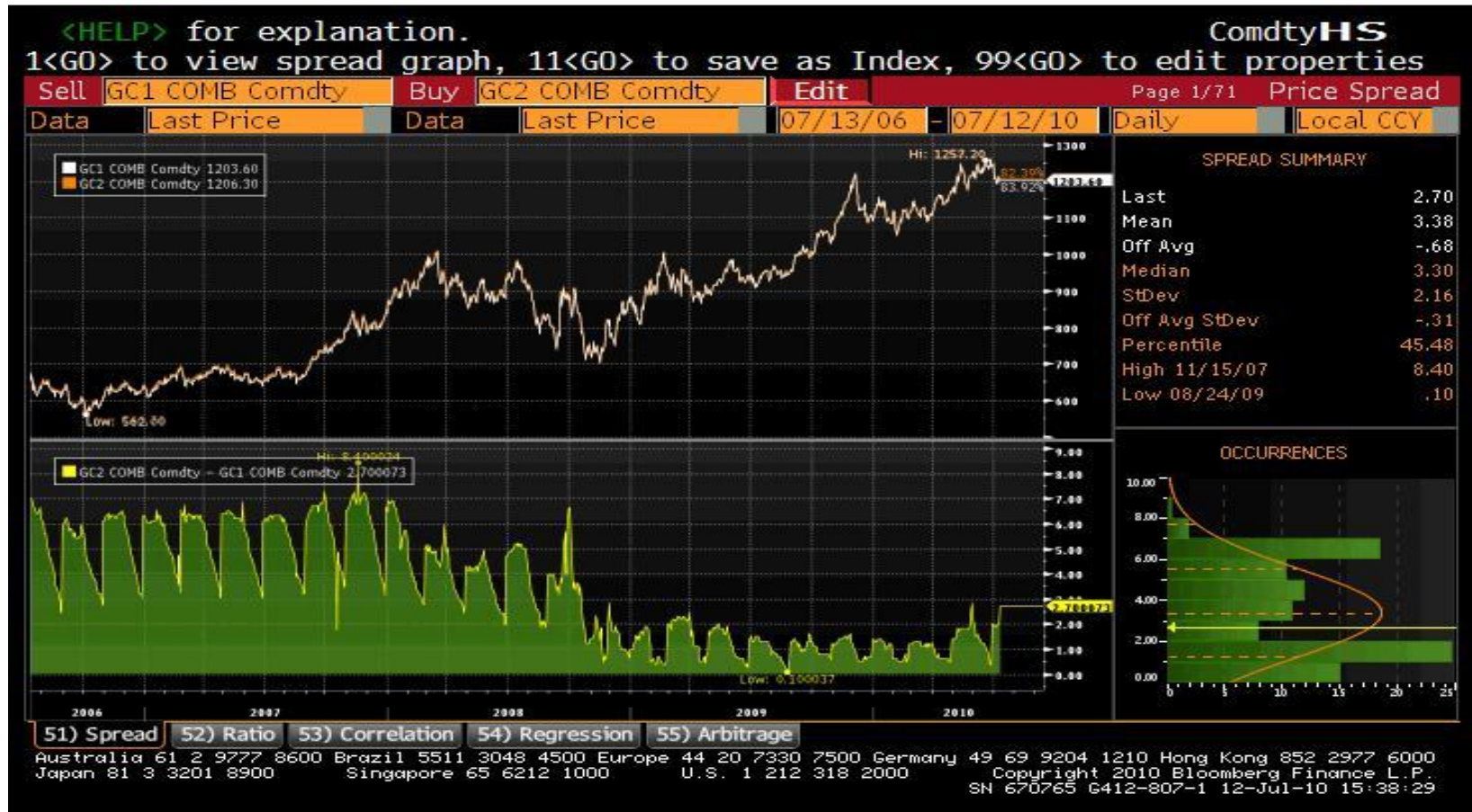
Utilizing futures contracts that have different expiration months in order to produce a bullish/bearish position that can be less volatile than an outright long/short position.

Because this strategy is less volatile, they also require less margin than outright futures positions, which results in a smaller buying power requirement and generally, a greater return on capital. It requires less margin because one contract is hedged by the other contract, which is great for traders with smaller accounts.

Spread intra commodity (calendar spread)

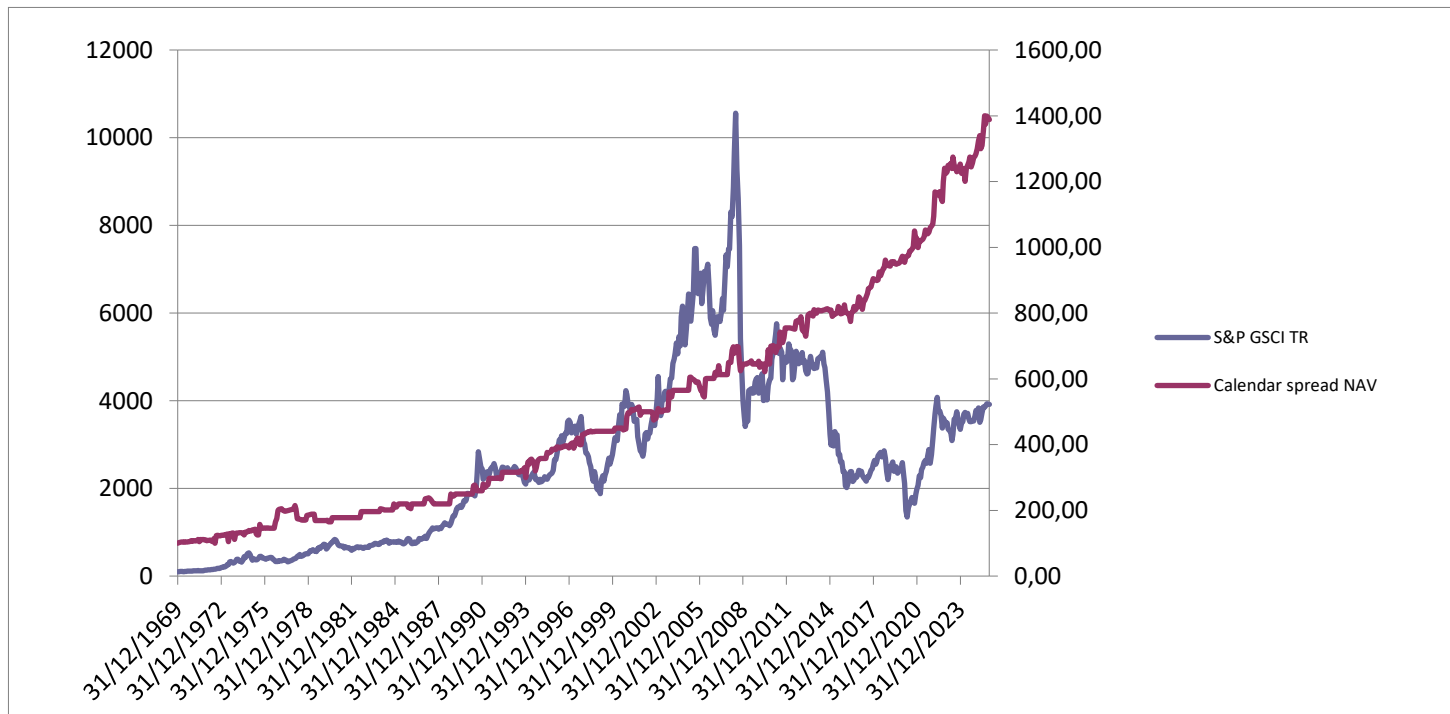


Gold futures nearby calendar spread



Calendar spread strategies NAV

(From 31-12-1969 to 31-12-2025)



Conclusion

Going forward, a tougher regulation for financial 'players' through the Commodity Futures Trading Commission (CFTC) rules, on November 5, 2013.

Annex 1 – VLLC ('Very Large Crude Carrier')



Investing in commodity markets

Annex 2 – S&P GSCI data – official web site

<https://www.spglobal.com/spdji/en/indices/commodities/sp-gsci/#overview>

Annex 3 – Information sources

Commodity	Source
Wheat	FAOSTATS
Corn	FAOSTATS
Soybeans	FAOSTATS
Lean Hogs	FAOSTATS and UN Data
Cattle	FAOSTATS and UN Data
Coffee	FAOSTATS
Sugar	U.S. Department of Agriculture (USDA)
Cocoa	FAOSTATS
Cotton	U.S. Department of Agriculture (USDA)
Gold	U.S. Geological Survey (USGS)
Silver	U.S. Geological Survey (USGS)
Crude Oil	UN Data and IEA
Natural Gas	UN Data and IEA
Aluminum	U.S. Geological Survey (USGS)
Copper	U.S. Geological Survey (USGS) and International Copper Study Group (ICSG)
Lead	U.S. Geological Survey (USGS)
Nickel	U.S. Geological Survey (USGS) and International Nickel Study Group (INSG)
Zinc	U.S. Geological Survey (USGS)

Annex 4 - Download data

It is also possible to integrate Bloomberg into Excel.

Useful for calculation and downloading data.

Practical application : how to add Bloomberg into Excel.

Close price = « PX_LAST »

Annex 5 – Electricity Map

<https://app.electricitymaps.com/map>